

Tree Facts and Folklore

Identification, Ecology, Uses (traditional and modern) and Folklore of Southeastern Trees

**by
Dan D. Williams**

Copyright 2014

**Possum Publications
Athens, Georgia**

Contents

Introduction-----	3
Tree Identification-----	4
How to Identify A Tree Using This Book-----	11
Tree Ecology And Forest Succession-----	12
Wood Properties-----	20
Trees With <u>Compound Leaves</u>-----	22
Native Shrubs With Compound Leaves-----	37
Trees With <u>Opposite Leaves</u>-----	38
Trees With <u>Simple Alternate Toothless Leaves</u>-----	57
Trees With <u>Simple Alternate Toothed Leaves</u>-----	71
Trees With Lobed Leaves-----	100
Pines and Other Conifers-----	124
Appendix I Shade Tolerance, Moisture Preference and Typical Canopy Position for Major Southeastern Trees-----	142
Appendix II Exotic Tree Key-----	149
Bibliography-----	151
Alphabetical Index -----	153

Introduction

This book is for folks interested in identifying Southeastern trees and learning about the growth and survival strategies that help them fit into the forest ecosystem. The book also tells how people have used trees over the years to make life better, and about interesting stories and folklore associated with some Southeastern trees.

The Tree Identification Primer teaches the reader three important but easy tree tips for identifying trees based on leaf arrangement characteristics:

simple leaves versus compound leaves

opposite leaves versus alternate leaves

leaf lobes versus leaf teeth

Major Southeastern trees are placed in one of six groups based on the above tree tips. A mnemonic (memory helper) for remembering the trees in each group is provided. This technique lets you quickly tell which group a tree belongs to, and gives you an easy way to remember the trees in the group. Once you know the tree's group, refer to the descriptions of trees in the group for more details.

There are many good tree identification guides available for more comprehensive information about identifying Southern trees. Native Trees of the Southeast, by L. Kathryn Kirkman, Claud L. Brown and Donald J. Leopold is one we recommend as an overall guide, and the one used as reference for this book.

The memory mnemonics in this book are both ridiculous and useful! Even an experienced dendrology (tree id) professor would have trouble naming the major trees in just one of the six tree groups without some kind of memory aid. Once you learn these silly sayings, you'll be able to rattle off the major trees in all six groups, a useful tool for tree identification as well as for verbal and written communication.

Tree Identification

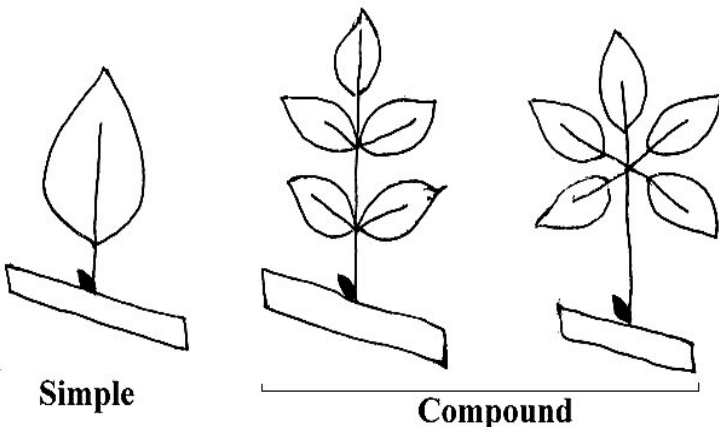
Once you **know** a tree well, you identify it just by looking at it. You visually take in its overall appearance, the same way you recognize your friends when you encounter them. They just have that “look”.

On the other hand, we **learn** to identify trees by using their growth characteristics to divide them into groups of similar trees; then, we match details of the tree's appearance with the descriptions in this and other books written for that purpose.

Here are the most important tree characteristics used to break trees into similar groups for identification:

Tip 1. Simple versus compound leaves

Tree leaves may be either simple or compound. **Simple** leaves are made of a single structure (a leaf) attached to a woody or semi-woody twig. **Compound** leaves are made of many identical leaflets attached to a (usually) flexible green leaf stem. Leaflets are almost always eye-shaped (elliptical).



It is not always easy to tell a leaf from a leaflet. Try this. Grab the possible leaf. Follow it back to where it attaches to the tree. If it attaches to a slender green flexible stem, then it is a leaflet. It will most likely be roughly eye-shaped, and one of many identical leaflets arranged opposite each other on the green flexible

leaf stem.

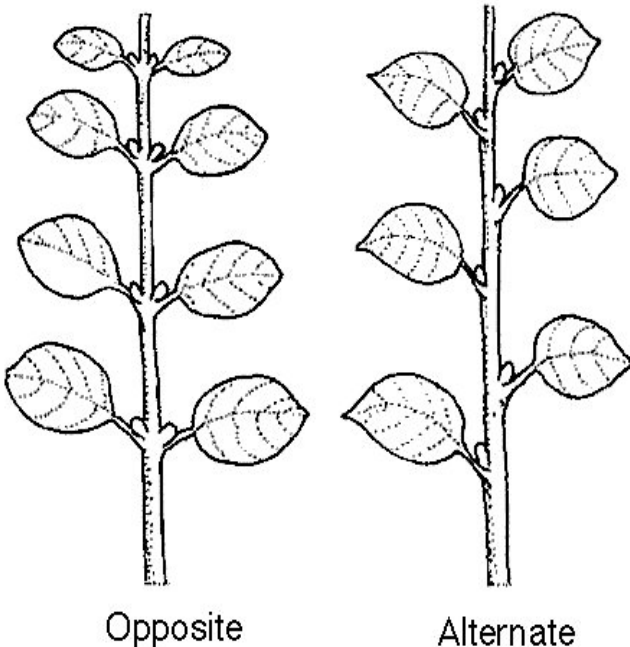
On the other hand, if the possible leaf attaches to a woody or semi-woody stem, then it really is a leaf. Simple leaves can have many shapes, including eye-shape.

Another clue to recognizing a leaf is its axillary bud. This is a bud situated at the base of the leaf where it attaches to the twig. If you find this bud, you know you have a leaf. Axillary buds can be confusing, though, because they are not always visible.

Some trees like ashes and hickories have swollen leaf bases, but most don't, so use this as a clue along with other information.

Tip 2. opposite versus alternate leaves

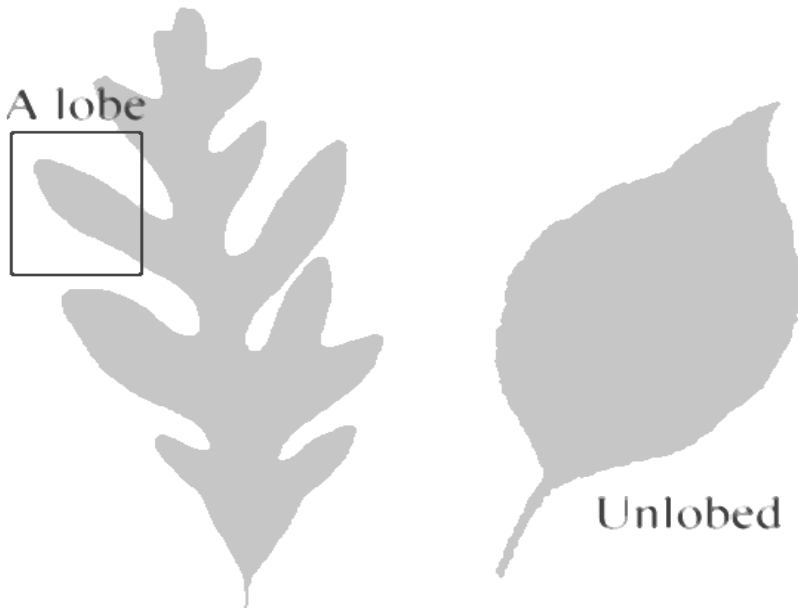
The leaves of all trees are arranged in one of two ways. **Opposite** leaves attach to the twig opposite each other. **Alternate** leaves attach to the twig one above the other (alternately). The drawing below illustrates the two arrangements. A tree's twigs and branches are also arranged in this way, a very helpful winter id tool.



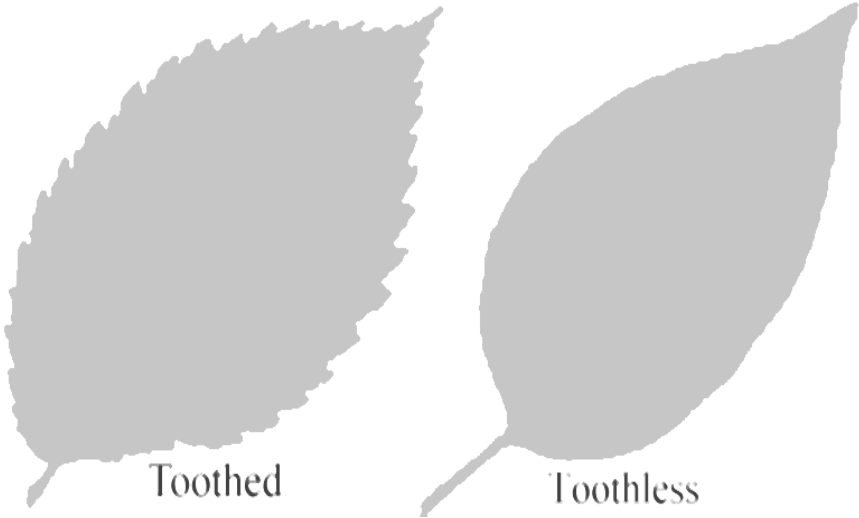
BEWARE! Leaflets on a compound leaf are almost always attached opposite each other on the flexible green leaf stem, but this does not mean the leaves are opposite. First determine if you are looking at simple or compound leaves, then decide if the leaves are opposite or alternate.

Tip 3. lobes and teeth

Some trees like oaks have **lobed** leaves. Lobes are large, usually rounded leaf projections separated by indentions. Other trees like dogwood have **unlobed** leaves.



Some trees like elms have **toothed** leaves with teeth located along the the leaf edge. Others like persimmon have **toothless** leaves.



Here is a list of Southern trees placed in 6 categories based on the tree tips just described:

Trees with Compound leaves (Page 22)

Hickory
Pecan
Black Locust
Honey Locust
Walnut
Boxelder
Ash

Useful Mnemonic:

"Hicks Peeking over a Low Wall at a Box of Ashes"

Trees With Opposite Leaves (Page 38)

Maple

Ash

Dogwood

Old Mans Beard (fringetree)

Buckeye

Catalpa

Paulownia

Useful Mnemonic:

“MADogs with Beards and Buckeyed Cats named Paul”

Trees with simple alternate toothless leaves (Page 57)

Pawpaw

Magnolia (including cucumbertree)

Red Bay

Redbud

Persimmon

Blackgum

Useful Mnemonic:

“My toothless Pawpaw found a Magic Red Persimmon that gave him Blackgums.”

Trees with simple alternate toothed leaves, group 1. (Page 71)

Elm

Birch

Hackberry

Hornbeam

Buckthorn

Silverbell

Serviceberry

Useful Mnemonic #1:

“Elmer is the son of a Birch who Hacked the Horn off a Buck with a Silver Service sword.”

Trees with simple, alternate toothed leaves, group 2.

(Page 87)

Willow

Alder

Holly

Sourwood

Cherry

Cottonwood

Basswood

Beech

Useful Mnemonic #2:

“Willy and Al Hollered and drank Sour Cherry wine when they caught a Cottonpickin' Bass at the Beach.”

Trees with lobed leaves (Page 100)

Sweetgum

Maple

Sassafras

Poplar

Mulberry

Sycamore

Oak

Useful Mnemonic:

“Sweet Mable, the Sassy and Popular Mulberry, got Sick in Oklahoma.”

Pines and other conifers (Page 124)

Pines with 5 needles per bundle:

only white pine

Pines with 2 and 3 needles per bundle:

slash pine has mostly 2 and 3 needles per bundle

shortleaf pine has mostly 2, rarely 3 needles per bundle

Pines with 3 needles per bundle:

“Lob a Long Pitch into the Pond.”

Loblolly, Longleaf, Pitch, Pond

Pines with 2 needles per bundle:

“Virginia will Spruce up the Table Shortly.”

Virginia, Spruce, Table-mountain, Shortleaf

Other Conifers (page 135)

Tree Bark Types

Smooth bark: Bark is smooth, lacking plates, ridges or scales. Beech, hornbeam and red maple are examples of trees with smooth bark.

Ridged bark: Bark has relatively flat-topped ridges separated by furrows. The ridges may sometimes be interwoven. Yellow poplar, sourwood and mockernut hickory have ridged bark.

Peeling Bark: Bark peels horizontally into thin strips. Yellow birch, river birch and pin cherry have peeling bark.

Platy bark: Bark consists of wide or narrow flat plates. Sweetgum, blackgum and winged elm have platy bark.

Shaggy/Flaky/Scaly bark: Bark separates into partially detached vertical strips, flakes or scales. Shagbark hickory, white oak and hophornbeam are examples.

Blocky bark: Bark breaks into squares or rectangles. Dogwood, persimmon and ash have blocky bark.

Fibrous bark: Bark forms thin loose strands that partially separate from the tree. Red-cedar, baldcypress and pond cypress have fibrous bark.

How To Identify A Tree Using This Book

When you encounter an unknown tree, first determine what is one leaf (page 4). This will tell you whether the tree has simple or compound leaves. Now decide if the tree has opposite or alternate leaves. Be sure to look at several leaf and twig examples in your investigation (not just one). Note whether the leaf has lobes and/or teeth. It's usually easy to tell if the tree is a pine or other evergreen conifer.

Now you have enough information to turn to the chapter in this book that best fits your tree's description so far:

Compound leaves	(page 22)
Opposite leaves	(page 38)
Simple alternate <u>toothless</u> leaves	(page 57)
Simple alternate <u>toothed</u> leaves	(page 71)
Lobed leaves	(page 100)
Pines and other conifers	(page 124)

Once in the appropriate chapter, simply thumb through the identification sections of the trees until you get a match. Notice the underlined portions of the tree description. They tell you the most important characteristics to look for to distinguish the tree from others, especially in winter. **Also notice the little gray squares beside each leaf silhouette.** They are one inch on a side and give you a general idea of the size of the leaf. Next, you might look the tree up in your favorite tree identification book.

If you're unable to identify the tree, it may be an introduced exotic. Check the table in the Appendix II, Southeastern Exotic Tree Key.

If this doesn't help, consult an expert:

email the author: williams@warnell.uga.edu
call your local county cooperative extension agent
consult your local botanical garden or nature center
look up your major state university and email them

Practice makes perfect and persistence pays off, so keep on trying and you will succeed!

Tree Ecology And Forest Succession

Each tree in this book is described in terms of its ecology. By this we mean the conditions of light, moisture, soil PH (acidic or basic) and soil nutrients the tree prefers as well as the mature tree's preferred vertical position in the forest. A tree's ecology determines its strategy for survival in the forest environment, and the interaction of trees based on their ecologies constitutes much of the study of forest ecology.

A tree's ecology can be broken down into the following factors:

Shade Tolerance (need for sunlight)

Shade Intolerant tree: Tree requires full sunlight from seedling to maturity. It will die if shaded for extended periods.

Shade Intermediate tree: Tree can germinate and grow to seedling stage in moderate or even full shade, but must have the moderate to full sunlight of an opening in the forest canopy in order to mature.

Shade Tolerant tree: Tree can germinate, grow to seedling stage and on to maturity in near full shade, though it can also prosper in full sunlight.

Soil Moisture Preference

Wet Site Tree: Tree can germinate and grow in soil that is saturated with water for extended periods during the year. Swamps and frequently-flooded river bottoms are wet sites.

Moist Site Tree: Tree grows best on sites that have a consistent supply of soil moisture, but also have good soil drainage so roots are rarely saturated. Lower slopes near streams and north-facing mid-slopes are usually moist sites.

Dry Site Tree: Tree can grow on dry sites with limited soil moisture and usually low levels of nutrients. Dry site trees prefer moist sites, but are out-competed by other trees on these sites. Ridge tops and exposed upper south-facing slopes are dry sites.

Soil PH and Soil Nutrients

Soil PH means whether the soil is acidic (like lemons and vinegar), neutral (like distilled water) or basic (like limestone and baking soda). A PH of 7 is considered neutral. Anything below 7 is an acidic PH, and anything above 7 is a basic PH. A soil PH between about 5.5 and 6.5 is considered ideal for plant growth.

Acidic Soil: Trees adapted to acidic soils can tolerate or may even prefer these soils. For example, members of the heath family like rhododendron, azalea, sourwood and blueberries prefer moderately acidic soils. Acidic soils are usually lower in nutrients than neutral and slightly basic soils because the acidity restricts the flow of nutrients to plants.

Most Southeastern soils are moderately acidic because the abundant rainfall leaches bases like calcium from the soil and carries them away. In the Western Blue Ridge Province of North Carolina, Tennessee and Georgia there are areas where sulfur-rich slate rocks such as those of the Anakeesta formation of the Great Smoky Mountains, release sulfuric acid into the soil, creating highly acidic soils that require some special adaptations for plants to thrive there.

Neutral to slightly Basic Soil: Most trees are well adapted to neutral and slightly basic soils which usually contain high levels of calcium. The calcium usually comes from limestone (or its metamorphic equivalent marble) the crushed and compressed bodies of shell and coral reef creatures that once lived in an ocean. Soils derived from basalt lava or its metamorphic equivalent amphibolite can also be fairly high in calcium. Neutral and slightly basic soils enhance the flow of nutrients to the tree, enabling it to prosper. Southeastern provinces like the Valley and Ridge, Cumberland Plateau and Coastal Plain are often geologically underlain by limestone, which produces neutral of slightly basic soils that are very good for plant growth.

Ultramafic Soil: Ultramafic soils contain high levels of magnesium and iron derived from ancient ocean crust that was thrust up on land during plate tectonic mountain-building events. They are usually found in narrow bands associated with geologic

thrust faults, and the high magnesium often gives them a fairly high PH. Ultramafic soils are usually thin and low in essential tree nutrients, so only trees and other plants adapted to high levels of magnesium and iron, and poor dry soil can survive. Pitch pine and blackjack oak are two tree examples.

Fire Tolerance Rating

Fire Intolerant trees are completely killed after one or more moderate fires. The tree is usually a weak sprouter.

Fire Resistant tree's above-ground parts may be damaged or killed by a moderate fire, but the tree will resprout readily from the roots and is not ultimately killed.

Fire Tolerant tree's above-ground parts may be scarred but are not killed by a moderate fire.

Fire Dependent trees require a moderate fire to open their fruit for seed dispersal and burn away forest litter, exposing mineral soil.

Vertical Position in the Forest Canopy

Dominant canopy trees like yellow poplar, most pines and white oak occupy the highest vertical position in the forest-- the canopy-- where they receive abundant sunlight. Typically, two or three dominant canopy tree species dominate the canopy in large numbers.

Dispersed canopy trees like black cherry, blackgum and white ash share the canopy with the dominants, but in fewer numbers and they are usually more widely dispersed throughout the forest. Both dominant and dispersed canopy trees can occupy lower vertical levels as they grow toward the canopy.

Subcanopy trees like sourwood and American holly reside just below the canopy dominants where they can survive on light that filters through the canopy.

Understory trees like dogwood and hornbeam live in the shade below the subcanopy. Subcanopy and understory trees don't move into the canopy. They are permanently adapted to the positions they occupy.

Forest Succession

Forests change as they get older. Young forests have a mix of tree types (also called tree species) that will slowly change as the forest ages. When the forest is finally mature it will have a different set of tree species from that of its younger self. This change in tree species over time is called forest succession, and it results from the changing conditions of light, moisture, nutrients and soil PH that occur as the forest matures. The trees themselves actually bring about these changes.

The following paragraphs describe five forests that help illustrate the stages of forest succession in Southeastern forests. Keep in mind that the time required for each stage can vary widely depending on site characteristics and past history.

The Developing Forest (0-50 years)

Forest succession begins when **pioneer** trees like pines, black locust, cherries and yellow poplar colonize abandoned land or re-sprout from logged land. Like the tough and enduring pioneers who first settled Southeastern forests, pioneer trees are a special breed adapted to the often harsh conditions found on abandoned, worn out land, including dry, nutrient-poor acidic soil. Pioneer trees require the full sunlight found on open sites to survive and grow, and they are called **shade intolerant** trees for this reason.

Pioneer trees grow fast. By about age 15 years their young crowns have grown together creating a closed, low canopy. The forest is quite dense with many small trees per acre composed of only a few different tree species, the vast majority of which are pioneer trees.

As the developing forest grows the pioneer trees grow taller, reaching mature height by age 50 years. During this time fierce competition for sunlight, water and nutrients has thinned the developing forest, creating a more open understory. Pioneer tree root systems have broken up the soil and brought nutrients and organic matter to the surface in the form of autumn leaves. Growing conditions are better than they were in the beginning, but not enough sunlight penetrates the canopy to support the sun-loving seedlings of pioneer trees. This ushers in the next stage of forest succession.

Pioneer Forest (50-100 years)

Pioneer trees continue to dominate the canopy of the pioneer forest, but another group of trees called **shade intermediate** trees begins to take dominance from the pioneers. Trees like oak, birch and red maple have become established as seedlings under the shade of pioneer trees, but they need the sunlight of forest openings to mature and grow into the forest canopy. Openings are provided by the death of pioneer trees due to competition, disease, flood, fire, wind and ice. Shade intermediate trees take advantage of the openings, rapidly growing into the canopy where they will eventually replace the pioneers.

This process occurs over the next 50 years as most of the short-lived pioneer trees die of old age. Yellow poplar, one of the few long-lived pioneer trees is an exception. A handful of poplars will remain to become giant monarchs in the old forest to come. After about 100 years, most of the forest canopy trees are shade intermediates with a few pioneer trees comprising the balance. The trees are fewer in number now, but average tree diameter is greater.

Intermediate Forest (100-200 years)

Trees intermediate in shade tolerance dominate the canopy during this stage, but another final wave of new trees slowly wrests dominance from them. **Shade tolerant** trees like hemlock, beech, sugar maple, buckeye and hickory become established as seedlings in the full shade of the forest floor. They can grow to maturity in the limited sunlight of the understory, though they do take advantage of openings not claimed by the faster growing intermediate trees. The slowest growers, and the longest lived members of the tree world, shade tolerant trees may remain in the understory for 100 years eventually rising into the canopy to share dominance with trees already there.

By now the forest has established a well developed subcanopy layer and an understory layer where trees specifically adapted to these environments grow, and where future canopy trees waiting for new openings bide their time. In addition, the soil has become deeper and more fertile as a result of the forest building process.

Mature Forest (200-250 years)

By age 200, shade tolerant trees equal or exceed intermediate trees

in the forest canopy. Shade tolerant trees will dominate the forest canopy from now on accompanied by a few intermediates and one or two old yellow poplars. The number of trees per acre has greatly declined as pioneer and intermediate trees have succumbed to competition and old age, but average tree size has increased considerably with many in the 30-inch plus diameter range. By now the forest floor has developed a thick layer of humus-rich soil that supports a flourishing non-woody plant community.

Climax Forest (250 + years)

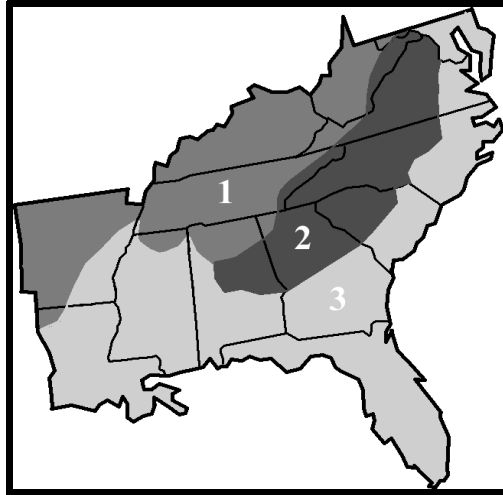
The climax forest is the ultimate expression of forest development. Old age and the struggle for survival have greatly thinned the climax forest, but the remaining trees are quite large and show signs of advanced age like lichen encrusted bark, dead limbs in the canopy, hollow trunks and lightening blasted tops. When the ancient giants fall their root masses come out of the ground leaving pits. When these root masses decay creating mounds, a pit and mound topography develops that is characteristic of the climax forest as is the abundance of decaying logs on the forest floor.

Openings created in the climax forest by natural disturbances let sunlight down into lower levels where forest succession begins anew. Small openings created by the death of one or two trees are usually filled by intermediate trees because they can grow into the gaps faster than shade tolerant trees. Medium sized gaps like those created by wind and ice allow enough space for shade tolerant trees to move up along with the intermediates. Large openings like those created by fire, drought and disease favor pioneer trees by providing plenty of sunlight for pioneer tree seedling establishment and growth. Over time this creates a patchwork of mostly ancient forest interspersed with patches of younger trees of various ages. This multiple-age structure is the hallmark of the climax forest.

Soil moisture has an extremely important effect on forest succession. Each soil moisture situation (wet, moist, dry) has its own particular series of tree species involved in forest succession on that site. For example, wet site forest succession usually begins with black willow and hazel alder, shade intolerant pioneer trees on wet sites. Later, American sycamore and river birch, shade intermediate wet site species take over, and finally shade tolerant trees like bitternut hickory dominate the mature wet site forest.

Appendix I (page 142) shows shade tolerance, moisture preference and typical canopy position for major Southeastern trees in table form. This table is useful in the forest for determining the successional stage (forest age) and site moisture characteristics of a forest you encounter.

Major Southeastern Provinces and Forest Types



This map shows the physiographic provinces of the Southeast. The medium gray area on the upper left (1) represents the Cumberland Plateau and Valley and Ridge provinces. These two provinces have long narrow ridges usually made of sandstone. The ridges are separated by long valleys made of limestone.

The dark gray area in the middle (2) represents the Blue Ridge and Piedmont Provinces. Mountainous terrain dominates the northwestern portion of this area (Blue Ridge), while rolling hills form the southeastern section (Piedmont). The Blue Ridge and Piedmont provinces are made of metamorphic rock like granite gneiss and quartz-muscovite schist. No limestone is found there. The light gray area on the right (3) represents the Coastal Plain province, an area once covered by ancient oceans. Large areas of limestone can be found here, but most of the soil is sandy or alluvial. The Georgia Coastal Plain includes the half-million-acre Okefenokee Swamp, home of the Cypress-Gum Forest.

We will use the following acronyms for the physiographic provinces in the tree descriptions in this book:

Cumberland Plateau (CUP), Valley and Ridge (V&R), Blue Ridge (BR), Piedmont (P), Coastal Plain (CP).

Here is a list of Southeastern forest types by province. The tree descriptions found later in the book will refer to these forest types by name.

Cumberland Plateau (CUP) and Valley and Ridge (V&R)

Oak-Pine Forest (*has more oak than pine*)

Oak-Hickory Forest (*oaks and hickories dominate mature canopy*)

Calcareous (calcium-rich) Forest (*usually underlain by limestone*)

Cove Forest (*found along creeks and large streams*)

Bottomland Forest (*hardwoods, found along rivers, swamps*)

Blue Ridge Mountains (BR)

Red Spruce-Fraser Fir Forest (*found on highest peaks in E. U.S.*)

Northern Hardwood Forest (*beech, birch, buckeye forest*)

Cove Forest

Acidic Cove Hemlock Forest (*hemlock-dominated forests*)

Bottomland Forest

Oak-Hickory Forest

Pine-Oak Forest (*has more pine than oak*)

Piedmont (P)

Pine-Oak Forest

Oak-Hickory Forest

Cove Forest

Bottomland Forest

Coastal Plain (CP)

Pine-Oak Forest

Evergreen Oak Forest (*scrub forests on dry sandy sites*)

Longleaf Pine-Wiregrass Forest

Bottomland Forest

Maritime Forest (*live oak-dominated*)

Cypress-Gum Swamp

Uses: Wood Properties

The wood physical properties of the trees in this book are listed in the “Uses” section for each tree, for example, strong wood, hard wood, flexible wood. The following paragraphs describe the major properties of wood and provide examples. Most of this information comes from Horace Kephart's book, *Camping and Woodcraft*.

Very Hard Wood: The wood is physically very hard like the proverbial rock, and is also usually heavy, strong and tough. It is difficult to work with tools, but can hold up under hard use including physical shocks without breaking or splitting. The wood becomes smooth and polished with use. Very hard wood makes good mauls and mallets, wooden machine parts like the gears shuttles and bobbins of textile mills and wooden wedges (gluts). This category includes dogwood, hornbeam, hophornbeam, black locust, Osage orange, post oak, overcup oak, sugar maple, persimmon, serviceberry, blackjack oak, chestnut oak, winged elm and hickory.

Strong Wood: The wood strongly resists bending and breaking. It is shock-resistant and hard enough to take a beating in use without failing. Strong/tough wood is good for a wide variety of uses including, furniture, flooring, tool handles, lumber and sports equipment. This category includes hickory, ash, Osage orange, serviceberry, yellow pines, red oaks, sugar maple, birches, beech, oaks, walnut, black cherry, honeylocust and elms.

Soft Wood: Soft wood is easily worked with hand tools and is usually light weight as well as soft. It may also be strong for its weight. Soft woods are good for paper pulp, lumber, veneer (thin sheets of wood cut with a rotary blade that come off the blade like paper towels off a roll), plywood, cabinets, carvings, artificial limbs, utensils like bowls, dugout canoes and toys. Included in this category are spruce, white pine, willow, yellow pine sapwood, pawpaw, Fraser fir, catalpa, buckeye, basswood, yellow poplar, cucumbertree, cottonwood, boxelder and magnolias.

Wood easy to split: Easily split woods are good for short boards, puncheon floors in cabins, split logs for cabin walls, wooden shingles, firewood and splits for baskets and chair bottoms. This category includes basswood, red cedar, chestnut, slippery elm (green), hackberry (if straight-grained), spruce, ash, beech (green), black birch (green), dogwood (green), Fraser fir and oaks.

Wood difficult to split: This category includes woods with interlocking grain that will not readily respond to the wedge. These woods are good for chopping blocks, anvil bases, wooden ship keels, wagon wheel hubs and spokes and steam-bent furniture. This category includes buckeye, elms, boxelder, black cherry, sweetgum, blackgum, hemlock and sycamore.

Flexible/Pliable Wood: Flexible/pliable woods can be permanently bent into useful shapes and remain strong without breaking. These woods make good bentwood furniture, barrel hoops, stone axe handles, snowshoes, rope, cordage and fiber for cloth. Indians and Pioneers typically used young tree sprouts for rope, cordage and cloth fiber. Trees in this category include basswood, mulberry, slippery elm, winged elm, hackberry, redbud and yellow poplar.

Springy Wood: Springy woods spring back to their original shape after being temporarily bent without losing their elasticity. Springy woods are excellent for hunting bows, tool handles and old-fashioned lathe poles. They include osage orange, hickory, white oak, dogwood, hornbeam, hop hornbeam, serviceberry, black locust, honey locust, redcedar and ash.

Rot Resistant Wood: Resists rotting when placed in the ground, under water or when constantly exposed to wet weather. Rot resistance usually refers to the heartwood of the tree, not the sapwood. Rot resistant woods are used for fence posts, railroad ties, posts for structures like barns, roof shingles (if the wood splits well), water pipes and underwater pilings. Included here are alder (under water only), catalpa, black cherry, elms (under water only), honeylocust, mulberry, chestnut oak, post oak, Osage orange, pitch pine (heartwood), redcedar, chestnut, cypress, black locust, hophornbeam, overcup oak, white oak, yellow pine (heartwood), sassafras and black walnut.

Trees With Compound Leaves

(hickory, pecan, black locust, honey locust, black walnut, boxelder maple, green ash, white ash, mountain ash)

"Hicks Peeking over a Low Wall at a Box of Ashes"

For native shrubs with compound leaves see page 37.

HICKORIES

There are about 20 species of hickory world wide, with 17 occurring exclusively in the eastern U.S., making hickory a truly American tree today. The fossil record tells a different story, however, documenting its former presence in Central Europe, China, the former Soviet Union, the Western U.S. and Alaska.

Hickory trees rank number 1 in several applications. The wood makes the best tool handles due to its toughness and shock resistance. Ash reigned supreme for axe handles until Europeans came to America and found hickory. Southeastern Indians preferred hickory for their powerful longbows due to its superior springiness. There is no better native wood for smoking and cooking meat than hickory, and no tastier or more valuable native American tree nut than the pecan.

Hickories arrive in the forest late in its long successional journey, co-dominating the mature and climax oak-hickory forests, of the eastern U.S. In fact, numerous large hickory trees in a forest usually indicate the forest is a century or more old. Our ancient oak-hickory forests were vital to the economy of the Southeastern Indians, providing a food bounty that included white oak acorns, hickory nuts, persimmons, mulberries, buckeyes (edible when roasted), pawpaws and beechnuts.

Bitternut Hickory (*Carya cordiformis*), Walnut Family

Identification (page 4):

Leaves: compound (7, 9 & 11 leaflets), alternate, leaflets toothed, lower surface fuzzy, fall color = yellow

Flowers: tiny, borne on catkins (slender drooping spikes), late spring

Fruit: nearly round bitter nut with a thin yellowish husk, mature in fall

Twigs: slender, twig end buds made of 2 sulfur yellow bud scales, bud scales valvate (in 2 halves like a clam shell)

Bark: smooth or slightly ridged, often with whitish patches

Ecology (page 12):

Shade tolerant, Dispersed Canopy tree

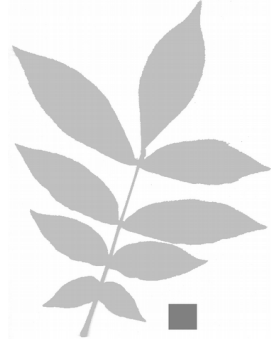
Moist sites along creeks & rivers

Fire Intolerant-Resistant, Prolific Sprouter

Mature and Climax forests

Bottomland F., Cove Hardwood F.

All Southeastern provinces



Uses (page 20):

Hickory wood is very hard, very strong, springy, shock-resistant, splits well only when green, lacks rot resistance, burns hot and long whether green or dry, making excellent coals for cooking or smoking food. The wood is excellent for tool handles, farm and sports equipment, wagon frames, hubs and spokes and firewood.

Bitternut hickory wood is considered more brittle than other hickories and inferior to them for tool handles and other uses, but it is considered superior for smoking meat.

Pioneers pressed oil from the bitter nuts they used to treat rheumatism and for burning in their primitive button lamps.

Comments:

Bitternut hickory is our only native hickory that grows primarily along streams. It is one of the fastest growing hickories, but is the most short-lived hickory, reaching only about 200 years.

Bitternut is closely related to the pecan tree (also a hickory), sharing many of its characteristics except its valuable edible nut.

Bitternuts are so bitter, many wildlife species won't eat them. This enhances their chance of survival.

Pignut Hickory (*Carya glabra*), Walnut Family

Identification (page 4):

Leaves: compound (5 leaflets per leaf), alternate, leaflets toothed, leaves hairless, fall color = yellow

Flowers: catkins (slender drooping spikes), late spring

Fruit: pear-shaped hickory nut with thin black husk that usually splits only partially open after falling in autumn.

Twigs: slender and hairless

Bark: ridged, with an irregular diamond pattern

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

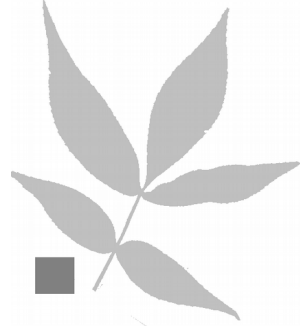
Dry to moist sites on upper slopes

Fire Intolerant-Resistant, Prolific Sprouter

Mature and Climax forests

Oak-Hickory Forest

All Southeastern provinces



Uses (page 20):

See bitternut hickory (page 23) for general hickory wood characteristics and uses.

Pioneers made wagon wheel hubs of hickory, and Daniel Boone, legendary woodsman and long hunter of the late 1700's, roasted venison over the fire on his rifle's ramrod made of hickory wood.

Pignut was once called broom hickory, and considered superior for sweeping and scrubbing brooms. A knife was used to shave thin 10-inch-long strips all around the end of a 5-foot-long, 1 1/2-inch-thick hickory sapling, leaving the upper end of the strips still attached. Next, the core from which the strips were shaved was cut off, creating a broom for sweeping or scrubbing cabin floors.

Pignut hickory nuts are small and hard, but the sweet meat is eaten by many wild animals. John Lawson, who traversed the Carolinas in 1700, wrote, "Hiccory nuts have very hard shells, but excellent sweet kernals with which in a plentiful year, the old hogs that can crack them fatten themselves and make excellent pork."

Comments:

The name pignut has 2 possible origins. The nuts were once considered fit only for pigs. Alternatively, the nut and its enclosing husk resemble a pig's head in shape. Take your pick, or both!

Red Hickory (*Carya ovalis*), Walnut Family

Identification (page 4):

Leaves: compound (5 & 7 leaflets), alternate, leaflets toothed, leaflets hairless, aromatic, fall color = yellow

Flowers: inconspicuous, borne on catkins (slender drooping spikes), late spring

Fruit: hickory nut with thin to medium-thick black husk that splits completely open after falling in autumn

Twigs: slender, hairless

Bark: shaggy, but much less so than shagbark and shellbark

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

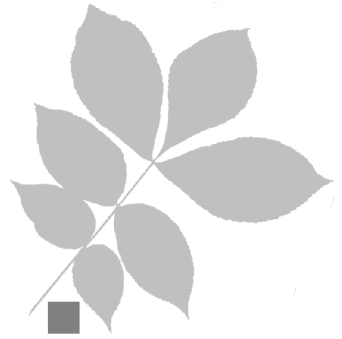
Moist to dry sites on slopes

Fire Intolerant-Resistant, Prolific Sprouter

Mature and Climax forests

Oak-Hickory Forest

All Southeastern provinces



Uses (page 20):

See bitternut hickory (page 23) for general wood characteristics.

John Lawson, who traversed the Carolinas in 1700 says, "There is another sort [of hickory] we call red hickory, the heart thereof being very red, firm and durable; of which walking sticks, mortars, pestils and several other fine turnery-wares are made."

Indians made hickory war clubs and 60-inch-long hickory bows that pulled about 50 pounds. In the 1530's, Spanish explorer Cabeza de Vaca saw an arrow shot from an Indian bow penetrate 6 inches into the base of a poplar tree.

Comments:

Red hickory is not as abundant as pignut and mockernut. Some authorities consider it a variety of pignut hickory, hence the alternate common name of false pignut hickory.

Red hickory nuts are between pignut and mockernut in size.

All Southern hickories and pecan are attacked in late summer by the hickory twig girdler (*Oncideres cingulata*), a type of longhorn beetle that lays eggs in a living twig and then cuts the twig off the tree by chewing around it, creating a distinctive circular cut pattern. The young beetle larvae use the severed twig as food.

Shagbark Hickory (*Carya ovata*), Walnut Family

Identification (page 4):

Leaves: compound (5 leaflets), alternate, leaflets toothed,
leaves hairless with a faint apple smell, fall color = yellow

Flowers: borne on catkins (slender drooping spikes) in late spring

Fruit: large hickory nut with thick black husk that splits
completely when mature in autumn

Twigs: thick, twig end bud long and hairy

Bark: very shaggy

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

Moist sites along streams & lower slopes

Fire Intolerant-Resistant, Prolific Sprouter

Prefers calcium-rich soil

Mature and Climax forests

Oak-Hickory F., Calcareous F.

All SE provinces except very rare in CP



Uses (page 20):

See bitternut hickory (page 23) for general wood characteristics.

Cherokee Indians used hickory bark to finish off baskets. They drank a hickory bark "tea" for strength during sacred ball games.

In 1640, British botanist William Strachey wrote of the hardness of the shell, the sweetness of the nuts, and described how the Indians made "pawcohicora" (hickory milk) from shagbark nuts by boiling the pounded nuts (shell-and-all) in water, and adding this liquid to venison broth, creating an oil-rich nutritious soup.

The shaggy bark of shagbark hickory has been made into a wild tasting syrup by heating bark chips, then boiling with sugar and water. It supposedly resembles maple syrup in flavor.

Comments:

Shagbark prefers calcium-rich soils, so it is more common in the Cumberland Plateau and Valley and Ridge where limestone is abundant than in the Blue Ridge and Piedmont. It is rare in the CP. A relatively rare Southeastern hickory, shellbark hickory (*Carya laciniosa*) has shaggy bark like shagbark, but it has oval instead of round nuts and 7-9 leaflets per leaf instead of the 5 leaflets of shagbark. Shellbark nuts are darker and its twigs much stouter.

Sand Hickory (*Carya pallida*), Walnut Family

Identification (page 4):

Leaves: compound (7, sometimes 9 leaflets), alternate, leaflets toothed, leaflets dotted beneath with tiny silvery scales (use a hand lense), leaf stem hairy, fragrant, fall color = yellow

Flowers: borne on catkins (slender drooping spikes) in late spring

Fruit: hickory nut with black husk covered with hairs and tiny yellow resinous scales (use a hand lense), mature in fall

Twigs: slender, end bud covered with small silvery scales (only visible in fall, use a hand lense)

Bark: interwoven ridges and furrows

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

Dry, nutrient-poor sandy sites

Fire Resistant, Prolific Sprouter

Intermediate & Mature Forests

Oak-Pine F.& Oak-Hickory F.

All SE provinces except very rare in CP



Uses (page 20):

See bitternut hickory (page 23) for general wood characteristics.

Sand hickory is little used today except for fire wood.

The small but sweet nuts along with other hickory nuts and walnuts were once an occasion for "nut crackings", where young people gathered to crack the nuts with flat irons (usually used for ironing clothes), and to play "kissing" games!

Comments:

Sand hickory is not abundant in most oak hickory forests, but it does dominate on thin, dry, nutrient-poor sites like those of the Cumberland Plateau underlain by sandstone. It is also found on dry rocky ridge tops and sandhills in the Piedmont.

Sand hickory has hairy leaf stems and aromatic leaves like mockernut hickory, but usually only 7 leaflets per leaf and the leaflets are smaller than those of mockernut.

The aromatic leaf smell of sand hickory is related to the tree's tetraploid (3 chromosome sets) nature, a condition it shares with shagbark, red and mockernut hickories. Other hickories are diploid (2 chromosome sets) and are far less aromatic.

Mockernut Hickory (*Carya tomentosa*), Walnut Family

Identification (page 4):

Leaves: compound (7 & 9 leaflets), alternate, leaflets toothed, hairy and with an aromatic smell, fall color = yellow

Flowers: borne on catkins (slender drooping spikes) in late spring

Fruit: large V. hard tan hickory nut with thick black husk that splits completely open when mature in autumn

Twigs: thick and hairy

Bark: thickly ridged and interwoven

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

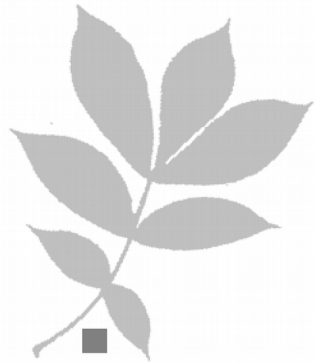
Moist to dry sites on lower & upper slopes

Fire Intolerant-Resistant, Prolific Sprouter

Intermediate, Mature & Climax forests

Oak-Hickory Forest

All Southeastern provinces



Uses (page 20):

See bitternut hickory (page 23) for hickory wood characteristics.

Mockernut hickory wood is stronger than all other hickories.

Naturalist William Bartram told how Southeastern Indians pounded mockernut hickory nuts (shell-and-all), put the pieces in boiling water and extracted an oil-rich creamy substance called hickory milk used as an ingredient in corn and hominy cakes.

Bartram saw over 100 bushels of hickory nuts in one lodge alone.

John Lawson, who visited the Carolinas in 1700, says, "They take these nuts and break them very small betwixt two stones till the shells and kernels are indifferent small, and this powder you are presented with in little wooden dishes. The kernal dissolves in your mouth, and the shell is spit out."

Comments:

Mockernut is probably the tree William Bartram referred to as "shellbarked hickory" in his Georgia Travels, since today's shellbark (*Cary laciniosa*) is rare in Georgia. His Latin name for "shellbarked" (really mockernut) was *Juglans exaltata*.

The name mockernut refers to the hard thick shell containing but little meat, a "mockery" to the hungry traveler! Another version says the name "moker-nut" is Dutch meaning "heavy hammer nut".

Pecan (*Carya illinoensis*), Walnut Family

Identification (page 4):

Leaves: compound (9 to 17 leaflets), alternate, leaflets toothed, leaflets are curved (sickle-shaped), terminal leaflet always present, fall color = dull yellow

Flowers: borne on catkins (slender drooping spikes) in late spring

Fruit: the familiar edible pecan, mature in fall

Twigs: medium thick

Bark: has flattened scaly ridges

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

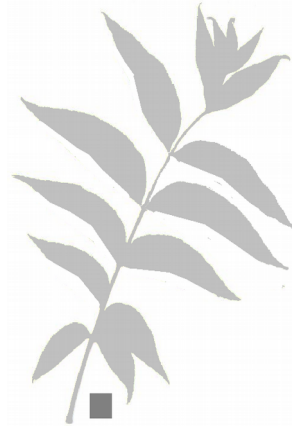
Moist sites along streams and river bottoms

Fire Intolerant, Prolific Sprouter

Mature and Climax forests

Planted in rows in commercial plantations

Now growing in all Southeastern provinces



Uses (page 20):

The commercial pecan and occasionally flooring and cabinets.

Hickories are divided into 2 groups based on genetic properties.

True hickories include shagbark, pignut, mockernut, shellbark and red. Pecan hickories include bitternut, pecan, nutmeg hickory and water hickory. True hickories have stronger wood.

Comments:

Pecan is not native to the Southeast. Its original range covers the Mississippi River valley from Illinois down to Texas. The valuable nuts quickly made it a desirable and frequently planted tree.

Hernando DeSoto encountered and described pecan trees and pecan nuts in 1541 after he crossed to the west side of the Mississippi River.

Both Thomas Jefferson and George Washington planted pecan trees on their estates. In those days the tree was called the "Illinois nut" tree due to its Mississippi River drainage origin.

In the days before pecan plantations, harvesters would actually cut down huge old wild trees to get at the nuts more easily. Many an ancient pecan tree was killed like this for just 1 crop of nuts!

Shellbark hickory (*Carya laciniosa*) hybridizes with pecan, producing an edible nut sold commercially as the 'Burton' hecan.

Black Locust (*Robinia pseudoacacia*), Bean Family

Identification (page 4):

Leaves: compound (7 to 21 oval leaflets), alternate, leaflets toothless, fall color = yellow

Flowers: large white, pea-like in late spring

Fruit: flattened brown bean-like pod 2-4 inches long, late summer

Twigs: zigzag with rosebush-size thorns

Bark: gray brown with thick scaly ridges

Ecology (page 12):

Shade Intolerant Pioneer

Dominant or Dispersed Canopy tree

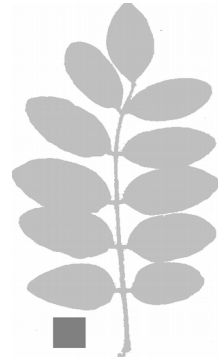
Dry-moist exposed mountain slopes

Fire Intolerant, Prolific Sprouter

Developing, Pioneer & Intermediate forests

Oak-hickory F., Oak-Pine F., Pine-Oak F.

All SE provinces, but rare in Coastal Plain



Uses (page 20):

Black locust wood is very hard, strong, springy, difficult to split green or dry and has very rot-resistant heartwood. It was once the primary wood for attaching glass electrical insulator pins to power poles due to its durability and all-weather stability.

Indians made excellent bows from the strong springy wood.

Pioneers used the rot-resistant heartwood for fence posts which were said to, “last twice as long as the hole”, and building posts.

The tree enriches the soil by fixing nitrogen and storing it in roots.

Comments:

Black locust trees often have a beat-up look. The bark is black with pieces sloughing off, and the crown is thin and brown. This condition is mainly attributable to the locust borer and the locust leaf miner. They attack the tree regularly, seldom killing it, but leaving its trunk deformed and its leaves brown. A drive through the mountains in late summer frequently reveals conspicuous groups of brown-leaved black locust trees suffering from effects of the locust leaf miner, at its peak this time of year.

Black locust is the pioneer tree on exposed Southern mountain mid-slopes. Yellow poplar does this job on lower slopes and in coves and pines do it on exposed upper slopes and ridge tops.

Honeylocust (*Gleditsia triacanthos*), Bean Family

Identification (page 4):

Leaves: twice compound (leaflets have leaflets), alternate, leaflets toothless, fall color = yellow

Flowers: small and greenish white to greenish yellow, spring

Fruit: long flattened twisted brown bean pod 12-18" long, fall

Twigs: zigzag with large round multiple thorns

Bark: scaly ridges with large round multiple thorns

Ecology (page12):

Shade Intolerant, Pioneer tree

along fences, roadsides and field edges.

Moist to dry sites, likes calcium-rich soil

Fire Intolerant, Prolific Sprouter

Developing forest and Pioneer forest

Scattered through all Southeastern provinces,



Uses (page 20):

Honeylocust wood is hard, strong, springy, hard to split if seasoned and rot-resistant.

Cherokee Indians made hunting bows from the springy wood.

They used pulp from the fruit pod as a sweetener, and ate the young pods and seeds. Cherokee Indian ball players bathed in bark tea to ward off tacklers, a reference to the tree's thorny trunk.

Pioneers used it for fence posts, and made locust beer (metheglin) from the pods. They used the long hard thorns as clothing pins called "confederate pins", during the Civil War hard times.

Thornless varieties are used as ornamental trees due to honeylocust's extreme hardiness in marginal urban settings.

Comments:

Honeylocust's name comes from the rattling of the pods that sounds like locust insects according to Donald Peattie.

American pioneers believed John the Baptist ate honeylocust fruit instead of grasshoppers when he lived in the Judean wilderness.

Unlike other bean family members, honeylocust does not fix nitrogen.

The natural range of honeylocust follows the Cumberland Plateau, where calcium-rich soils are found, but Indians, Pioneers and hungry animals introduced it to the rest of the Southeast.

Black Walnut (*Juglans nigra*) Walnut Family

Identification (page 4):

Leaves: compound (15 to 23 leaflets), alternate, leaflets not sickle-shaped, end leaflet often absent, fall color = yellow

Flowers: inconspicuous, early spring

Fruit: a round black walnut in a thick green husk slightly smaller than a tennis ball, late summer

Twigs: have chambered pith (twig center has tiny black lined chambers)

Bark: black and deeply ridged and furrowed

Ecology (page 12):

Shade Intolerant, Dispersed Canopy tree

Moist, well-drained sites

Fire Resistant, Prolific Sprouter

Developing, Pioneer & Intermediate forests

River Cove F., Bottomland F.

All Southeastern provinces



Uses (page 20):

The finest cabinet wood in North America due to its strong durable rich-grained wood that is easily sawn, planed and chiseled, takes stain well and warps and splits very little in drying.

Indians prized the oil-rich edible nuts and used the inner bark and nut husk as a laxative and tapeworm treatment.

Pioneers used the nut oil as hair oil, and made a yellow clothing dye from green husks and bark. They made brown dye from the green husks they used in their hair to cover gray, and they made a leaf infusion (material soaked in water) to get rid of "bedbugs".

Comments:

Black walnut roots produce the chemical juglone that inhibits growth of competing tree roots, an adaptation called allelopathy.

Naturalist William Bartram consistently found black walnut growing in abandoned Indian fields. He was convinced the Indians cultivated black walnut and other useful trees and shrubs.

Along with black walnut, Bartram also found persimmon, honeylocust, Chickasaw plum, French mulberry (*Callicarpa americana*), red mulberry and mockernut hickory which Bartram called "shellbarked hickory, *Juglans exaltata*".

Boxelder Maple (*Acer negundo*), Maple Family

Identification (page 4):

Leaves: compound (3-5 leaflets), opposite, leaflets slightly lobed & toothed, looks like poison ivy, fall color = brown

Flowers: inconspicuous, early spring

Fruit: abundant paired winged seeds (“helicopters”), autumn

Twigs: very green and hairless

Bark: gray-brown, smooth

Ecology (page 12):

Shade Tolerant, but acts like a Pioneer
on moist open sites

Subcanopy tree in the forest

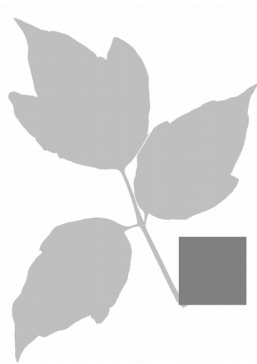
Fire Resistant, Prolific Sprouter

Moist to wet sites

Pioneer and Intermediate forests

River Cove F., and Bottomland F.

All Southeastern provinces except CP



Uses (page 20):

Boxelder wood is soft, weak and brittle, with no rot resistance.

Indians tapped the tree for sugar, and made an inner bark tea to induce vomiting. They made boxelder wood charcoal used for ceremonial painting and tattooing.

Boxelder has been planted as a shade tree because of its fast growth rate, but it is short-lived, falling apart in 50 years, and it has little autumn leaf display. These drawbacks preclude its use as a practical ornamental tree.

The tree is surprisingly hardy in the Midwest, where it is used as a shade tree and was once tapped for sugar as a substitute for that of the sugar maple whose range excludes this region.

Comments:

Boxelder is our only native maple with compound leaves.

Donald Peattie says, “The box elder probably takes its name from its elder-like leaves and its wood like that of the box tree.”

Boxelder along with red maple and silver maple are considered soft maples by the wood using industry because the wood of these trees is softer and considered inferior to that of sugar maple and black maple (*Acer nigrum*), both considered hard maples.

Green Ash (*Fraxinus pennsylvanica*), Olive Family

Identification (page 4):

Leaves: compound (5-9 (usually 7) leaflets), opposite, leaflets toothed, fall color = yellow

Flowers: small in compound clusters that appear in early spring

Fruit: two-inch-long narrow yellowish single-winged seed, wing extends about **half way up** seed body (see white ash)

Twigs: greenish and hairless, axillary bud is on top of twig leaf scar (*axillary bud is the bud located where the leaf attaches*), twig buds are brown and fuzzy

Bark: gray-brown, with square plates that a finger nail can dent

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

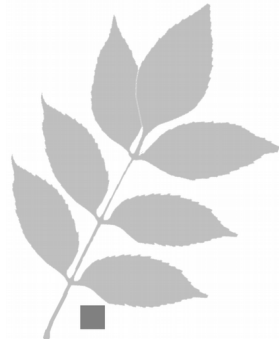
Moist & wet sites

Fire Resistant, Prolific Sprouter

Pioneer, Intermediate and Mature forests

River Cove F., Bottomland F.

All Southeastern provinces



Uses (page 20):

Wood is tough and springy, but less so than white ash. Green ash splits easily green or dry and burns readily green or dry. It is light, but slightly heavier than white ash, and has many of the same uses. Swamp-grown green ash is probably the "swamp ash" guitar makers tout as an excellent wood for electric guitar bodies.

Comments:

Green ash is abundant in the Coastal Plain and Piedmont. It is less abundant in the mountains where white ash is by far more common above about 2500' elevation.

Both green ash and white ash are members of the olive family in common with the cultivated edible olive.

The ash tree has been mistakenly associated with Norse mythology which actually refers to the European Yew (*Taxus baccata*).

Ash trees sprout prolifically after cutting, producing abundant leaves and long straight root sprouts. Since Neolithic times

European peoples have used the leaves for animal fodder, the long straight shoots for spears, bows and arrows and the larger stems as an excellent renewable fuel source.

White Ash (*Fraxinus americana*), Olive Family

Identification (page 4):

Leaves: compound (5-9 (usually 7) leaflets), opposite, leaflets toothed, fall color = yellow

Flowers: small in compound clusters that appear in early spring

Fruit: two-inch-long narrow yellowish single-winged seed, wing extends from the end of the seed body, summer

Twigs: greenish and hairless, axillary bud is at least partially enveloped by leaf scar (*axillary bud is the bud located where the leaf attaches*), twig buds are black and hairless

Bark: gray-brown, with square plates that a fingernail can dent

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

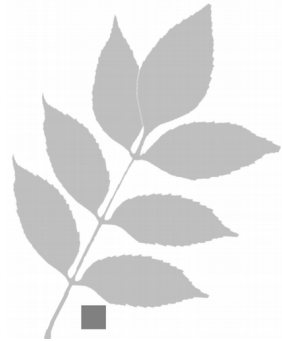
Moist to wet sites

Fire Resistant, Prolific Sprouter

Pioneer, Intermediate and Mature forests

River Cove F., Cove Hardwood F.,
Bottomland F.

All Southeastern provinces, but mostly in BR
and CUP, and rare in Coastal Plain



Uses (page 20):

White ash wood is strong, lightweight, springy, splits easily green or dry and burns readily green or dry. It was considered the best all around campfire fuel in any season, green or dry by Horace Kephart, author of Camping and Woodcraft (1917).

Modern uses include baseball bats (ash is the best), bentwood furniture, paddles, oars, tennis racquets, hockey sticks, snowshoes, tool handles, veneer and even steering wheels!

Ash is not as strong as hickory, so it is less used than hickory for ax and hammer handles.

Indians made bows from the straight-grained, springy wood, and drank a tea made from the buds to treat snakebite.

Comments:

The Asian emerald ash borer beetle was introduced into the U.S. and has destroyed thousands of ash trees in the North. It is now (2014) working its way down into the Southeast. This exotic beetle readily kills both green and white ash.

Mountain-Ash (*Sorbus americana*), Rose Family

Identification (page 4):

Leaves: compound (9-17 leaflets), **alternate**, leaflets toothed,
lower leaf surface whitish, fall color = yellow-bright red

Flowers: small white and borne in dense clusters, early summer

Fruit: orange-red berry-like about 1/4-inch diameter, late summer

Twigs: stout and reddish brown

Bark: thin gray and scaly on mature trunks

Ecology (page 12):

Shade Tolerant, Subcanopy tree

Moist sites above 5000 feet elevation

Fire Intolerant, Moderate Sprouter

Mature and Climax forests

Spruce-fir F. where it is a diagnostic tree

Blue Ridge province only



Uses (page 20):

Mountain ash wood is light, soft and weak.

Pioneers added the fruit to meat stews and made jelly and wine.

Often planted as an ornamental within its range for its bright red berries and bright autumn foliage.

Many birds eat the fruit, including grouse, grosbeaks and cedar waxwings, as do squirrels and chipmunks.

In the northeast U. S. the fragrant inner bark of mountain-ash is an important winter food for moose.

Comments:

A member of the Rose family which includes rose, cherry, apple, hawthorn and plum, mountain-ash is unrelated to green and white ash which belong to the olive family along with fringetree.

The fruit of mountain ash is a pome, and is structured like the fruit of other rose family members including apples and roses. Like rose hips, mountain ash pomes are edible but not very palatable.

Tennessee pioneers once called the tree “Peruve”, due to its use as a substitute for Peruvian bark (the bark of the quinine tree) which was used to treat malaria.

In Europe, *Sorbus* genus trees are called rowan trees. In English folklore, the European rowan tree (*S. aucuparia*) is fabled as the tree upon which the devil hanged his mother!

Native Shrubs With Compound Leaves

Sumac (*Rhus spp.*)

Sumacs have compound alternate leaves with toothed leaflets and stout but hollow branches filled with spongy pith easily removed. The fruit is a large cluster of fuzzy red berries. The plant often forms clonal colonies, providing a beautiful autumn display.

Indians mixed sumac leaves and willow leaves with tobacco. Indians and Pioneers poured cold water over the berry clusters to make a refreshing beverage sometimes called pink lemonade. Northern Pioneers used the hollowed sumac branches as collecting tubes for maple sugar.

Elderberry (*Sambucus canadensis.*)

Elderberry shrubs usually grow near water. The leaves are compound and opposite with 5 to 9 toothed leaflets per leaf. Flowers are tiny but grow in large white clusters that turn into bunches of small dark purple berries. Elderberry stems are hollow with a spongy pith easily removed.

Indians and Pioneers ate elderberries. Pioneers made jelly and wine from them, and they made the hollow stems into flutes and whistles.

Northern Pioneers used the hollowed elderberry branches as collecting tubes for maple sugar.

Devil's Walkingstick (*Aralia spinosa*)

The trunk of this tall shrub is covered with short thorns, suggestive of the name. The leaves are twice compound (leaflets have leaflets) and alternate, and their size makes them the largest compound leaves in North America. Tiny white flowers form a huge cluster at the top of the plant, and they turn into a large cluster of inedible purple berries.

Pioneers clipped the very young spring leaves and cooked them as a potherb.

Trees With Opposite Leaves

(maple, ash, dogwood, fringetree, buckeye, catalpa, paulownia)

See also boxelder maple (page 33) and ash (page 34)

“MADogs with Beards and Buckeyed Cats named Paul”

MAPLES

Maples are loosely divided into two groups mainly based on commercial value. Hard maples have toothless leaves and relatively hard valuable wood and include sugar maple, southern sugar maple and chalk maple in the Southeast. Soft maples have toothed leaves and relatively soft, less valuable wood and include red maple, silver maple, boxelder maple, striped maple and mountain maple in the Southeast.

Maple sugar and syrup from the sugar maple tree usually come first to mind when this group of trees is considered. Pure maple syrup is a truly healthful addition to pancakes, waffles and French toast because it contains minerals like calcium, magnesium, manganese and vitamins A, B2, B5 and B6, folic acid niacin, biotin and protein, elements maple trees and human beings need to stay fit. Just be sure to buy the real thing, not a "simulated " or watered-down cane sugar version.

Interestingly, early botanist Andre Michaux wrote that a tastier sugar is made from the soft maples like red maple and silver maple than from sugar maple, but it takes twice the sap to make the same amount of sugar and syrup.

The Iroquois Indians to whom the Cherokees are closely related have a legend about four brothers who arduously stalked a great bear to the top of a high mountain where they finally struck him down. They built a huge fire, cooked the bear and ate. After this feast, they looked down toward the valley only to see thousands of sparkling stars, for they were no longer on the mountain but up in the sky. The killed bear's bones suddenly articulated and began to run, so the 4 brothers grabbed their bows and followed him across the sky, becoming stars in the constellation Ursa Major. Every autumn when they kill him, the bear's fat and blood drip down to color the maple leaves in bright hues of yellow and red.

Southern Sugar Maple (Florida maple) (*Acer floridanum*)
(*Acer barbatum*) Maple Family

Identification (page 4):

Leaves: simple, opposite, lobed but not toothed, lower leaf surface whitish and slightly fuzzy; shaped like sugar maple leaves, fall color = yellow-bright red

Flowers: small yellowish, appearing in early spring

Fruit: paired winged seeds (helicopters), mature in fall

Twigs: slender and reddish brown

Bark: light gray and smooth, breaking up on older trunks

Ecology (page 12):

Shade Tolerant, Understory tree

Moist, calcium-rich sites along streams

Fire Intolerant, Moderate Sprouter

Mature & Climax forests

Oak-hickory F., River Cove F., Bottomland F.

Piedmont, V&R and lower Chattahoochee

River in the CP



Uses Page 20):

Though related to sugar maple, southern sugar maple is too small to tap for sugar. It is, however, sometimes planted as an ornamental for its small size, attractive shape and beautiful autumn leaves.

Wildlife eat the winged seeds.

Comments:

Southern sugar maple is often confused with chalk maple (*Acer leucoderme*). Both have slightly fuzzy leaf undersides. In fact, the Latin name *barbatum* means "bearded" or fuzzy. Check this by folding the leaf underside over your finger and holding it up to the light. The 2 trees can be distinguished, however, by the whitish leaf underside of southern sugar maple. Chalk maple leaves are yellow-green underneath. One would expect chalk maple to have the whitish leaf underside, but this is not the case! To remember this difference, call southern sugar maple, "chalk **leaf** maple".

Southern sugar maple prefers rich stream coves as its habitat, while chalk maple prefers rocky ravines and slopes above the streams.

Southern sugar maple has a wider range and is more abundant.

Chalk maple (*Acer leucoderme*) Maple Family

Identification (page 4):

Leaves: simple, opposite, lobed but not toothed, shaped like sugar maple leaves but smaller, lower leaf surface yellow-green and slightly fuzzy, fall color = yellow-bright red

Flowers: small yellowish, appearing in early spring

Fruit: paired winged seeds (helicopters), mature in fall

Twigs: slender and reddish brown

Bark: light gray, smooth, often chalky white, with multiple trunks

Ecology (page 12):

Shade Tolerant, Understory tree

Moist calcium-rich sites, ravines & rocky slopes

Fire Intolerant, Moderate Sprouter

Mature and Climax forests

Oak-hickory F., River Cove F., Bottomland F.

Primarily in Piedmont, rarely in CUP



Uses (page 20):

Though related to sugar maple, chalk maple is too small to tap for sugar.

Wildlife eat the winged seeds.

Chalk maple was planted along Georgia streets in the early 1900's, and is sometimes still planted for its spectacular autumn colors.

Comments:

Chalk maple is often confused with southern sugar maple (*Acer barbatum*). Both are understory trees with sugar maple-like leaves. Both have slightly fuzzy leaf undersides. They can be distinguished by the whitish leaf underside of southern sugar maple. Chalk maple leaves are yellow-green underneath. One would expect the opposite to be true; however, the "chalk" part of chalk maple's name actually refers only to the trunk (*leucoderme* means "white skin"). So call southern sugar maple, "chalk **leaf** maple", and call chalk maple, "chalk **bark** maple".

Chalk maple's range is slightly smaller than S. sugar maple's, and it is usually less common in the forest though locally abundant.

Dr. Charles Wharton, author of Natural Environments of Georgia, found higher-than-normal soil fertility (high calcium levels) where chalk maple, red mulberry and Eastern redbud grew together.

Striped Maple (*Acer pensylvanicum*) Maple Family

Identification (page 4):

Leaves: simple, opposite, 3-lobed like a goose foot, leaf lobes are finely toothed, fall color = yellow

Flowers: yellowish, appearing in late spring

Fruit: paired winged seeds (helicopters) on a drooping stalk,
mature in late summer-early fall

Twigs: stout with whitish-green stripes

Bark: marked on young trunks and limbs by distinctive stripes

Ecology (page 12):

Shade Tolerant, Understory & Subcanopy tree

Moist cool sites in mountain coves

& protected slopes between

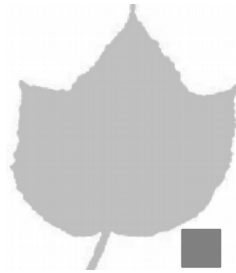
2500 and 4500 feet elevation

Fire Intolerant, Poor Sprouter

Mature and Climax forests

Cove Hardwood F., Oak-hickory F.

Primarily in Blue Ridge and scattered in Cumberland Plateau



Uses (page 20):

Pioneers made whistles from the branch bark which slips easily in summer. They called the tree whistlewood.

Deer browse the twigs and ruffed grouse eat the buds.

The tree is called moosewood up north where moose, deer and beaver eat the inner bark as an important winter survival food.

Comments:

Striped maple and mountain maple (*Acer spicatum*) both grow in the southern Appalachian mountains and may be confused. Striped maple is usually a larger tree than mountain maple, sometimes ascending into the subcanopy, while mountain maple is always an understory tree. Striped maple has distinctive bark stripes that mountain maple lacks. Striped maple flowers and fruit are borne on a drooping stalk, while those of mountain maple are usually borne on an ascending stalk. Striped maple leaves are larger than those of mountain maple and have 3 finely-toothed lobes (goose feet), while mountain maple leaves have 5 roughly-toothed lobes, and finally, striped maple is abundant between 2500 and 4500 feet elevation, while mountain maple rarely descends below 4500 feet.

Red Maple (*Acer rubrum*), Maple Family

Identification (page 4):

Leaves: simple, opposite, 3-5 lobed, toothed, leaf shape variable,
fall color = yellow-bright red

Flowers: bright red, appearing in early spring before leaves

Fruit: bright red paired winged seeds (helicopters), spring

Twigs: reddish

Bark: smooth and gray, becoming slightly scaly on older trees

Ecology (page 12):

Shade Intermediate, Dominant to Dispersed Canopy

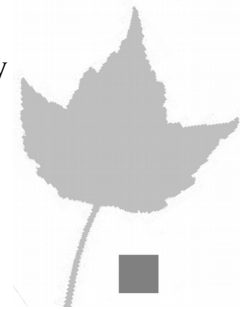
Wet to dry sites (a truly cosmopolitan tree)

Fire Intolerant, Prolific Sprouter

Pioneer, Intermediate, Mature & Climax forests

All forest types except Fraser Fir F.

All Southeastern provinces



Uses (page 20):

Red maple is a soft maple, with wood inferior to that of hard maples like sugar maple. The wood is easily worked with tools. Modern uses include veneer and low-grade furniture.

Indians made lye from red maple ashes and used it to render oil from white oak acorns. They used the inner bark to treat blindness.

Pioneers used the wood to make spinning wheels, furniture legs and other “turnery”. They added sulfate of iron to liquid extracted from the bark to make black ink. If alum was added instead, a cinnamon-colored dye was produced. They also made a blue dye from the bark for dyeing wool.

A favorite ornamental tree due to its hardiness on a variety of sites and its bright autumn colors. There are many available cultivars.

Comments:

Red maple is by far the most common maple in America. It outstrips all other maples on most sites, yielding to sugar maple in the richest mountain coves and to mountain maple above 5000 feet. Its tremendous ability to adapt to a variety of environments is attributed to polyploidy (multiple chromosome sets). The tree can draw from this huge gene reserve to adapt to its surroundings.

Some part of the red maple tree is red in every season:

winter twigs, spring flowers & fruit, summer twigs, fall leaves.

Sugar Maple (*Acer saccharum*) Maple Family

Identification (page 4):

Leaves: simple, opposite, 5 lobed, leaf lobes toothless,
leaves are hairless, shaped like Canada flag,
fall color = bright red-yellow

Flowers: yellowish-green, appearing in spring with the leaves

Fruit: greenish paired winged seeds (helicopters) mature in fall

Twigs: reddish-brown

Bark: smooth and gray, becoming quite scaly on some older trees,
often stained black due to "bleeding" sapsucker holes

Ecology (page 12):

Very shade Tolerant, Dominant Canopy tree

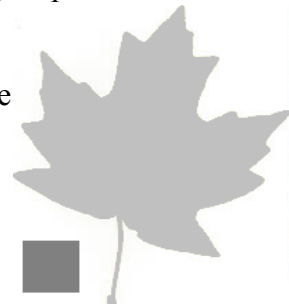
Moist rich, high-calcium stream-side sites

Fire Intolerant, Poor Sprouter

Mature and Climax forests

Cove Hardwood F., Bottomland F.

CUP, V&R, BR; rare in P; absent in CP



Uses (page 20):

The wood is denser and of higher quality than other maples, and is called "hard maple" in the wood industry. Modern uses include fine furniture, flooring, musical instruments and decorative woodwork.

Sugar maple wood burns down to excellent coals for cooking.

The sap is exceptionally high in sugar and has been extensively tapped for maple sugar and syrup within the tree's range.

Pioneers preferred sugar maple ashes for soap making because of their high potash (lye) content.

Sugar maple is widely planted as an ornamental due to its glorious autumn colors and hardiness. Though a mountain native, it seems to grow quite well when planted in other Southeastern provinces.

Comments:

Dr. William M. Harlow says, "To taste the cold sweet sap as it drips from the tree is every American's birthright." It takes about 30 gallons of that cold sweet sap to make one gallon of maple syrup, and sweetness varies with individual trees.

Two forms of sugar maple wood, bird's-eye maple and curly maple have been popular for specialty furniture, mantle pieces and guitar tops. The cause of these wood deformations is not yet known.

Silver Maple (*Acer saccharinum*) Maple Family

Identification (page 4):

Leaves: simple, opposite, toothed, 5-lobed with long lobes
& deep indentions, leaf undersides whitish or silvery,
fall color = yellow-red

Flowers: yellowish-green, appearing in spring before the leaves

Fruit: yellowish-brown paired winged seeds (helicopters) mature
in spring, one of the wings usually aborts

Twigs: reddish-brown

Bark: smooth and dark gray, becoming scaly on older trees

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

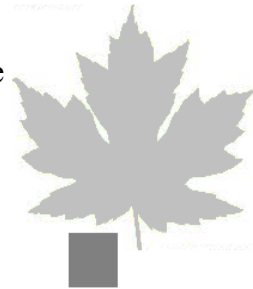
Moist rich stream-side sites

Fire Intolerant, Moderate Sprouter

Pioneer & Intermediate forests

Bottomland F.

CUP, BR, V&R, sporadically in P and CP



Uses (page 20):

Silver maple wood is soft brittle and undesirable for woodworking. Horace Kephart says soft maples (including silver maple) will burn even when green.

Cherokee Indians drank a hot tea made from silver maple bark to treat measles.

Silver maple is not recommended as an ornamental tree due to its brittle wood and susceptibility to a variety of diseases.

Comments:

Silver maple is the fastest growing native maple, but unlike other maples it is not a long-lived tree.

The leaves have long flattened leaf stems (petioles) that twist in the wind, flashing the white leaf underside.

In the Southeast, the leaves usually turn a uniform pale yellow color in autumn, usually lacking the reds, oranges and bright yellows of sugar maple.

The early botanist Francois Michaux found that Mid-western farmers tapped silver maple and got a finer grade of sugar from it than from sugar maple, though it took much more sap to make a gallon of syrup.

Mountain Maple (*Acer spicatum*) Maple Family

Identification (page 4):

Leaves: simple, opposite, 5-lobed, toothed, fall color = yellow-red

Flowers: small, yellowish-green, borne on a slender erect stalk

Fruit: paired winged seed maturing in summer

Twigs: slender and hairy

Bark: reddish-brown

Ecology (page 12):

Shade Tolerant, Understory tree

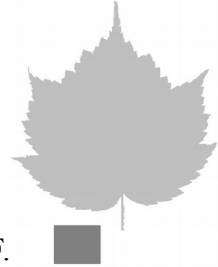
Moist rich streams, found only in high
mountains above about 4500' elevation

Fire Intolerant, Moderate Sprouter

Mature forest & Climax forest

Northern Hardwood F., Spruce-fir F., Fraser Fir F.

Blue Ridge province



Uses (page 20):

Mountain maple bark has been used to treat intestinal worms and eye infections, and mountain folk made a rose-tan dye from the bark.

Deer browse the twigs. Ruffed grouse eat the winter buds.

Though the autumn leaves of this tree can display brilliant shades of scarlet and yellow, its restricted habitat and range prevent its wide use as an ornamental.

Comments:

Mountain maple and striped maple (*Acer pensylvanicum*) both grow in the southern Appalachian mountains (for example, Great Smoky Mountains National Park) and may be confused. Striped maple is usually a larger tree than mountain maple, occasionally ascending into the subcanopy, while mountain maple is always an understory tree. Striped maple bark has the distinctive stripes that mountain maple lacks. Striped maple flowers and fruit are borne on a drooping stalk, while those of mountain maple are usually borne on an ascending stalk. Striped maple leaves are larger than those of mountain maple and have 3 finely-toothed lobes (goose feet), while mountain maple leaves have 5 roughly-toothed lobes, and finally, striped maple is abundant between 2500 and 4500 feet elevation, while mountain maple rarely descends below 4500 feet.

Ashes, see page 34

Alternate-leaf dogwood (*Cornus alternifolia*) Dogwood
Family

Note: This tree, the only dogwood with alternate leaves, is included here with other members of its primarily opposite-leaved family.

Identification (page 4):

Leaves: simple, **alternate**, toothless, leaf veins run parallel to leaf margin (arctuate venation), leaves narrower than those of flowering dogwood, fall color = bright red

Flowers: small white and in clusters

Fruit: blue-black berry-like drupes on red stalks

Twigs: distinctive greenish color

Bark: dark reddish brown

Ecology (page 12):

Shade Tolerant, Understory tree

Moist sites along streams

Fire Intolerant, Moderate Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Cove F.

Blue Ridge & Piedmont provinces, but more
common in Blue Ridge mountains



Uses (page 20):

The fruit is eaten by at least 11 bird species, (and formerly by the now extinct passenger pigeon) as well as black bear.

Leaves and bark are eaten by white-tailed deer, beaver and rabbit.

Occasionally used as an ornamental (pagoda dogwood cultivar) for its showy flowers and autumn colors.

Comments:

Other common names for this tree include pigeonberry, green osier, and umbrella dogwood, a reference to the tree's appearance.

The Latin name *Cornus* refers to the cornelian cherry (*Cornus mas*), a European dogwood with edible fruit (but not a cherry!).

Alternte-leaf dogwood is happily not affected by the exotic dogwood anthracnose fungus that is lethal to flowering dogwood.

Flowering Dogwood *Cornus florida*) Dogwood Family

Identification (page 4):

Leaves: simple, opposite, toothless, leaf veins run parallel to leaf margin (arctuate venation), fall color = bright red

Flowers: large 4-petaled white "flower"(actually leaf bracts)

Fruit: bright red oval drupe (one-seeded berry-like fruit)

Twigs: greenish with purplish tinge, flower buds round

Bark: breaks into small black squares ("alligator bark")

Ecology (page 12):

Shade Tolerant, Understory tree

Moist to dry sites on streams and lower slopes

Fire Resistant, Prolific Sprouter

All forest types except those in swamps
and on high mountains

All Southeastern provinces



Uses (page 20):

Dogwood is very hard, and strong with a shock resistance second only to hickory. It wears very smooth in use, is hard to split when dry, but splits easily when green. It makes excellent firewood.

Indians drank a tea made from the petals for colds and chewed the bark for headache. They bathed in bark tea to treat poison ivy.

Pioneers made a tea from the flowers, fruit and root bark to treat fevers and malaria. A root infusion was used as a cure for worms.

Pioneers made wooden wedges of dogwood called gluts to use in splitting logs, and wooden mallets from the very hard root wood.

The wood was once widely used in textile mills for shuttles, bearings and spools due to its hardness and shock resistance.

Dogwood is very widely planted as an ornamental tree for its hardness, beautiful flowers and red fall colors, but many trees have been killed by the exotic dogwood anthracnose fungus.

Symptoms include small purple leaf spots with tan centers.

Comments:

Country folks say Christ was crucified on a dogwood tree. Ever since then, this once tall canopy tree has become stunted, and bears the signs of crucifixion in its flowers and fruit. The white petals represent Christ's purity, and their tips bear red-stained nail prints. The bright red fruit represents His blood shed for mankind.

Swamp Dogwood (*Cornus stricta*) Dogwood Family

Identification (page 4):

Leaves: simple, opposite, toothless, leaf veins run parallel to leaf margin (arctuate venation), leaves smaller than those of flowering dogwood, fall color = bright red

Flowers: small white and in clusters

Fruit: dark blue to purple drupe (one-seeded berry-like fruit)

Twigs: green to reddish-brown, hairless, pith (inside twig) is yellowish

Bark: older bark is streaked with thin reddish-brown plates

Ecology (page 12):

Shade Tolerant, Understory tree/shrub

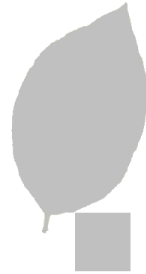
Wet to moist sites along swamps and streams

Fire Intolerant, Good Sprouter

Pioneer, Intermediate & Climax forests

Bottomland F., Cove F., Maritime F.

Piedmont and Coastal Plain



Uses (page 20):

Swamp dogwood probably has the same wood characteristics as flowering dogwood (page 47) and may have been used similarly.

Comments:

Swamp dogwood is also called stiff cornel dogwood.

Swamp dogwood and **silky dogwood** (*Cornus amomum*) look very much alike and grow in similar habitats. They can be distinguished by the hairless twigs and yellowish pith of swamp dogwood, versus the silky twigs and brownish pith of silky dogwood.

Cornus sanguinea is the dogwood of England and Europe.

Its common name, “dog-tree” entered the English language in 1548 and had become “dogwood” by 1614. The fruit was called “dogberry”. Another Old English word dagwood, which has Celtic roots, referred to use of the hard wood for skewers or daggers.

Still another early English name for the dogwood tree was “whipple-tree”, which alludes to its use as part of the harness used in hitching a horse to a cart, an apt job for this strong hard wood.

The European cornelian cherry (*Cornus mas*) is also a dogwood (not a cherry). It has edible fruit, still eaten in Eastern Europe.

Fringetree (Old-mans-beard) (*Chionanthus virginicus*) Olive Family

Identification (page 4):

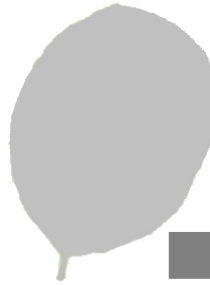
Leaves: simple, opposite, unlobed, toothless, wide oval, leaf stems purplish, fall color = yellow

Flowers: a mass of white strap-like petals (looks like an old man's beard), spring

Fruit: bluish-black oval single-seeded berry (drupe) appearing in mid to late summer

Twigs: stout, gray-brown

Bark: gray-brown



Ecology (Page 12):

Shade Tolerant, Understory tree/shrub

Rich moist woods

Fire Intolerant, No Sprouts

Mature and Climax forests

Oak-Hickory F.

All provinces except the highest elevations of the Appalachians

Uses (page 20):

Indians used crushed roots and bark of fringetree to treat skin inflammations, sores and wounds.

Fringetree is planted as an ornamental due to its small stature, attractive flowers and fruit, yellow autumn foliage and its hardiness. British consider it the finest American flowering tree.

Comments:

Other names for this tree include old man's beard and grancy graybeard, which refer to the mass of white flower petals it produces in spring.

The first part (*Chio*) of the genus name *Chionanthus* means "snow" in Latin and also refers to the snowy white flowers.

Like hollies and persimmons, fringetrees are dioecious, meaning there are male trees and female trees. The male trees provide the best display of the white strap-like flowers (old man's beard), but only the female trees produce the attractive dark purple fruit.

An Olive family member, along with the commercial olive and the ashes, fringetree produces olive-like fruit, edible but not tasty. But even the edible olive is terrible until it is soaked in brine and lye.

BUCKEYES

In America we call trees of the genus *Aesculus* buckeyes, a reference to the seed's resemblance to the eye of a buck deer. In Europe they are called horse chestnuts. This interesting name originated in the 1550's when the Flemish ambassador to Turkey visited the palace of Turkish Potentate Suliman the Magnificent and noticed Turkish soldiers feeding the fruit of a strange tree to their horses. The Flemish ambassador, Ogier de Busteq, sent a young specimen of the tree to the Paris Botanical Gardens, from where it eventually spread to all of Europe as the horse chestnut.

The idea of feeding these reportedly poisonous *Aesculus* seeds to horses boggles the imagination, since symptoms noted in animals and people eating the seeds include mucous membrane inflammation, vomiting, stupor, twitching and paralysis. But American Indians also ate buckeye seeds after roasting or boiling them which removes the poison. During World War II rationing in England, people used the roasted seeds as a bitter coffee substitute.

According to Fred Hageneder, British children call horse chestnuts "conkers", and play an interesting game of the same name whereby two children, each with a horse chestnut seed on a string try to bash the other's seed. Contestant 1 must hold his seed still while contestant 2 does the bashing, then vice versa. Whomsoever obliterates his opponent's conker wins!

Hageneder also reports that horse chestnut seeds contain saponins and were used in Europe to make soap. Further, he says creams and ointments made from horse chestnut seeds were actually used effectively to treat hemorrhoids and varicose veins.

The Ohio buckeye (*Aesculus glabra*) which grows north of the Southeastern states is, in addition to our yellow buckeye, one of the only 2 tree-size buckeyes in America. It differs from yellow buckeye in having its buckeye seeds enclosed in a capsule with spines, whereas, yellow buckeye's fruit capsule is smooth.

Ohio Pioneers wove sturdy Ohio buckeye fiber hats by plaiting thin strips of buckeye wood shaved off a branch with a sharp wood plane, then pulled into fibers across a nail-studded board.

Yellow Buckeye (*Aesculus octandra*) (*Aesculus flava*) Buckeye Family

Identification (page 4):

Leaves: palmately compound (5-7 toothed leaflets radiate from a palm-like center), opposite, fall color = yellow-dark red

Flowers: large, yellowish with red streaks, April-May

Fruit: the familiar shiny brown "buckeye", enclosed in a smooth tan-brown leathery capsule, late summer

Twigs: stout with big buds

Bark: thin plates resembling jigsaw puzzle pieces

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

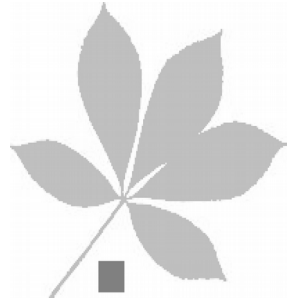
Moist rich mountain coves

Fire Intolerant, Poor Sprouter

Mature and Climax Forests

Cove Hardwood F., Northern Hardwood F.

Cumberland Plateau & Blue Ridge provinces



Uses (page 20):

Buckeye wood is very soft, weak, smells bad when green, hard to split, shrinks greatly on drying and lacks rot-resistance.

Modern uses include crates, drawing boards, boxes and toys.

Crushed seeds were once used for Library paste (book glue) due to the bug-resistant poisons they contain.

Cherokee Indians call this tree “Os kwa da”, which means buckeye in their language. Though buckeye fruit contains a poison called glucoside, the Indians used heat and leaching to remove it and ate the nutritious starch they contain. The Indians also used crushed buckeyes thrown into rivers and lakes to poison fish. Cooking supposedly removed the poison, rendering the fish edible.

Appalachian mountain folk carried the seeds to cure rheumatism, and made a tincture (alcohol extract) from crushed seeds to treat hemorrhoids. They carried buckeyes in their pockets for good luck, passing them down the generations.

Comments:

Yellow buckeye reaches full growth potential in the southern Appalachians where Earth's largest specimens grow in rich coves and on protected slopes between 1500 and 4500 feet elevation.

Red Buckeye (*Aesculus pavia*) Buckeye Family

Identification (page 4):

Leaves: palmately compound (5-7 toothed leaflets radiate from a palm-like center), opposite, fall color = yellow-red

Flowers: bright red, April-May

Fruit: the familiar shiny reddish-brown "buckeye", enclosed in a tan-brown leathery capsule, late summer

Twigs: stout with big buds

Bark: grayish

Ecology (page 12):

Shade Tolerant, Understory shrub

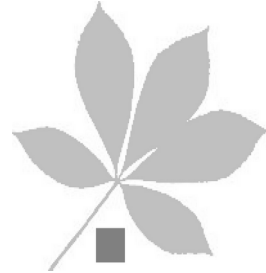
Moist rich slopes & coves

Fire Intolerant, Poor Sprouter

Intermediate, Mature and Climax forests

Pine-Oak F., Oak-Hickory F., Bottomland F.

Coastal Plain & lower Piedmont



Uses (page 20):

Indians made a salve for sores from the mashed nut meat of red buckeye, and chewed very small pieces of the nut for "colic".

Indian women drank a bark tea to induce delivery.

Red buckeye is planted as an ornamental for its small size and showy red flowers that attract hummingbirds. It is frost hardy north of the Coastal Plain.

Comments:

Red buckeye sometimes hybridizes with painted buckeye (*A. sylvatica*), a Piedmont species, where their ranges overlap. The offspring have reddish-yellow flowers.

In May, 1775, William Bartram was on the banks of the Keowee River in the upper Piedmont province of South Carolina only 25 miles from the Blue Ridge mountains where he mentions seeing, "... a new and singularly beautiful species of *Aesculus pavia*...."

Francis Harper, editor of the Naturalist Edition of The Travels of William Bartram says this was most likely a red-flowering form of painted buckeye (*Aesculus sylvatica*). We will never know for sure if this was really a hybrid, but what a treat it would be to have rambled with Bartram through this barely touched sylvan paradise one year before America's official birth!

Bottlebrush Buckeye (*Aesculus parviflora*) Buckeye Family

Identification (page 4):

Leaves: palmately compound (5-7 toothed leaflets radiate from a palm-like center), opposite, fall color = yellow

Flowers: tiny white on several slender erect stalks, summer

Fruit: the familiar shiny brown "buckeye", enclosed in a tan-brown leathery capsule, late summer-early fall

Twigs: stout with big buds

Bark: grayish

Ecology (page 12):

Shade Tolerant, Understory shrub

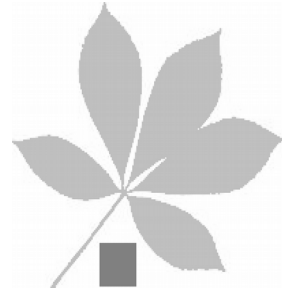
Moist rich slopes

Fire Intolerant, Moderate Sprouter

Intermediate, Mature and Climax Forests

Oak-Hickory Forest

Native to the Central Alabama Piedmont



Uses (page 20):

Planted as an ornamental for its showy white stalked flowers that attract butterflies in summer, this shrub is easy to grow from seed (squirrels usually leave the toxic seeds alone), and it grows and flowers well in shade or sun. It produces root suckers that allow it to expand into an attractive shrub clump.

Comments:

Bottlebrush buckeye is named for the flower stalks that look like white soapy (but gorgeous!) bottle brushes.

William Bartram was the first botanist to describe this lovely shrub in his rambles through the Southeast between 1773 and 1778. He apparently brought seeds back to his garden in Philadelphia, for an ancient descendant of the more ancient original shrub still grew in the famous Bartram's Garden as late as 1930.

Bartram called this shrub *Aesculus alba: nomen nudum*. Nomen nudum means "naked name" and denotes the unofficial (unregistered) status of the Latin name he gave to the shrub.

The official species name, *parviflora*, is Latin for small flowers. This buckeye does, in fact, have smaller flowers than other buckeyes, but it makes up for this by massing them on 10-inch long flower stalks.

Painted Buckeye (*Aesculus sylvatica*) Buckeye Family

Identification (page 4):

Leaves: palmately compound (5-7 toothed leaflets radiate from a palm-like center), fall color = yellow-dark red

Flowers: yellowish-green with red streaks, April-May

Fruit: the familiar shiny brown "buckeye", enclosed in a tan-brown leathery capsule, late summer

Twigs: stout with big buds

Bark: grayish

Ecology (page 12):

Shade Tolerant, Understory shrub

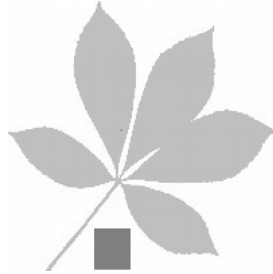
Moist rich slopes

Fire Intolerant, Moderate Sprouter

Mature and Climax forests

Oak-Hickory F., Cove F.

Piedmont province



Uses (page 20):

The seeds contain the same toxic compounds of all the buckeyes, and were probably used for the same purposes by Indians and Pioneers (see yellow buckeye, page 51).

Lacking the showy flowers and desirable form of red buckeye and bottlebrush buckeye, painted buckeye is less frequently planted as an ornamental.

Comments:

Painted buckeye grows in mature hardwood forests where time and dead leaves have built a thick dark layer of humus. It blooms early attracting the hiker's attention with its diminutive form and bright oddly-shaped yellow flowers.

Painted buckeye sometimes hybridizes with red buckeye, a Coastal Plain species, where their ranges overlap. The offspring have reddish-yellow flowers.

Sylvatica means "of the woods" in Latin, and oddly, the genus name *Aesculus* comes from the Latin word for an edible acorn!

Other common names for this delightful shrub include Georgia buckeye and Carolina buckeye.

William Bartram called this tree "*Pavia sylvatica*" when he found it growing along the Broad River in Georgia in the 1770's.

Southern Catalpa (*Catalpa bignonioides*) Bignonia Family

Identification (page 4):

Leaves: simple, opposite, toothless, heart-shaped, lower surface only slightly fuzzy, smell unpleasant, fall color = yellow

Flowers: large white with purplish blotches, late spring

Fruit: distinctive long very narrow cigar-shaped black capsule, capsule matures in fall and remains most of the winter

Twigs: stout and brittle

Bark: gray

Ecology (page 12):

Shade Intolerant, tree of fields and hedgerows

A variety of soils in open areas

Fire Resistant, Good Sprouter

Originally native to central Alabama



Uses (page 20):

The wood of catalpa is very rot-resistant and was once widely used for fence posts. It is easily worked but too scarce to be valuable.

Catalpa is regularly defoliated by the catalpa worm, a 2-inch-long black and yellow caterpillar specifically adapted to eat only catalpa tree leaves. The worms defoliate the tree year after year with no apparent ill effect on the tree.

Indians and Pioneers used the worms as fish bait. They made medicine from the seeds to induce vomiting and expel intestinal worms. They made a bark tea used as an external antiseptic and to treat snakebites. A tea made from seeds was used as a sedative.

Catalpa is planted as an ornamental for its showy white flowers and unusual fruit. Though a Southern tree, it is hardy throughout most of the U.S. and southern Canada. The seeds reportedly germinate easily.

Comments:

This tree's "range" has been expanded by planting due to its role as a fish bait host. Europeans first encountered the tree in Cherokee Indian fields.

Bignonioides refers to the Trumpet Creeper Family (Bignoniaceae) of which the tree is a member.

Also called cigar tree and Indian cigar, catalpa's long slender seed pods were once smoked for fun and as a cure for asthma.

Royal Paulownia (*Paulownia tomentosa*) Paulowniaceae
Family

Identification (page 4):

Leaves: large, simple, opposite, toothless, heart-shaped, very fuzzy, fall color = yellow

Flowers: large, light purple with yellowish centers, early spring

Fruit: small cottony seeds released from a black oval capsule

Twigs: stout but brittle due to chambered and hollow pith

Bark: gray-brown often with intricate markings

Ecology (page 12):

An introduced exotic native to
eastern and central China
that has spread throughout
the Southeast.

Fire Resistant, Prolific Sprouter



Uses (page 20):

Wood of this tree has woodworking characteristics equal to those of black walnut. It has been grown in plantations in the Southeast for export to Japan and China where it is in demand for specialty wood products like plaques, carvings and toys. American plantations have not proven widely profitable, however.

The cottony light-weight seeds were once used in China as packing material for shipping porcelain ware.

Paulownia is planted as an ornamental tree for its rapid growth and showy flowers, but the tree can grow quite large in America, and its large leaves become a nuisance on lawns and patios in autumn.

Comments:

In China it was once fashionable to plant a princess tree (the Asian common name) when a baby girl was born. When the girl reached marriage age, the tree was cut down and its wood carved into articles for her dowry.

Country folk in the Southeast sometimes call this tree "cottonwood" for its cottony seeds, but it bears no relation to true Eastern cottonwood (*Populus deltoides*).

The genus name *Paulownia* refers to Anna Pavlova, daughter of Paul I of Russia (1795-1865). The Latin name *tomentosa* *refers* to the fuzzy leaves.

Trees With Simple Alternate Toothless Leaves

(pawpaw, magnolias, redbud, redbay, cucumbertree,
persimmon, blackgum)

*“My toothless Pawpaw found a Magic Red Persimmon that gave
him Blackgums.”*

Have you ever wondered why some trees have toothless leaves and others have toothed leaves? This difference in leaf configuration seems too pronounced to be a random occurrence.

In fact, botanists believe tree species that evolved in tropical and subtropical environments are far more likely to have toothless leaves than those that evolved in cooler climates and usually have toothed leaves.

Why is this so? The teeth of toothed-leaved trees provide additional surfaces for photosynthesis. This becomes important in cool climates with short growing seasons. Toothed-leaved trees can use the extra surface area to get a jump start on photosynthesis in spring so they can maximize sugar production during the short growing season.

Trees with toothless leaves growing in tropical and subtropical environments don't need this boost. Instead they need ways to shed water from the leaf efficiently, like long pointed leaf tips.

If we look at the ranges and species distributions of both toothless and toothed-leaved trees, we find this idea verified. For example, most members of the Ebony (persimmon) and Custard-apple (pawpaw) families (both with toothless leaves) are tropical and most *Nyssa* (blackgum) species (also toothless) are subtropical, while most members of the Elm and Birch families (toothed leaves) are found in the north temperate zone.

Paleontologists and geologists actually look for the presence or lack of teeth on fossil leaves to help determine the climate and possibly global position (via plate tectonics) of lands where these plants grew.

Pawpaw (*Asimina triloba*) and (*Asimina parviflora*) Custard Apple Family

Identification (page 4):

Leaves: simple, alternate, toothless, smell like bell pepper, fall color = yellow

Flowers: maroon with 6 small leathery petals

Fruit: edible pawpaw (4-inch yellowish banana-like berry), rarely develops to edible maturity in the Southeast

Twigs: Twig buds are brown and fuzzy

Bark: dark brown

Ecology: (page 12)

Shade Tolerant, Understory tree/shrub

Moist sites

Fire Resistant, Good Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Cove F.

Piedmont & upper Coastal Plain



Uses (page 20):

American pawpaws are first mentioned in Desoto's chronicles of 1541. Desoto's soldiers, Indians and Pioneers ate pawpaw fruit. Peattie says the fruit is edible only when it turns black in late fall.

Fox, opossum and racoons eat them today.

The inner bark of pawpaw makes excellent fiber and was used by Indians and Pioneers for cordage, fishing nets and fish stringers.

The powdered seeds (which are poisonous) have been used to kill head lice, and the leaves are considered insecticidal.

Comments:

There are two species of American pawpaw; *Asimina triloba*, reaching 20 feet tall, and *Asimina parviflora*, shorter and with smaller leaves and flowers. Both are clonal shrubs like the aspens and sumacs, and both may grow near each other.

The “bell pepper” smell of pawpaw leaves comes from acetogenin a chemical the plant uses to repel insects. Only zebra swallowtail butterfly larvae successfully feed on them, and chemicals from the tree ingested by the butterfly give it protection from predators throughout its life.

The word pawpaw is a corruption of papaya, an unrelated species.

MAGNOLIAS

Evergreen Magnolias

Southern Magnolia, Page 60

Sweetbay Magnolia, Page 61

Deciduous Magnolias

Deciduous Magnolias with Ear-shaped Leaf Base

Frasier Magnolia, Page 62

Bigleaf Magnolia, Page 63

Deciduous Magnolias without Ear-shaped Leaf Base

Cucumbertree, Page 64

Umbrella Magnolia, Page 65

The Magnolia Family has about 12 genera (families consist of one or more genera, each of which consists of one or more species) world-wide, only 2 of which are in North America - Magnolia and Liriodendron.

Magnolias are considered ancient in origin because their flower and fruit structures resemble those of fossil plants more than those of modern plants.

Experts say the magnolias evolved long before conventional pollinators like bees and wasps; consequently, beetles and flies had to do the job, and they have been doing it ever since, relishing the protein-rich pollen the flowers make in lieu of nectar.

The most popular exotic ornamental magnolia in the Southeast is likely the saucer magnolia (*Magnolia × soulangeana*), bred in France in 1820 as a hybrid between 2 Asian magnolias. This small tree displays large saucer-shaped purplish-pink blossoms in early spring before the leaves emerge.

Magnolias were first cultivated by the Chinese. They used magnolia buds to flavor rice and made various medicines from magnolia bark and leaves.

Southern Magnolia (*Magnolia grandiflora*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, evergreen, leathery, glossy green above, greenish or brown wooly below

Flowers: very large showy white flowers with a lemony fragrance, early summer

Fruit: woody cone with many bright red berry-like follicles, early fall

Twigs: often rusty and fuzzy, twigs encircled by rings (stipular rings) where leaf attaches

Bark: gray and fairly smooth

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

Swamps and river bottoms

Fire Resistant, Good Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Maritime F., Swamp F., Pine-Oak F.

Coastal Plain native, but planted in Piedmont



Uses (page 20):

Magnolia wood is moderately hard and heavy with straight grain very similar to that of Cucumbertree (*M. acuminata*), and used for the same purposes, including furniture, veneer, paneling and cabinets.

Due to its resistance to twisting and warping, southern magnolia wood was once in demand as a material for venetian blinds, which are less stylish today and now made of plastic.

Associated with the classic Southern plantation era, this tree is widely planted as an ornamental for its showy flowers, fast growth, evergreen habit and hardiness. It continually drops dead leaves, but these are hidden if lower branches are left unpruned.

Comments:

Other common names for the tree include bullbay, bat tree and big laurel. The flower is the state flower of Louisiana and Mississippi. The Magnolia genus is named for Pierre Magnol (1638-1715), director of the botanical garden at Montpellier, France.

President Andrew Jackson planted a southern magnolia at the White House in honor of his wife Rachel, which still stands.

Sweetbay Magnolia (*Magnolia virginiana*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, evergreen, leathery, dull green above (not shiny), whitish beneath

Flowers: white, fragrant, smaller than *M. grandiflora*

Fruit: woody cone with many dark red berry-like follicles

Twigs: with stipular rings (ring around the stem where leaf attaches)

Bark: grayish, smooth

Ecology (page 12) :

Shade Tolerant, Subcanopy tree

Swamps and river bottoms

Fire Resistant, Moderate Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Maritime F., Swamp F., Pine-oak F.

Coastal Plain, lower Piedmont, rarely in Valley & Ridge



Uses (page 20):

Sweetbay shares wood characteristics with southern magnolia and cucumbertree, but is generally too small for use in timber products.

Indians made tea from the bark, cones and seeds for rheumatism.

Pioneers made a bark and alcohol infusion used as a cure-all.

Widely planted as an ornamental for its evergreen habit, fragrant flowers and whitish leaf undersides. Its narrow crown width makes it a good planting choice where horizontal space is limited.

Comments:

Sweetbay magnolia has several other common names including bay, beaver-tree, castor wood, Indian bark and swampbay.

Mainly a Coastal Plain tree, sweetbay has been documented in the Georgia Valley and Ridge in limestone sinkhole "sag ponds".

The Latin species name *virginiana* means from Virginia, U.S.A., as do the species names *virginianum* and *virginianus*. In contrast, the species name *virginicus* applied to a plant means, of the Virgin Islands. Apparently Latin names also confer male, female or neuter character to the named plant. *Virginiana* refers to a female plant, whereas *virginianus* refers to a male plant, and *virginianum* is a neuter designation. These gender names have nothing to do with the plant's reproductive system!

Fraser Magnolia (*Magnolia fraseri*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, deciduous, leaf base is ear-shaped (looks like it has ear lobes), leaves to 20 inches, fall color = yellow-brown

Flowers: white & scented like southern magnolia but smaller, bloom in mid-spring

Fruit: woody cone (greenish when young) with many red berry-like follicles, mature in mid-summer

Twigs: with stipular rings (ring around the stem where leaf attaches), twig end bud is purplish and 1 1/2 inches long

Bark: smooth and grayish, tree often has multiple trunks

Ecology (page 12):

Shade Intermediate, Dispersed Canopy & Subcanopy

Moist mountain streams & protected slopes

Fire Resistant, Prolific Sprouter

Intermediate, Mature and Climax forests

Cove F., Acid Cove Hemlock F., Oak-Hickory F.

Blue Ridge province



Uses (page 20):

A few birds eat the red follicles, and deer browse the twigs.

Occasionally used as an ornamental for its unique appearance.

Comments:

William Bartram, first American botanist to explore the Georgia Appalachians, found this tree growing near present-day Clayton, Georgia on Pinnacle Knob, which he called Mount Magnolia. In his classic book, The Travels of William Bartram, he describes this tree as, "a new and beautiful species of the family of flowering trees which here at the cascades of Falling Creek (Martin Creek today) grows in a high degree of perfection...."

A few lines later he says, "The leaves of this tree....when fully expanded at maturity are frequently above two feet in length.."

Modern botanists agree the longest Fraser magnolia leaves don't exceed 20 inches. Consequently, some folks have mistaken

Bartram's description for that of *M. macrophylla*, a Piedmont tree.

Bartram failed to officially register the tree, so discovery credit went to Scottish botanist John Fraser (1750-1811) who came later.

Bigleaf Magnolia (*Magnolia macrophylla*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, deciduous, leaf base is ear-shaped, leaves 20-30 inches long, fall color = yellowish-brown

Flowers: white like southern magnolia but smaller, mid-spring

Fruit: woody cone (greenish when young) with many red berry-like follicles, mid-summer

Twigs: with stipular rings (ring around the stem where leaf attaches), twig end buds fuzzy and up to 2 inches long

Bark: smooth grayish

Ecology (page 12):

Shade Intermediate, Subcanopy & Understory tree
moist streams and streamside bluffs

Fire Resistant, Prolific Sprouter

Intermediate, Mature and Climax forests

Cove F., Oak-Hickory F.

Piedmont & upper Coastal Plain provinces



Uses (page 20):

Brittle branches and large leaves make this tree susceptible to wind and ice damage, limiting its ornamental use. It is site demanding.

Comments:

Bigleaf magnolia is fairly rare within its range. At least some of this rarity can be attributed to collectors who transplant the tree from the woods to their gardens, where its tropical appearance adds a unique touch to the landscape.

This tree is rather short-lived and does not reproduce well from seed, but often reproduces by sprouting.

Bigleaf magnolia leaves (and flowers) are the largest simple leaves in North America, often exceeding two feet in length. Devil's walkingstick (*Aralia spinosa*), a thorny Southeastern native shrub sports the largest compound leaves in North America.

Botanist Andre Michaux was the first to describe bigleaf magnolia when he found a specimen growing near Charlotte, North Carolina in the month of June in the early 1790's. He saw it again in the Cumberland Mountains just 19 years after Daniel Boone helped cut the Wilderness Trail, and yet again near Knoxville, Tennessee.

Cucumbertree (*Magnolia acuminata*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, deciduous, wide eye-shaped, leaves without ear-shaped base, fall color = yellow

Flowers: yellowish, smaller than southern magnolia, mid-spring

Fruit: woody cone (greenish when young) with many red berry-like follicles, mature mid-summer

Twigs: have stipular rings (ring around the stem where leaf attaches)

Bark: gray-brown with scaly ridges similar to yellow poplar bark

Ecology (page 12):

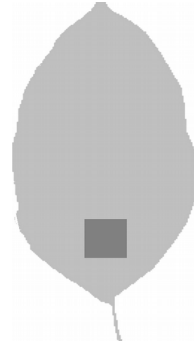
Shade Intolerant Pioneer, Dispersed Canopy tree
rich moist coves

Fire Intolerant, Prolific Sprouter

Pioneer, Intermediate and, Mature forests

Cove Hardwood F., Bottomland F.

CUP,V&R, BR, P; most common in mountains



Uses (page 20):

Like yellow poplar, cucumbertree wood is soft, light-weight, pliable, easily worked, shrinks and warps significantly on drying, is not rot-resistant, will barely burn green and is poor fuel when dry. Modern uses include boxes, crates, paneling veneer and toys.

Indians and Pioneers made bowls, troughs and canoes from it.

Pioneers used the bark as a substitute for quinine in treating malaria and typhoid. They chewed the bark to quit smoking.

At least five bird species eat the fruit.

Botanist Michaux said, " The inhabitants of the remotest parts of Pennsylvania and Virginia pick the cones when green and infuse with whiskey which makes a pleasant bitter.....much esteemed as a preventative against intermittent fevers, but I doubt whether it would be used if it had the same qualities when mixed with water."

Comments:

The tree's name comes from the immature fruit's supposed resemblance to the garden cucumber, though it is inedible.

Cucumbertree is slightly less shade intolerant than yellow poplar.

Unlike yellow poplar, cucumbertree is not long-lived, surviving only to about 150 years of age.

Umbrella Magnolia (*Magnolia tripetala*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, deciduous, leaves clustered in whorls around twig (like an umbrella), leaves without ear-shaped base, fall color = yellowish-brown

Flowers: white, smaller than southern magnolia, with unpleasant odor, mid-spring

Fruit: woody cone (greenish when young) with many red berry-like follicles, matures late summer

Twigs: have stipular rings (ring around the stem where leaf attaches), end buds purple

Bark: gray and smooth, tree often multi-trunked

Ecology (page 12):

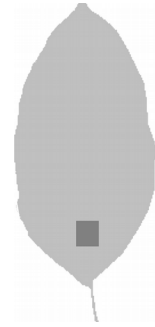
shade Tolerant, Understory tree
moist sites along mountain streams, usually
between 1500 & 2500 feet elevation

Fire Intolerant, Good Sprouter

Intermediate, Mature and Climax forests

Cove Hardwood F., Bottomland F.

Cumberland Plateau and Blue Ridge mountains, less
common in upper Piedmont



Uses (page 20):

The wood of this small understory tree has no commercial value. It is occasionally planted as an ornamental, and is purportedly easy to grow from seed.

Comments:

The tree was first described in 1743 by naturalist Mark Catesby in his book, Natural History of Carolina. Along with other American magnolias, it was exported to Europe where it was popular in gardens of the time.

Umbrella magnolia is abundant along low elevation mountain streams in Great Smoky Mountains National Park, where its large umbrella-like leaves stand out against the surrounding foliage.

The Latin name *tripetala* refers to the 3 petal-like sepals of the flower.

The Magnolia family is named in honor of Pierre Magnol, French botanist (1638-1715), originator of the plant family classification.

Eastern redbud (*Cercis canadensis*) Bean Family

Identification (page 4):

Leaves: simple, alternate, toothless, heart-shaped, leaf stem conspicuously swollen at base, fall color = yellow-red

Flowers: small pea-like, purplish pink, bloom in early spring

Fruit: flat bean pod about 6 inches long, often persists into winter

Twigs: slender and very zigzag

Bark: dark gray to brown

Ecology (page 12):

Shade Tolerant, Understory tree

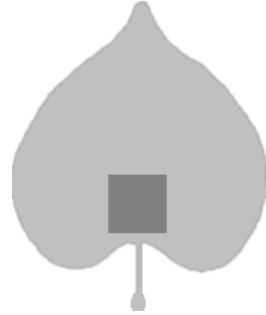
Moist sites, prefers calcium-rich soils

Fire Resistant, Prolific Sprouter

Intermediate and Mature forests

Bottomland F., Oak-Hickory F., Oak-Pine F.

All provinces except rare in Coastal Plain



Uses (page 20):

Redbud wood is hard, flexible and springy. Western Indians made bows from the wood of western redbud (*C. occidentalis*).

Redbud flowers are edible raw or cooked. Tender young bean pods may be eaten sauteed.

Cherokee Indians made a bark tea to treat whooping cough.

Pioneers used the bark to treat diarrhea.

Redbud is frequently planted as an ornamental tree for its showy flowers that bloom quite early in spring before the leaves emerge.

Comments:

Unlike most bean family members, redbud does not fix nitrogen from the air and store it in its roots.

The alternate common name Judas tree refers to the legend claiming Judas Iscariot hanged himself on this tree after betraying Jesus. Since then, the once mighty forest giant grows only as a stunted understory tree. Its formerly white flowers turned forever reddish and bloom near Easter to remind us of the crucifixion.

President George Washington's diary frequently mentions transplanting redbud trees from the surrounding forest to his Mt. Vernon estate grounds.

A thick patch of redbud often indicates calcium-rich (fertile) soil.

Oddly, the Cherokee name for this tree means "like peaches".

Red Bay (*Persea borbonia*) Laurel Family

Identification (page 4):

Leaves: simple, alternate, toothless, evergreen, aromatic when crushed, fuzzy beneath, leaves often have insect galls

Flowers: small yellow, appear in late spring

Fruit: dark blue oblong berry-like drupe

Twigs: twigs and buds are fuzzy

Bark: dark reddish-brown

Ecology (page 12):

Shade Tolerant, Understory tree

Moist to dry woods and on dunes

Fire Intolerant, Prolific Sprouter

Intermediate, Mature & climax forests

Pine-Oak F., Oak-Hickory F., Evergreen Oak F.



Uses (page 20):

The red colored wood of red bay is hard and strong and has been used for boats, cabinets and furniture.

The fragrant leaves are sometimes used in soups as a substitute for commercial bay leaves.

Seminole Indians used red bay as an emetic, to induce vomiting.

Birds (including the fish crow), turkey and quail eat red bay fruit.

Red bay's evergreen habit and attractive blue fruit make it a desirable landscape tree, but see below.

Comments:

In 2002 the red bay ambrosia beetle (*Xyleborus glabratus*), an Asian beetle was discovered on red bay trees in Savannah, Georgia. The beetle transports the laurel wilt fungus (*Raffaelea lauricola*) which infects and kills red bay trees. The disease, which wilts both leaves and stems, is currently (2014) spreading through the Southern Coastal Plain.

Sassafrass, another member of the Laurel family has also been effected by the disease, but to a lesser extent.

Leaf galls caused by the red bay psyllid plant louse are usually abundant on red bay leaves and help in identifying the tree.

Other Laurel family members include the cinnamon tree, the camphor laurel, the avocado and the bay leaf laurel, all economically important exotic trees.

Cucumbertree (*Magnolia acuminata*) Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, deciduous, wide eye-shaped, leaves without ear-shaped base, fall color = yellow

Flowers: yellowish, smaller than southern magnolia

Fruit: woody cone (greenish when young) with many red berry-like follicles

Twigs: have stipular rings (ring around the stem where leaf attaches)

Bark: gray-brown with scaly ridges similar to yellow poplar bark

Ecology (page 12):

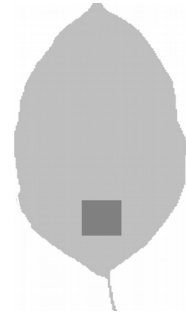
Shade Intolerant Pioneer, Dispersed Canopy tree
rich moist coves

Fire Intolerant, Prolific Sprouter

Pioneer, Intermediate and, Mature forests

Cove Hardwood F., Bottomland F.

CUP,V&R, BR, P; most common in mountains



Uses (page 20):

Like yellow poplar, cucumbertree wood is soft, light-weight, flexible, easily worked, shrinks and warps significantly on drying, is not rot-resistant, will barely burn green and is poor fuel when dry. Modern uses include boxes, crates, paneling veneer and toys.

Indians and Pioneers made bowls, troughs and canoes from it.

Pioneers used the bark as a substitute for quinine in treating malaria and typhoid. They chewed the bark to quit smoking.

At least five bird species eat the fruit.

Botanist Michaux said, " The inhabitants of the remotest parts of Pennsylvania and Virginia pick the cones when green and infuse with whiskey which makes a pleasant bitter.....much esteemed as a preventative against intermittent fevers, but I doubt whether it would be used if it had the same qualities when mixed with water."

Comments:

The tree's name comes from the immature fruit's supposed resemblance to the garden cucumber, though it is inedible.

Cucumbertree is slightly more shade tolerant than yellow poplar.

Unlike yellow poplar, cucumbertree is not long-lived, surviving only to about 150 years of age.

Persimmon (*Diospyros virginiana*) Ebony Family

Identification (page 4):

Leaves: simple, alternate, toothless, leaves often have black spots, fall color = bright red & yellow

Flowers: small, yellow urn-shaped, in late spring

Fruit: orange-colored edible persimmon (looks like a tiny pumpkin), ripe in late fall (unripe fruit is very astringent)

Twigs: zigzag, twig buds are black

Bark: black, with distinctive square blocks larger than dogwood's

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

Moist to dry sites

Fire Resistant, Prolific Sprouter

Intermediate, Mature and Climax forests

All forest types except high mountains & swamps

All Southeastern provinces



Uses (page 20):

Persimmon wood is very hard, heavy, strong and smooth. Modern uses include textile shuttles, cue sticks and golf club heads.

Persimmon leaves make a full-bodied tea rich in vitamin C.

Cherokee Indians dried the fruit and made persimmon bread.

Desoto was offered persimmon loaves by Indians near present-day Memphis, Tennessee. He confiscated dried persimmons (he called them prunes) from Indian villages in Arkansas.

Cherokees chewed the bark for heartburn and used the wood to carve "disease stampers", used somehow in healing? Carolina Indians used the seeds as dice or coins in a game of chance.

Pioneers made a tea from the astringent unripe fruit to treat diarrhea, uterine hemorrhages and sore throat.

Confederate soldiers ground the seeds as a coffee substitute.

Comments:

Diospyros means "fruit of Zeuss".

Captain John Smith of early Virginia said of unripe persimmons, "If they be not ripe, it will draw a man's mouth awrie in torment."

Some folks say persimmons are not fit to eat until they are wrinkled; some say they must fall off the tree first, and still others say don't eat them until a hard frost has ripened them.

Blackgum, Pepperidge (*Nyssa sylvatica*) Tupelo Family

Identification (page 4):

Leaves: simple, alternate, toothless but occasionally with 3 "horns" on the leaf end, fall color = bright red & yellow

Flowers: tiny, greenish-yellow, spring

Fruit: blue-black drupe with a ridged seed, mature in late summer

Twigs: Distinctive uniform tan color, twig pith has dark bands, branches often emerge from the trunk at near 90 degrees

Bark: gray, platy, looks similar to sweetgum

Ecology (page 12):

Shade Intermediate, Subcanopy tree

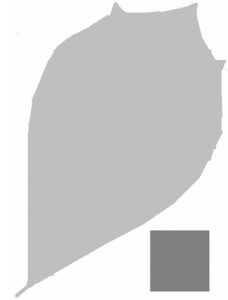
Moist to dry sites, a "cosmopolitan" tree

Fire Tolerant, Prolific Sprouter

Pioneer, Intermediate, Mature & Climax forests

Most forests except highest mountains & swamps

All Southeastern provinces



Uses (page 20):

Wood is moderately hard, very strong, lacks rot-resistance and is very nearly impossible to split due to interlocked wood grain.

Kephart calls it "unwedgeable". The wood will scarcely burn green, making it good for fire backlogs and andirons. Modern uses include pulp for paper, veneer and plywood.

Cherokee Indians used blackgum twigs, bark and roots in multi-ingredient medicines for a variety of ailments. Cherokee women drank a root tea to "facilitate" delivery.

Pioneers said the wood was, "as easy to split cross-ways as long-ways." They used it for wooden mauls, wagon wheels, gun stocks and mule sled runners. Pioneers dipped the frayed twigs in whiskey for tooth brushes.

Heart rot frequently hollows the blackgum trunk which Pioneers used as "bee-gums" to house honey bees.

Comments:

The genus name *Nyssa* was the name of a mythical Greek water nymph, while tupelo resembles the Creek Indian word for swamp.

Blackgum is misnamed, producing no black wood and no gum!

Blackgum is the longest-living non-clonal flowering plant in Eastern North America, capable of ages exceeding 650 years.

Trees With Simple Alternate Toothed Leaves

The trees in this large group are presented in two sections based on the two mnemonics used to remember them.

First Group:

elm	page 72
birch	page 75
hackberry	page 78
hornbeam	page 81
buckthorn	page 83
silverbell	page 84
serviceberry	page 85

"Elmer is the son of a birch that hacked a horn off a buck with a silver service sword."

Second Group:

willow	page 87
alder	page 88
holly	page 89
sourwood	page 92
cherry	page 93
cottonwood	page 96
basswood	page 97
beech	page 98

"Willy and Al hollered and drank sour cherry wine when they caught a cottonpickin' bass at the beech."

Winged Elm (*Ulmus alata*) Elm Family

Identification (page 4):

Leaves: simple, alternate, small, double toothed (teeth have teeth), leaves have slightly uneven bases, fall color = yellow

Flowers: small, brownish, present in v. early spring before leaves

Fruit: seed enclosed in an oval papery wing, mature in spring

Twigs: very slender and zigzag, often with corky wings

Bark: grayish-brown with scaly ridges

Ecology (page 12):

Shade Intermediate (but often acts as a pioneer)

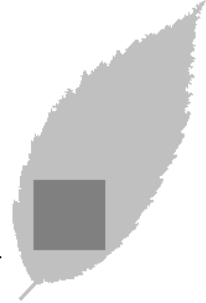
Subcanopy, sometimes Dispersed Canopy tree

Moist to dry sites (prefers high-calcium soils)

Pioneer & Intermediate forests

Cove F., Pine-Oak F., Oak-Pine F., Oak-Hickory F.

All Southeastern provinces, but rare in CP



Uses (page 20):

Winged elm wood is hard, heavy, flexible, lacks rot-resistance and has interlocking grain which makes it nearly impossible to split.

Indians made excellent cordage from the fibrous inner bark. They used the flexible stems and root runners for fish trap hoops and as hafts for stone axes.

Pioneers used the inner bark to tie up cotton bales. The interlocking grain made winged elm useful for wagon wheels and hubs, wooden mauls and mule skid runners. An elm stump was preferred for the traditional chopping block and as a pedestal for the blacksmith's anvil.

In the days of great sailing ships, elm (all elm species) was preferred for keels because it did not readily split on impact like oak, and it lasts forever when continually submerged.

Comments:

Another name for this tree, "Wahoo" comes from its Creek Indian name.

Winged elm is susceptible to Dutch elm disease, but has not been as devastated by it as has American elm (*Ulmus americana*).

Since the Stone Age, European elms have been considered the very best livestock fodder, better than basswood or ash.

The old English folk name for elms is "elfin wood".

American Elm (*Ulmus americana*) Elm Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves smooth above and below, leaf base very uneven,
fall color = yellow

Flowers: small, greenish, present in v. early spring before leaves

Fruit: seed enclosed in an oval papery wing, mature in spring

Twigs: slender and zigzag, hairless

Bark: grayish with scaly plates, inner bark striped

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

Moist to Dry sites (prefers calcium-rich soils)

Fire Intolerant, Moderate Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Cove F., Oak-Hickory F.

All southeastern provinces



Uses (page 20):

American Elm wood is hard, heavy, flexible and usually has straight grain, lacks rot-resistance but last forever submerged.

In the days when most products were shipped in barrels, American elm was considered the best wood for slack cooperage (wooden barrels used for shipping dry goods like flour and gun powder).

American elm made the best barrel staves and the best barrel hoops for securing the staves in the barrel shape.

Northern pioneers made very durable ox whips of American elm by stripping off bark and braiding it into a long supple whip shape.

An elm stump was preferred for the traditional firewood chopping block and as a pedestal for the blacksmith's anvil.

Comments:

Once considered the finest of street trees for its lovely urn-shaped growth pattern and arching branches, American elm has been devastated by Dutch elm disease. This exotic killer, caused by at least 3 Asian fungi of the genus *Ophiostoma*, and transported by the elm bark beetle, was first isolated in the Netherlands. It appeared in Ohio in 1928 and has destroyed millions of American elm trees. Winged elm and slippery elm are affected by Dutch elm disease, but to a much lesser degree than American elm.

Slippery Elm (Red Elm) (*Ulmus rubra*), Elm Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth), leaves very rough above and densely fuzzy below, leaves may have slightly uneven bases, fall color = yellow

Flowers: small, greenish, present in very early spring before leaves

Fruit: seed enclosed in an oval papery wing, mature in early spring

Twigs: thick, zigzag, grayish, twigs and buds are hairy

Bark: grayish with scaly ridges

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

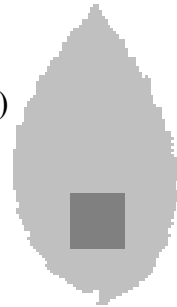
Moist sites along streams (prefers calcium-rich soils)

Fire Intolerant, Moderate Sprouter

Intermediate, Mature & Climax forests

Bottomland F., Cove F.

All Southeastern provinces except the
Atlantic CP of Georgia



Uses (page 20):

The wood is strong, flexible, lacks rot-resistance, lasts forever submerged, is easy to split green, but difficult to split dry.

Cherokee Indians gave a mixture of slippery elm bark along with other plants to pregnant women to help deliver the baby.

Indians used the inner bark for cordage. They first boiled bark strips in lye water (made from wood ashes) to soften the bark and make it pliable.

Pioneers made splits for basket weaving and chair bottoms simply by pounding 4-inch-diameter saplings with a wooden maul. The wood readily separated into usable layers.

Old timey ball players chewed the inner bark and rubbed it on their gloves to help the ball stick to their fingers (the “spit ball” pitch).

Comments:

Slippery elm inner bark has a mucilaginous (slimy like okra) quality that distinguishes it from all other trees. This slime was used to treat sore throats and wounds and to produce a stool-softening beverage similar to modern-day Metamucil.

Yellow Birch (*Betula alleghaniensis*) Birch Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves often looked paired on the twig, fall color = yellow

Flowers: catkins (small spike-like projections)

Fruit: nutlets in catkins

Twigs: slender, zigzag, greenish brown, smell like wintergreen

Bark: shiny gold to silver, peeling into thin strips

Ecology (page 12):

Shade Intermediate, Dominant & Dispersed Canopy

Cool moist soils above 3000 ' elevation

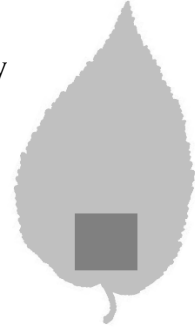
Fire Intolerant, Poor Sprouter

Intermediate, Mature & Climax forests

Spruce-Fir F., N. Hardwood F., Cove F.,

Acid Cove Hemlock F.

Blue Ridge province



Uses (page 20):

Yellow birch wood is hard, very strong, stiff, resinous, hard to split dry, easier to split green and burns better green than dry. Modern uses include furniture, paneling, bowls and cabinets.

Yellow birch bark is full of resinous oil and makes excellent kindling and torches according to Horace Kephart.

Indians used the resinous punk wood (soft rotten wood) as tinder for fire by friction.

Pioneers used the hard strong but light wood for wagon wheel hubs, oxen yokes and snow sled frames.

Comments:

Yellow birch seeds often fall on old rotting logs where they germinate and grow into mature trees. After the host log rots completely away, the birch tree appears to be perched on legs.

Yellow birch is a diagnostic tree of the northern hardwood forest found in the U.S. Northeast and Lake States. Other trees in this forest include American beech, sugar maple, white ash, Eastern hemlock and white pine. In the southern Appalachian mountains, however, northern hardwood forests include yellow buckeye in the mix but exclude white ash, hemlock and white pine.

Yellow birch was formerly called by the Latin name *Betula leutea*.

Black Birch (*Betula lenta*) Birch Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves often look paired on the twig, fall color = yellow

Flowers: catkins (small spike-like projections)

Fruit: nutlets in catkins

Twigs: slender, zigzag, shiny reddish, smell like wintergreen

Bark: dark gray with black vertical cracks (no peeling)

Ecology (page 12):

Shade Intermediate, Dominant & Dispersed Canopy

Moist mountain sites between 1500' and 4500'

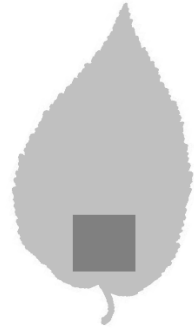
Fire Intolerant, Poor Sprouter

Intermediate, Mature & Climax forests

Cove F., N. Hardwood F., Oak-Hickory F.,

Bottomland F., Acid Cove F.

Cumberland Plateau & Blue Ridge provinces



Uses (page 20):

Black birch wood is hard, very strong, very stiff, resinous, hard to split dry, easier to split green. It exceeds yellow birch as excellent firewood, "doing its own blowing," according to old Carolina mountaineers, which refers to its flammable wintergreen oil.

Modern uses include furniture, plywood, cabinets and toys.

Indians and Pioneers made a bark and twig infusion (pounded material soaked in water) to stimulate urine flow, purify the blood, help the stomach and to treat rheumatism. Indians boiled the inner bark, making a plaster for wounds.

Pioneers made birch beer by fermenting a mix of black birch sap and corn. They also made a tea by steeping the twigs in hot water.

Oil of wintergreen was once distilled from crushed black birch twigs and bark, which contain far more oil than yellow birch. It took about 100 birch saplings for each quart of oil! Now made synthetically, it is used as a fragrance, flavoring and an anti-inflammatory drug (contains the active ingredient in aspirin).

Comments:

Black birch, a mountain tree, is most abundant along streams in the low elevation zone (1500'-2500'), but it also climbs slopes, following the moisture gradient up as high as 4500' elevation.

River Birch (*Betula nigra*) Birch Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves triangular in shape, fall color = yellow

Flowers: catkins (small spike-like projections)

Fruit: nutlets in catkins

Twigs: slender, zigzag

Bark: pinkish, peeling into thin strips

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

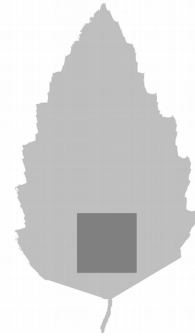
Moist sites along rivers and large creeks

Fire Intolerant-Resistant, Moderate Sprouter

Intermediate, Mature and Climax forests

Bottomland F., Cove F.

CUP, Piedmont & Coastal Plain provinces



Uses (page 20):

River birch wood is similar to other birches and has been sparingly used for the same purposes (see yellow birch page, 75). Its bark lacks the flammable oils that make good tinder and emit the wintergreen odor.

Cherokee Indians used the inner bark in a remedy for milky urine, and they drank bark tea to "check the bowels".

Long ago, Carolina rice planters made river birch hoops for barrels used to ship rice.

River birch is widely planted as an ornamental for its attractive form, peeling pinkish bark, and surprising hardness.

Spring buds were a favorite food of the extinct Carolina paroquet.

Comments:

River birch is well adapted to the riverbank where it grows.

Intermediate in shade tolerance, it requires moderate sunlight.

Openings are provided by the death of adjacent trees. If these are lacking, river birch will lean far out in the river channel into sunlight sufficient for its needs. This trait, however, works against the tree when great floods cut away the bank anchoring its roots, and it is washed into the river and away. The tree may lose the battle against the raging flood, but in the long run, its great roots stabilize the bank, providing excellent protection against erosion.

Hackberry (*Celtis occidentalis*) Elm Family

Identification (page 4):

Leaves: simple, alternate, toothed with large teeth (but leaf base usually toothless), leaves often rough above, leaf veins prominent, fall color = yellow

Flowers: small, greenish

Fruit: dark red or purple berry-like drupe

Twigs: slender, zigzag, often with woody galls

Bark: gray with prominent warts

Ecology (page 12):

Shade Intermediate, Subcanopy or Dispersed Canopy

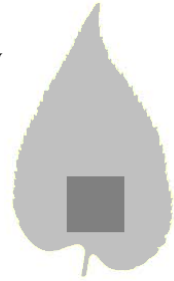
Moist to dry sites, may prefer calcium-rich soils

Fire Intolerant, Poor Sprouter

Pioneer, Intermediate and Mature forests

Cumberland Plateau and Valley &

Ridge provinces, rare in the Piedmont



Uses (page 20):

Hackberry wood is hard like the elms but not as hard as winged elm. Unlike winged elm, hackberry usually splits fairly easily. It is tough and flexible and was once used for barrel hoops.

Roy Underhill says, when lumbermen find a hackberry with interlocked grain, they sell it as elm. When the grain is straight, they sell it as ash!

Indians flavored their food with the sugary fruit pulp.

Illinois pioneers made cabin floor boards from split hackberry.

At least 25 species of birds eat the berry-like drupes, which are also edible by people.

Comments:

Hackberry trees are regularly attacked by gall-producing insects of the genus *Pachypsylla*. The galls form on the leaves and twigs and actually help identify the tree by their presence. Forming 1/2-inch diameter woody growths, the galls disfigure leaves and twigs but do not permanently damage the tree.

Hackberry and Georgia hackberry rarely overlap in range, but if so, hackberry leaves are 2 1/2 -5 inches long, whereas Georgia hackberry leaves are usually no longer than 2 inches. Hackberry is a medium-large tree, but Georgia hackberry is always shrubby.

Georgia Hackberry (*Celtis tenuifolia*) Elm Family

Identification (page 4):

Leaves: simple, alternate toothed with large teeth (but leaf base toothless), leaves rough above, leaf veins prominent,
fall color = yellow

Flowers: small greenish

Fruit: orange berry-like drupe

Twigs: slender, zigzag, often with woody galls

Bark: gray with prominent warts

Ecology (page 12):

Shade Intolerant, Pioneer tree

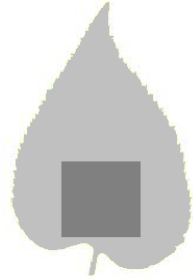
Dry sites, prefers calcium-rich soils

Fire Resistant, Poor-Moderate Sprouter

Pioneer and Intermediate forests

Pine-Oak F., Oak-Hickory F.

Piedmont province



Uses (page 20):

Georgia hackberry is a small tree with virtually no commercial value and little ornamental appeal, but its presence in the forest adds species variety.

Like other hackberries, the orange fruits are eaten by birds and small mammals, and can also be eaten by people.

Comments:

The genus name *Celtis* is an old Greek name applied to the genus by the botanist Linnaeus. The species name for this tree, *tenuifolia*, means slender leaf in Latin.

Other states call this tree dwarf hackberry. The tree is on Canada's Species At Risk list.

A Michigan State University study showed dwarf (Georgia) hackberry was absent on sites supporting abundant Heath family members like blueberries, huckleberries and wintergreen (*Chimaphila* sp.). This emphasizes its preference for high-calcium soils, since all Heath family members prefer acidic soils.

Apparently hackberry trees looked more like cherries than elms (their rightful family) to Pioneers, for the name hackberry comes from the Scottish word "hagberry" which refers to the Scottish cherry tree (*Prunus avium*)!

Sugarberry (*Celtis laevigata*) Elm Family

Identification (page 4):

Leaves: simple, alternate, toothed (sometimes toothless), shiny and smooth above, leaf tip long and pointed, often with 3 prominent leaf veins, fall color = yellow

Flowers: small greenish

Fruit: brownish to orange-red berry-like drupe

Twigs: slender, zigzag

Bark: gray with prominent warts

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

Moist sites along streams

Fire Intolerant, Poor-Moderate Sprouter

Intermediate and Mature forests

Bottomland F., Cove F., Oak-Hickory F.

All SE provinces, most abundant in

lower Piedmont and Coastal Plain



Uses (page 20):

Sugarberry wood is hard like the elms but not as hard as winged elm. It is tough and flexible. Unlike winged elm, sugarberry usually splits fairly easily. Modern uses include furniture, veneer and plywood.

Pioneers used sugarberry for barrel hoops and barrel staves for slack cooperage (barrels for holding dry materials).

Birds and other creatures including people eat the berries.

Comments:

Donald Peattie says the early French people of Louisiana (where the tree is abundant in bottoms) called sugarberry *bois inconnu*, which means the "unknown tree," just as early Englishmen called hackberry the "unknown tree." This predicament resulted because the trees look somewhat like elms, but have completely different fruit structures. The warty bark probably caught their attention, but caused further confusion in reconciling these trees to the elm family. The lack of major use by Indians probably also helped hackberry and sugarberry remain "unknown" trees for a while.

Distinguish sugarberry from the hackberries by its nearly toothless leaves that are shiny and smooth instead of rough like hackberry.

American Hornbeam (*Carpinus caroliniana*) Birch Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves hairless below, fall color = yellow

Flowers: small greenish female catkins (spike-like
projections)

Fruit: pairs of mitten-shaped bracts surrounding a nutlet (look
like praying hands)

Twigs: slender, zigzag

Bark: blue-gray, smooth and muscular-looking

Ecology (page 12):

Shade Tolerant, Understory tree

Moist sites along streams and rivers

Fire Intolerant, Moderate Sprouter

Intermediate, Mature & Climax forests

All Southeastern provinces



Uses (page 20):

Hornbeam wood is nearly as hard as dogwood, hence its alternate name "ironwood". It is considered very strong, stiff, tough, and difficult to split. It is nearly as springy as hickory and ash, and has low rot resistance. The wood makes excellent mallet and axe handles. Diffuse porous wood and high compression strength make hornbeam a good choice (but heavy) for longbows.

The name hornbeam is reportedly derived from the Celtic word for yoke, and refers to the wood's use in yoking oxen. A beam of this strong wood was lashed across the horns of a pair of oxen and hitched to loads pulled by them.

In 1639 William Wood, early New England writer said hornbeam, "...is a tough kind of wood that requires much paines in riving.....being the best to make bolles and dishes not subject to cracke or leake."

Cherokee Indians used the inner bark to treat milky urine. They also used it in a medicinal formula to treat "navel yellowness".

Comments:

American hornbeam is always an understory tree found growing along a stream. In the southern Appalachians, it follows streams up to about 2500 feet elevation where cold stops its progress.

Eastern Hophornbeam (*Ostrya virginiana*) Birch Family

Identification (page 4):

Leaves: simple, alternate, double toothed (teeth have teeth),
leaves fuzzy below, fall color = yellow

Flowers: small greenish female catkins (slender rod-like
projections)

Fruit: nutlets in a fuzzy papery sac (looks like oat or hop seed),
fruit fuzz can sting or itch if touched

Twigs: slender, zigzag

Bark: brown with narrow flakes

Ecology (page 12):

Shade Tolerant, Understory tree (rarely subcanopy)

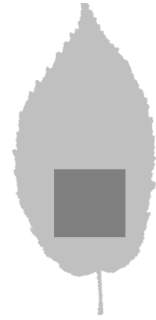
Moist to dry sites on banks & bluffs above streams

Fire Resistant, Good Sprouter

Intermediate, Mature and Climax forests

Cove F., Bottomland F., Oak-Hickory F.

All Southeastern provinces, but sparse in CP



Uses (page 20):

Hophornbeam wood is hard, strong, stiff, tough, springy and difficult to split. It is second only to hickory in fuel value. Diffuse porous wood and high compression strength make hophornbeam an good choice (but heavy) for longbows.

Cherokee Indians used hophornbeam and cucumbertree inner bark to make a decoction (bark boiled in water) for treating tooth ache. The patient held the mixture in his mouth until it cooled, then spit it out.

Pioneers used the same decoction as a laxative and as a tonic. In colonial times a fluid extract was used to treat malaria. Pioneers used the wood for bowls, axe and mallet handles and post packers. Hophornbeam buds and catkins are important winter food for ruffed grouse and other birds, but catkins can irritate human skin.

Comments:

Hophornbeam usually grows just above the stream bottom on adjacent bluffs, while its sib hornbeam grows in the bottom beside the stream or river.

Hophornbeam is one of the few trees with bark that breaks into thin vertical flakes, making it easy to identify in winter.

Carolina Buckthorn (*Rhamnus caroliniana*) Buckthorn Family

Identification (page 4):

Leaves: simple, alternate, toothed (small teeth), leaves shiny with conspicuous parallel veins, fall color = yellow

Flowers: small, white

Fruit: berry-like drupe, first red, then black at maturity in autumn

Twigs: twig end buds are hairy, twigs stink when crushed

Bark: gray with dark blotches

Ecology (page 12):

Shade Tolerant, Understory tree

Moist rich woods, prefers limestone soils

Fire Intolerant, Poor Sprouter

Intermediate, Mature and Climax forests

Cove F., Oak-Hickory F.

All SE provinces except rare in Blue Ridge
and absent in Coastal Plain



Uses (page 20):

Buckthorn fruit is greedily eaten by many birds.

The tree is fairly easy to grow from seed, and could be used as an ornamental tree in backyard wildlife plantings

Comments:

Some botanical references call this tree *Frangula caroliniana*, but this author will stick to the original Latin name of this and other recently “renamed” trees (like the cypresses) until genetic mapping confirms the changes or until time renders them acceptable.

Buckthorn (which is really thornless) is a fairly rare tree in the wild, growing in scattered locations. In Georgia, the author has seen it along the shores of Lake Hartwell on the Georgia-South Carolina border, growing as an understory tree in a mature oak-hickory forest. It was autumn and the leaves were bright yellow. Another tree member of the Buckthorn family, cascara buckthorn (*Rhamnus purshiana*) is native to northern California, Washington and Oregon. Known by early Spanish explorers as “sacred bark” due to its outstanding laxative properties, the tree's bark and berries have been marketed by American companies for years as “cascara”.

A European native, *Rhamnus cathartica* was long used as a laxative, though its toxicity makes it too risky by today's standards.

Carolina Silverbell (*Halesia carolina*) Storax Family

Identification (page 4):

Leaves: simple, alternate, toothed (very small teeth), fall color = yellow

Flowers: white bell-shaped flowers, hence the name, spring

Fruit: long narrow four-winged dry capsule

Twigs: slender

Bark: young trunks and branches have conspicuous stripes, older bark is reddish and flaky

Ecology (page 12):

Shade tolerant, Understory tree

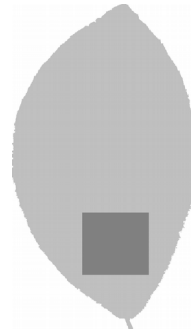
Moist sites along rivers & their tributaries

Fire Resistant, Good Sprouter

Intermediate, Mature and Climax forests

Bottomland F., Cove F.

Blue Ridge and Piedmont, rarely in CP



Uses (page 20):

Donald Peattie says the tree is (was) called bellwood by loggers in reference to the flowers, and its wood makes a beautiful bird's-eye figure when cut as veneer.

This tree makes a pretty ornamental on moist sites.

Comments:

Mountain silverbell (*Halesia carolina* var. *monticola*) grows to be a large size dispersed canopy tree in the southern Appalachian mountains. Here it was called “peawood” by mountain folks due to the supposed resemblance of its flowers to the garden pea.

When the southern Appalachian forests were initially logged during the early 1900's, mountain silverbell was considered a desirable lumber tree and was cut along with the many other species in this botanically rich region.

Great Smoky Mountains National Park now protects many acres of virgin mountain silverbell, such as those in Cucumber Gap near Elkmont campground. Here, at around 3000 feet, the silverbells dominate the canopy along with large cucumbertrees for which the gap was named. Mountain silverbell also grows as high as 5000 feet elevation in the northern hardwood forest and the high elevation northern red oak forest on exposed slopes and ridge tops.

Downy Serviceberry (*Amelanchier arborea*) Rose Family

Identification (page 4):

Leaves: simple, alternate, toothed, leaf base slightly heart-shaped, leaves slightly fuzzy below, fall color = red, orange, yellow

Flowers: white, appear very early in spring before leaves

Fruit: berry-like reddish-purple pome (like a small apple), mature in early summer

Twigs: slender, twig end buds are long, slender and pointed, chewed twigs have a faint bitter-almond taste

Bark: black with distinctive dark vertical slits on trunk

Ecology (page 12):

Shade tolerant, Understory tree

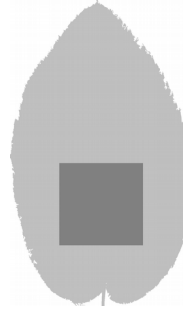
Dry to moist sites

Fire Intolerant, Good Sprouter

Pioneer, Intermediate, Mature
& Climax forests

Oak-Hickory F., Pine-Oak F.

All SE provinces, but rare in CP



Uses (page 20):

Serviceberry wood is considered heavy, very hard, strong, springy and lacks rot-resistance. Horace Kephart says it is virtually inflammable when green and so makes good backlogs, andirons and side logs for camp cooking fires.

Cherokee Indians made bows from the springy wood when hickory was scarce. They ate the fruit raw and dried it for use in winter.

They made a tea from the leaves in combination with other herbs to aid digestion, and bathed children in the tea to treat worms.

Pioneers used the same tea made by the Cherokees to slow menstrual bleeding, and they made pies and bread from the berries.

Forty bird species and numerous animals (including bears) eat the fruit which is edible and palatable to man. Ralph Waldo Emerson said the Cree Indians made a delicious pudding from the fruit.

Comments:

According to folklore, the name serviceberry relates to traveling preachers who arrived in Appalachian settlements each spring when winter snows began to melt and serviceberry bloomed to conduct wedding and funeral services delayed until then.

Allegheny Serviceberry (*Amelancier laevis*)Rose Family

Identification (page 4):

Leaves: simple, alternate, toothed, leaf base slightly heart-shaped,
leaves smooth below, fall color = red, orange, yellow

Flowers: white, appear in spring after the leaves

Fruit: berry-like reddish-purple pome (like a small apple),
mature in early summer

Twigs: slender, twig end buds are long, slender and pointed

Bark: black with distinctive dark vertical slits on trunk

Ecology (page 12):

Shade Tolerant, Understory tree

Dry to Moist sites above 2500' elevation

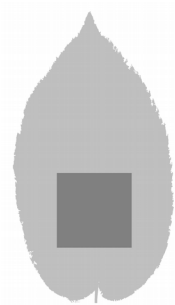
Fire Intolerant-Resistant, Moderate Sprouter

Pioneer, Intermediate, Mature & Climax forests

Oak-Hickory F., Acid Cove F.,

Northern Hardwood F., Spruce-Fir F.

Blue Ridge province



Uses (page 20):

The wood has the same properties as downy serviceberry (page 85).

Cherokee Indians used the roots of Allegheny serviceberry in a recipe for “bad diarrhea”.

Allegheny serviceberry fruit is eaten by many birds and animals.

In the southern Appalachians where this tree replaces downy serviceberry at higher elevations, bears often climb the small trees to get the fruit, breaking many branches in the process.

The fruit of this tree is considered better tasting than downy serviceberry fruit. Early botanist Francois Michaux found them for sale in Philadelphia, but adds they were only bought by children.

Comments:

Both downy serviceberry and Allegheny serviceberry occur in the southern Appalachian mountains. Downy serviceberry is rarely found growing above 2500 feet elevation, and Allegheny serviceberry is rarely found below 2500 feet. Also, Allegheny serviceberry leaves are said to have a more tapered and pointed leaf tip than those of Downy serviceberry.

“Willy and Al Hollered and drank Sour Cherry wine when they caught a Cottonpickin' Bass at the Beach.”

Willow, Alder, Holly, Sourwood, Cherry, Cottonwood, Basswood, Beech

Black Willow (*Salix nigra*), Willow Family

Identification (page 4):

Leaves: simple, alternate, toothed, long and narrow, fall color = yellow

Flowers: small catkins

Fruit: cottony seeds that float on the wind, early spring

Twigs: slender, reddish, cut twigs will root, forming a new tree

Bark: reddish-brown with scaly plates

Ecology (page 12):

Shade Intolerant, Pioneer tree

Wet sites along streams and swamps

Fire Resistant, Good Sprouter

Developing forest and Pioneer forest

Bottomland F.

All Southeastern provinces



Uses (page 20):

The wood is flexible and very shock resistant. It has been used for artificial limbs, boxes, woodenware, and wicker furniture because it holds nails very well and does not easily split.

Cherokee Indians used willow bark to make baskets.

Other Indians made water-tight baskets from willow splits (thin wood strips split from the trunk). They also used a bark infusion (chopped bark soaked in cold water) to treat pain, colds and fever.

Horace Kephart says willow twigs (withes) can be gathered in late spring or early summer, soaked in water and woven while wet into baskets.

Donald Peattie says black willow wood made a fine charcoal used in black powder rifles during the American Revolution.

Comments:

Willow leaves and bark contain salicin, a chemical similar to aspirin and with the same pain-killing properties.

Hazel Alder (*Alnus serrulata*), *Birch Family*

Identification (page 4):

Leaves: simple, alternate, finely toothed, fall color = yellow

Flowers: long slender pollen-filled male catkins in late winter

Fruit: tiny conelet (looks like a tiny pine cone) remaining on tree

Twigs: brownish-gray

Bark: grayish, sometimes muscular-looking like hornbeam

Ecology (page 12):

Shade Intolerant, Pioneer tree

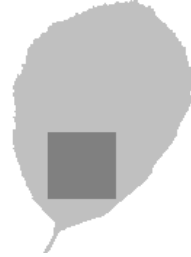
Wet to Moist sites, streams, lakes, swamps

Fire Intolerant, No Sprouts

Developing forest, Pioneer forest

Bottomland F.

All Southeastern provinces



Uses (page 20):

Alder wood is very durable underwater, making good pilings for primitive docks and bridges and for water pipes. Horace Kephart says the wood burns readily giving considerable heat but is not long lasting.

The Cherokee Indians used alder extensively for medicine:

- root infusion along with other plants for pain

- root decoction with other plants to induce vomiting

- root decoction with other plants for indigestion

- bark infusion for sore eyes

- cold bark tea makes urine flow

- alder bark tea is given children for “thrush”(mouth infection)

- cold bark tea thins the blood and reduces blood pressure

- warm bark tea reduces menstrual flow

- infusion of the inner bark to treat poison ivy

Infusion is made by soaking pounded or shredded parts in water.

Decoction is made by boiling the plant parts in water.

Comments:

Hazel alder fixes nitrogen from the air and stores it in its roots, enriching the soil where it grows.

The tiny female cones (look like small pine cones) of alder stay on all year, making the tree easy to identify.

Mountain Holly (*Ilex montana*) Holly Family

Identification (page 4):

Leaves: deciduous, simple, alternate, toothed, leaf veins look sunken into leaf surface, leaf looks wrinkled, fall color = pale yellow

Flowers: small inconspicuous

Fruit: red berry-like drupe that stays on into the winter

Twigs: grayish with short spur-like shoots attached along twig

Bark: grayish

Ecology (page 12):

Shade Tolerant, Understory tree

Moist sites above 4000' elevation,
prefers acidic soils

Intermediate, Mature & Climax forests

High elevation Oak-Hickory F.,

N. Hardwood F., Spruce-Fir F.

Blue Ridge, Cumberland Plateau



Uses (page 20):

Since the berries persist on the tree into winter, they provide valuable food for many birds, including grouse, turkey, bluebirds, catbirds, mockingbirds, robins, thrushes, bluejays, cedar waxwings, and thrashers. Small mammals like chipmunks and squirrels also eat them. Mountain holly twigs are browsed by deer in winter.

Comments:

The alternate name, winterberry for this tree refers to the persistent berries.

Though early botanists combed the Appalachians for a century looking for new plant species, this inconspicuous little tree was not discovered and named until 1840 in the Catskill Mountains.

Possumhaw (*Ilex decidua*), a small deciduous holly of the Piedmont has smaller leaves than mountain holly and its leaves lack sunken veins and the wrinkled look of mountain holly leaves.

Holly trees are either male or female, and only female trees produce berries. Persimmon, boxelder maple, ginko, Eastern red-cedar, cottonwood and willow all share this characteristic, designating them as dioecious in reproductive arrangement.

American Holly (*Ilex opaca*)Holly Family

Identification (page 4):

Leaves: evergreen, simple, alternate, with sharp spines, glossy

Flowers: small and inconspicuous

Fruit: red berry-like drupe

Twigs: thick and gray

Bark: smooth and gray

Ecology (page 12):

Shade Tolerant, Subcanopy & Understory tree

Dry to Moist sites, prefers acidic soils

Fire Intolerant-Resistant, Good Sprouter

Intermediate, Mature & Climax forests

Oak-Hickory F., Pine-Oak F., Acidic Cove F.,

Evergreen oak F.

All Southeastern provinces



Uses (page 20):

Holly wood is hard, very close grained, has ivory-white wood and makes excellent fuel. Modern uses include inlays in fine furniture.

Cherokee Indians scratched themselves with the spiny leaves to relieve muscle cramps. They carved the wood into spoons and used holly berries to make dye. They wore holly berries to ensure success in battle. Their name for the tree means “shaking the top”.

Pioneers chopped fresh bark and roots, added alcohol and water, aged the mixture for 8 days and used it for a general “cure-all”.

They made a leaf tea to induce sweating and break a fever.

Boughs and berries have long been harvested for Christmas wreaths and decorations, threatening the tree's survival in the wild.

Hollies of many varieties are widely planted as ornamentals.

Eighteen bird species eat the berries which persist through winter.

Comments:

A sticky substance for trapping birds called birdlime has been made by boiling the bark of holly species into a sticky paste.

Washington Irving said George Washington's false teeth were made of American holly, which is probably true!

Folklore says hollies first began to grow in Christ's footsteps. The thorns represent His crown and the berries, His blood.

Holly is witch-proof, and a holly stick will turn the meanest dog!

Yaupon Holly (*Ilex vomitoria*) Holly Family

Identification (page 4):

Leaves: small, evergreen, simple, alternate, toothed, leaves thick and shiny above, leaves without spines

Flowers: small and inconspicuous

Fruit: shiny red berry-like drupe

Twigs: light gray

Bark: gray, but on mature trees it's red-brown and scaly

Ecology (page 12):

Shade Tolerant, Understory tree

Moist to Dry sites

Fire Intolerant-Resistant, Moderate Sprouter

Intermediate, Mature & Climax forests

Maritime Forest, Coastal Dunes

Coastal Plain province



Uses (page 20):

Yaupon holly is the only native North American plant whose twigs and leaves contain caffeine. Southeastern Indian tribes used the plant extensively for a beverage called the "black drink", and they lucratively traded its leaves with Indians far from its natural range. According to Dr. Charles Hudson, former University of Georgia professor of Anthropology, "Indians drank black drink for three different purposes: (1) as a stimulating social beverage, much as we drink coffee or tea...; (2) as a medicine to improve their physiological and psychological well-being; and (3) possibly as an emetic to induce vomiting" during cleansing ceremonies. Twigs and leaves were prepared by pounding in a mortar, then parching them black in the fire to concentrate the caffeine. This material was then boiled to make a tea which was usually drunk hot from a conch shell cup. Only males were allowed to drink the tea, and only females were allowed to make it, but this author suspects the girls got their share while the braves were "at work"!

Comments:

Yaupon holly tolerates Southeastern Piedmont climate conditions well, and its evergreen habit, small size and attractive foliage and berries make it ideal and widely used for landscaping in the Southeast, with several varieties including a weeping form.

Sourwood (*Oxydendrum arboreum*) Heath Family

Identification (page 4):

Leaves: simple, alternate, toothed (V. small teeth), sour-tasting,
fall color = red, orange, yellow

Flowers: small, white, urn-shaped, clustered on a drooping spray
that looks like a gracefully-posed hand, early-summer

Fruit: small woody capsules on a drooping spray, persisting into
autumn

Twigs: slender, zigzag, reddish, often forming very straight
vertical shoots

Bark: reddish with tall flat-topped ridges, trunk always crooked

Ecology (page 12):

Shade Intermediate to Tolerant, Subcanopy tree

Dry sites, prefers acidic soils

Fire Resistant, Good Sprouter

Developing, Pioneer, Intermediate & Mature forests

Pine-Oak F., Oak-Hickory F.

All SE provinces except Coastal Plain



Uses (page 20):

Indians used the straight shoots for arrow shafts. They stewed bark scraped from young shoots in winter for a diarrhea medicine. They believed burning sourwood for fuel caused sickness.

Pioneers used the very hard wood for tool handles and sled runners (takes a high polish). It turns well on a lathe for making into chair legs and bowls. Horace Kephart says sourwood shoots make excellent carveable walking sticks. "They grow exceptionally straight and slender and season lightweight, strong and hard."

For childhood asthma, Mountaineers put a lock of the child's hair in a hole drilled in a sourwood tree just above head height. When he grew taller than the hole, he would be cured. If he died, so would the tree.

Sourwood honey is famous for its excellent taste.

Comments:

Though sourwood shoots are quite straight, the tree itself presents a very crooked growth pattern easily identified in the forest.

Sourwood, rhododendrons, mountain laurel and blueberries are all members of the acid-loving Heath family.

Black Cherry (*Prunus serotina*) Rose Family

Identification (page 4):

Leaves: simple, alternate, toothed, crushed leaves smell like bitter almond, there are 2 tiny glands (dots) on the leaf stem,
fall color = red, orange, yellow

Flowers: small, white, clustered on drooping stalks, early spring

Fruit: small round cherry, red, then black when mature, summer

Twigs: scratched twigs smell like bitter almond

Bark: smooth and silvery on young trees, flaky like "burnt potato chips" on older trees, black knot fungus cankers often abundant on trunk, aiding in identification

Ecology (page 12):

Shade Intolerant Pioneer, Dispersed Canopy tree

Dry to Moist sites

Fire Intolerant-Resistant, Prolific Sprouter

Developing, Pioneer, Intermediate, Mature & Climax

Pine-Oak F., Oak-Hickory F., Cove F., Bottomland F.,

N. Hardwood F.

All SE provinces



Uses (page 20):

Black cherry wood is hard, strong, smooth-grained, difficult to split, easily worked, with moderately rot-resistant heart wood. Kephart says it makes only fair fuel.

Cherry wood makes excellent furniture and cabinets and is very valuable for this purpose in northern states where it grows large, in abundance and is managed for this purpose.

Cherokee Indians ate wild cherries. They used the straight woody shoots for arrow shafts. They drank an inner bark decoction (material boiled in water) for fever, colds and coughs.

Pioneers made a cordial drink called cherry bounce by adding brandy or rum to the crushed fruits, and put the cherries in pies, jams, cherry wine and ice cream flavoring.

Many birds and animals, including bears, relish wild cherries.

Comments:

If black cherry leaves die during the growing season, they are poisonous to livestock due to their hydrocyanic acid (also called prussic acid) content. Autumn leaves don't contain the poison.

Carolina Laurelcherry (*Prunus caroliniana*) Rose Family

Identification (page 4):

Leaves: simple, alternate, toothed (only a few small teeth),
evergreen, crushed leaves smell like almond, leaf underside
has 2 tiny glands near the leaf base

Flowers: small white, in drooping clusters, early spring

Fruit: cherry, green when young and black when mature in fall

Twigs: green at first, then becoming gray

Bark: gray

Ecology (page 12):

Shade Intolerant Pioneer, Dispersed Canopy tree

Moist sites

Fire Intolerant-Resistant, Moderate Sprouter

Developing, Pioneer and Intermediate forests

Bottomland F., Maritime F.

Coastal Plain, naturalized in Piedmont



Uses (page 20):

Though a Coastal Plain native, Carolina laurel cherry grows well in the Piedmont and its evergreen habit, moderate size and attractive flowers and fruit make it a popular ornamental, fairly easily grown by planting seeds immediately after ripening. It makes a carefree evergreen screen that provides food for birds.

Comments:

Like black cherry, the wilted leaves of this tree contain hydrocyanic acid (also called prussic acid), fatal to humans and animals. Green leaves are browsed by deer with impunity.

In 1773, botanist William Bartram found this tree growing along the St. Marys River in Georgia along with live oak, water tupelo and bald cypress. He used the Latin name *Prunus laura cerapa* for it here, but later called it *Prunus nemoralis* when he encountered it in Alabama.

Cherry seeds along with those of apples and peaches contain hydrocyanic acid, a form of cyanide. If you swallow (don't chew) these seeds they won't hurt you. If you chew and swallow just a few, you are safe, but a large handful, chewed and swallowed could very likely kill a human.

This tree is usually called cherrylaurel, but that makes no sense!

Pin Cherry (Fire Cherry) (*Prunus pensylvanica*) Rose
Family

Identification (page 4):

Leaves: simple, alternate, toothed, with long narrow tips,
fall color = red, orange, yellow

Flowers: white in clusters, blooming when leaves first appear

Fruit: round cherry, bright red when mature in late summer

Twigs: shiny red with orange lenticels (horizontal spots)

Bark: smooth, fiery coppery color with peeling plates and
lenticels

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

Moist high mountain sites, common on burned areas

Fire Tolerant (seeds), Moderate Sprouter

Developing & Pioneer forests

Spruce-Fir F., Northern Hardwoods F.

High elevations in the southern Appalachian Mts.



Uses (page 20):

Pin cherry fruits are edible and were once made into jelly.

Twenty three kinds of birds and many animals eat the cherries.

The tree is harvested for neither paper pulp nor lumber.

Comments:

Unlike black cherry, pin cherry is a short-lived tree, achieving little more than 50 years. Likewise, it never attains black cherry's height, but it serves an important purpose in the forest. When fire or disease (Fraser fir wooly adelgid) lays a site bare, pin cherry comes in to reclaim the soil, protecting it from erosion on steep mountain slopes, and paving the way for the mature and climax forests to come.

The seeds can survive in the ground for almost a century where they form a seed bank, protected by moist soil. Fire burns away duff, exposing the seeds and applying a thin ash layer as fertilizer so they can germinate into a thick ground-protecting stand of trees.

The largest Southeastern pin cherries are found on the western slope of Great Smoky Mountains National Park. Here, they are found growing at around 5000 feet elevation along with yellow birch, northern red oak, red spruce and Fraser fir and Mt. ash.

Eastern Cottonwood (*Populus deltoides*) Willow Family

Identification (page 4):

Leaves: simple, alternate, toothed, triangular with long flattened leaf stems that make the leaf "tremble" in the wind, fall color = yellow

Flowers: pollen-bearing flowers in slender catkins on male trees, small female flowers on female trees, early spring

Fruit: tiny seeds in cottony tufts floating on the wind, early summer

Twigs: yellowish-brown, stout with prominent lenticels (lines)

Bark: mature bark gray with ridges

Ecology (page 12):

Shade Intolerant Pioneer, Dispersed Canopy tree

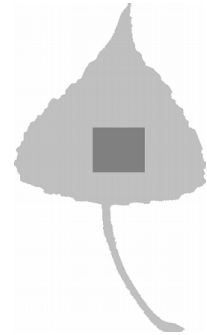
Moist and Wet sites

Fire Intolerant, Weak Sprouter

Developing and Pioneer forests

Bottomland Forest

Piedmont and upper Coastal Plain



Uses (page 20):

Cottonwood is soft, non-resinous and straight grained.

These characteristics led the the Indians to employ the wood for hearths and spindles used in making fire by friction.

The wood has also been used for pulp in paper-making, boxes, crates, excelsior (packing material) and matches.

Cottonwood grows fast and sprouts readily, consequently it has been used in coppice forestry where saplings are cut at 5 to 10 years. Root sprouts then grow rapidly, producing another usable crop in another 5 to 10 years. The coppice wood is used for fuel. Cottonwood burned in a woodstove is reputed to clean the chimney of soot.

Comments:

The name comes from the cottony seeds dispersed by wind that cover the ground like snow in early spring.

Eastern cottonwood ranges across the Mississippi River to South Dakota, then south to Texas, where it grows below the prairie along river bottoms. Prairie pioneers used it for fuel, cabins, corncribs, wind breaks, ox yokes, saddletrees and, finally, coffins.

Basswood (*Tilia heterophylla*) Linden Family

Identification (page 4):

Leaves: simple, alternate, toothed, heart-shaped with unequal leaf bases (lopsided, see silhouette), fall color = yellow

Flowers: small yellowish, early summer

Fruit: small round greenish balls (nuts) in clusters attached to a leaf-like structure extending from the leaf base, late summer, often persists into winter

Twigs: slender, zigzag, twig end bud (terminal bud) absent

Bark: gray with thin platy ridges that can be slightly scaly, trunk of older trees usually surrounded by numerous sprouts

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

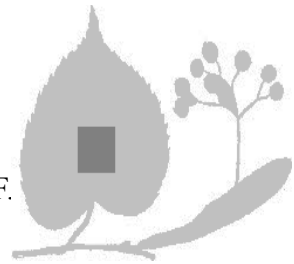
Moist sites along streams and lower slopes

Fire Resistant, Prolific Sprouter

Intermediate, Mature and Climax forests

Cove F., Bottomland F., Acid Cove Hemlock F.

All Southeastern provinces



Uses (page 20):

Basswood is soft, light-weight, tough, splits easily, flexible and easily worked with tools. The wood will scarcely burn green, so it makes good fire back logs. Dry basswood burns quick, but leaves no good coals for cooking. Modern uses for this soft light wood include wood carvings, artificial limbs, toys and string marionettes. Iroquois Indians carved gruesome "false-face" masks on basswood trees, then cut them free for use in ceremonies.

Indians and Pioneers considered basswood inner bark excellent material for cordage and ropes. They stripped bark from the numerous sprouts usually surrounding older trees, soaked it in water for a month (called retting), then twisted the strands into cords and ropes they claimed were softer than hemp on the hands when wet and would not kink when dry.

Basswood flowers make an excellent tea.

Comments:

In the Smoky Mountains in June, the trail is often covered with small round greenish basswood fruit. Basswood follows mountain streams up to about 4500' elevation where cold limits it.

American Beech (*Fagus grandifolia*) Beech Family

Identification (page 4):

Leaves: simple, alternate, toothed, with a thin papery feel and prominent leaf veins, tan leaves stay on tree all winter, fall color = yellow, green, tan

Flowers: tiny and inconspicuous, early spring

Fruit: a triangular nut enclosed in a spiny bur or husk in fall

Twigs: slender, zigzag with long pointed cigar-shaped end buds

Bark: smooth and gray all ages, often carved with initials

Ecology (page 12):

Shade Tolerant, Dispersed Canopy tree

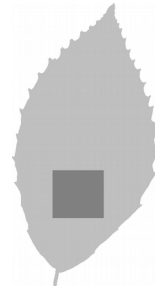
Moist sites along & just above streams

Fire Intolerant, Moderate-Poor Sprouter

Mature and Climax forests

Cove F., Bottomland F., Oak-Hickory F.,
Northern Hardwood F., Beech Gap F.

All Southeastern provinces, but limited in
lower Coastal Plain



Uses (page 20):

Beech wood is hard, strong, easily split green or dry, hard to season, lacks rot resistance, makes good coals. Modern uses include furniture, wooden cookware, clothespins.

Cherokee Indians believed beech trees were never struck by lightning. Statistics verify this, though the cause is unknown.

Ohio Pioneers stuffed mattress ticks with beech leaves because they lasted years and did not mat up or get musty. Beech wood was the very best for wooden bodies of woodworking planes.

Our ancient ancestors carved Sanskrit characters on beech bark.

Daniel Boone marked an old beech tree in Tennessee as follows: "D. Boone Cilled A Bar On Tree In Year 1760." The tree finally fell in 1916.

Comments:

In the southern Appalachians between 4500' and 5500' elevation, the very rare beech gap forest is found on exposed slopes. Stunted beech trees comprise 75% of the low canopy, with a carpet of mountain wildflowers below. Unfortunately, feral hogs and exotic beech bark disease are currently destroying these rare forests.

American Chestnut (*Castanea dentata*) Beech Family

Identification (page 4):

Leaves: simple, alternate, toothed with jagged teeth, leaves about 8 inches long, hairless below

Flowers: Creamy white on long upright stalks, distinctive odor, blooming in early summer

Fruit: edible nut about the size of a large marble, enclosed in pairs or triplets in a very sharp-spined husk

Twigs: zigzag, covered with lenticels (slit-like pores)

Bark: gray to brown with flattened plates when mature

Ecology:

Intermediate in shade tolerance, Dominant Canopy

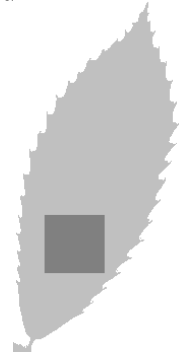
Dry-Moist sites on upper slopes and ridge tops

Fire Resistant, Good Sprouter

Intermediate and Mature forests

Oak-Hickory F., Oak-Pine F., Acid Cove Hemlock F.

Cumberland Plateau, Blue Ridge, upper Piedmont



Uses (page 20):

American chestnut wood splits easily, is light, strong, easily worked, with rot-resistant heartwood. Before this tree was wiped out by the exotic chestnut blight fungus during the early 1900's, it was used for furniture, fence posts and rails and caskets. The bark was a major source of tannin for curing animal skins.

Cherokee Indians boiled year-old chestnut leaves for heart trouble, and used the bark in a steep with 3 other trees for female troubles.

Indians and mountain folk ate the nuts, fed them to their livestock and sold them to buyers for city markets. It has been said the rise of mountain moon-shining resulted from the loss of chestnuts as a money crop for mountain farmers.

Chestnuts were an extremely important mast crop for a host of wild animals. Wildlife populations have not recovered from the loss of this valuable food source.

Comments:

The chestnut blight did not initially kill chestnut roots. Even today the roots send up sprouts that the blight eventually finds and kills. Blight-resistant chestnut trees have recently been bred, and may one day result in the return of our American chestnut forests.

Trees With Lobed Leaves

“Sweet Mable, the Sassy and Poplar Mulberry, got Sick in Oklahoma.”

Sweetgum	page 101
Maples	page 38
Sassafras	page 102
Poplar	page 103
Mulberries	page 104
Sycamore	page 106
Oaks	page 107

Sweetgum (*Liquidambar styraciflua*) Witch Hazel Family

Identification (page 4):

Leaves: simple, alternate, toothed, 5-lobed, leaves star-shaped with a resinous odor, fall color = red, orange, yellow

Flowers: greenish balls (male) in clusters, blooming around April

Fruit: the familiar "sweetgum ball" (female)

Twigs: stout, often with corky wings along the stem

Bark: grayish and platy

Ecology (page 12):

Shade Intolerant , Dominant Canopy tree

Wet to Dry sites

Fire Intolerant-Resistant, Good Sprouter

Developing, Pioneer and Intermediate forests

Bottomland F., Cove F., Oak-Hickory F.,

Oak-Pine F., Maritime F.

All Southeastern provinces



Uses (page 20):

Sweetgum wood is dense, tough and very difficult to split.

Modern uses include veneer, furniture, paper pulp and even chop sticks for export to China! Lumber is shipped to Europe as "satinwood", where it is made into furniture and shipped back as "Circassian walnut."

Indians and Pioneers chewed the aromatic sap as chewing gum, and used it to treat sores and skin troubles. Cherokees drank an inner bark tea to calm the nerves. Confederate soldiers used the sap and leaves boiled in milk to treat dysentery. Whiskey-soaked twigs were chewed to clean the teeth. When Spanish conqueror Cortez met Aztec emperor Montezuma in 1519, they smoked ceremonial pipes containing tobacco and sweetgum sap.

Sweetgum's hardness and lovely red autumn colors have led to its use as an ornamental, but sweetgum balls can be troublesome.

Comments:

This tree thrives in a variety of habitats from wet swamps to dry hillsides, as a pioneer and can live to 150 years. Girdling of the tree by man or beaver will not kill it, but repeated burning will.

Oriental sweetgum (*L. orientalis*) produces gum (just like our tree's) burned for centuries as incense in churches and temples.

Maples, page 38

Sassafras (*Sassafras albidum*) Laurel Family

Identification (page 4):

Leaves: simple, alternate, toothless, leaves are unlobed, two-lobed or three-lobed, fall color = red, orange, yellow

Flowers: small yellowish, clustered at twig ends, early-mid spring

Fruit: dark blue oblong berry-like drupe with a red stalk, early summer, only female trees bear the fruit

Twigs: slender, greenish, with a spicy flavor when chewed

Bark: brown with narrow fibrous ridges broken by horizontal slits

Ecology (page 12):

Shade Intolerant, Dispersed Canopy

Moist-Dry sites, fence-rows

Fire Tolerant, Prolific Sprouter

Pioneer and Intermediate forests

Oak-Pine F., Oak-Hickory F.

All Southeastern provinces



Uses (page 20):

Sassafras wood is light weight and rot resistant.

Cherokee Indians drank the root tea for headaches and colds.

Pioneers ate the raw flowers in spring considering them "very cleansing to the blood." They made bowls from the wood.

In Colonial times sassafras roots and bark were shipped in quantity to Europe for use as a tasty tea extolled as a cure-all, especially for venereal disease. Later, oil of sassafras was distilled from bark, twigs and roots and used in flavoring extracts, scented soaps and pharmaceuticals. The leaves were once chewed for their spicy mucilaginous flavor, and they were added to soups to thicken them similarly to those of the tree's relative redbay.

More recently, all parts of the tree have become suspect as containing a cancer-causing agent. This author grew up with a friend whose mother drank sassafras tea daily. She died of stomach cancer around age 40.

Comments:

Red mulberry leaves can also have mitten shapes, but the leaves are toothed as opposed to the toothless leaves of sassafras.

Yellow Poplar (Tulip Poplar)(*Liriodendron tulipifera*)
Magnolia Family

Identification (page 4):

Leaves: simple, alternate, toothless, lobed (4-6 lobes), leaf silhouette shaped like a tulip flower or a cat's face, fall color = orange, yellow

Flowers: greenish yellow tulip-like with orange petals

Fruit: cone made of many winged seeds attached to a central stalk, looks like a small woody chandelier

Twigs: brownish with stipular rings (rings that encircle the twig), twig end bud is flattened like a ducks bill

Bark: light gray with narrow ridges, often with whitish spots

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

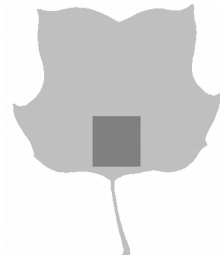
Moist rich, well-drained sites

Fire Resistant-Tolerant, Prolific Sprouter

Pioneer, Intermediate, Mature & Climax forests

Cove, Bottomland, Oak-Hickory & Hemlock Fs.

All Southeastern provinces



Uses (page 20):

Yellow poplar is soft, easily split and worked, lacks rot-resistance, won't burn green, pops when burned dry, leaving few coals. Uses include rotary-cut veneer, furniture and even chopsticks!

Cherokee Indians, whose name for the tree means canoe, carved the straight trunks into dugout canoes. They folded the bark into food containers, and used dried powdered bark as a preservative in medicines. Pioneers made bowls, spoons, dippers and dough mixing troughs because the wood is easily worked and does not taint food. It was a favorite wood for the hewed logs of log cabins and hand-made toys.

Yellow Poplar is widely planted as an ornamental.

Comments:

The prominent pioneer tree on moist Southeastern sites, yellow poplar's winged seeds colonize open land by the thousands, creating a thick pioneer forest. Unlike other pioneer trees, however, poplar remains into the Climax forest as a giant monarch. At nearly 200 feet, it is the tallest hardwood tree in the East.

Red Mulberry (*Morus rubra*) Mulberry Family

Identification (page 4):

Leaves: simple, alternate, toothed; leaves heart-shaped, mitten-shaped or 3-lobed; leaf underside fuzzy, fall color =yellow

Flowers: small & green

Fruit: looks like a long blackberry, reddish to black when mature

Twigs: slender, zigzag, leaf stem leaks milky sap when cut

Bark: tan to brown with scaly plates, often with flaky swellings

Ecology (page 12):

Shade Tolerant, Understory tree

Moist well-drained sites to dry sites

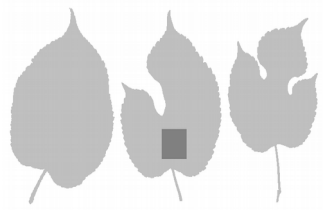
Fire Intolerant, Prolific Sprouter

Pioneer, Intermediate and Mature forests

Oak-Hickory F., Calcareous F.,

Bottomland F., Cove F., Maritime F.

All Southeastern provinces



Uses (page 20):

Mulberry wood is hard (heartwood), rot-resistant (heartwood), splits OK green and burns long with good coals.

Indians relished the fruit and made a drink from it to relieve fever. They used the inner bark for cordage and rope. Hernando Desoto's expedition to the South in 1541 encountered Indian women who wove excellent cloth from the inner bark of mulberry sprouts. His soldiers appropriated this cloth when their clothes wore out.

Naturalist William Bartram reported red mulberry among the plants he regularly encountered on old Indian village sites in Georgia in 1773. The others were persimmon, honeylocust, Chicasaw plum, French mulberry, black walnut and shell-bark hickory, his name for mockernut hickory (*Carya tomentosa*).

Pioneers used the rot-resistant wood for fence posts and tree nails (pegs used as nails) and ate the berries.

The milky sap and green fruit are said to be intoxicating and mildly hallucinogenic.

Many animals eat the berries including 21 bird species.

Comments:

Redbud trees have heart-shaped leaves like mulberry, but they are toothless. Basswood 's heart-shaped leaves have unequal bases.

White Mulberry (*Morus alba*) Mulberry Family

Identification (page 4):

Leaves: simple, alternate, toothed, leaves are shiny and hairless, leaves can be heart-shaped, mitten-shaped or 3-lobed, leaves exude milky sap when crushed, fall color = yellow

Flowers: small and green

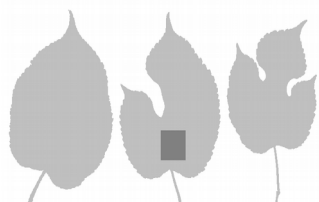
Fruit: like a long blackberry; purple, white or pink when mature

Twigs: slender, zigzag

Bark: tan to brown with scaly plates

Ecology (page 12):

White mulberry is native to China, but has spread throughout the U.S. and is most abundant on moist sites along streams.



Uses (page 20):

General James Oglethorpe, founder of Colonial Georgia, brought 500 white mulberry trees to Fort Frederica on St. Simons Island in 1733 along with silk worms to establish the silk industry in the fledgeling colony. The trees prospered and were rapidly spread by hungry birds, but the industry failed due to the South's hot climate and Asia's ability to produce cheaper silk.

General Oglethorpe was an innovative administrator who established the "Trustees Garden" in Savannah. Here he planted plants from across the world he thought might become agricultural successes in Georgia. These plants were dispensed to colonists who tried them on their farms, reporting the results to him.

Comments:

A tea made from white mulberry leaves has been drunk to relieve headache and soothe coughs.

The mulberry family includes figs (native to southwest Asia and the Mediterranean), Osage orange, the best native American wood for bows, the breadfruit tree and rubber tree, all with milky sap.

Paper mulberry (*Broussonetia papyrifera*), another member of this family but of a different genus, is native to eastern Asia where its fibrous bark has long been used in paper-making. South Seas natives made their tapa cloth for clothing from this tree which has been spread by birds throughout the eastern U.S. This tree's leaves are papery and very fuzzy above and below.

American Sycamore (*Platanus occidentalis*) Sycamore
Family

Identification (page 4):

Leaves: simple, alternate, toothed, 3-5 lobed, fall color = brown

Flowers: small, inconspicuous

Fruit: a tight tan-colored one-inch ball made of fuzzy nutlets

Twigs: slender, zigzag, grayish

Bark: stark white upper trunk and flaky brown down lower

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

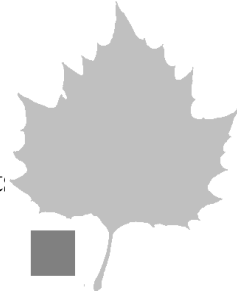
Always found along rivers and streams

Fire Intolerant, Prolific Sprouter

Pioneer, Intermediate, Mature and Climax forest

Bottomland F. Cove F. below 2500' elevation

All Southeastern provinces



Uses (page 20):

Sycamore wood is hard, difficult to split, lacks rot-resistance, is very tight-grained, hard to burn green, burns well seasoned. The tight grain has led to its use in butcher blocks for meat cutting and steam-bent furniture. The tight grain also gives lumber a characteristic figure used in ornamental woodwork.

Botanist Andre Michaux saw a 65-foot long sycamore dugout canoe made by French voyageurs that carried 9000 pounds of cargo.

In 1820, John James Audubon the famous bird artist saw hundreds of chimney swifts emerge with a mighty roar from a hole in a hollow sycamore, one dawn near Louisville, Kentucky .

Major Long's expedition to the Rocky Mountains found sycamores in Ohio 14 feet in diameter whose seeds were a favorite food of the Carolina parakeet, a now extinct American parrot once native to the eastern, midwest and plains states.

Comments:

George Washington reported a sycamore tree along the Ohio River measuring 13 feet in diameter. Huge old trees are often hollow.

In 1775 the skeleton and initialed powder horn of John Stewart, Daniel Boone's hunting partner, were found in a hollow sycamore where he had hidden after being mortally wounded by Indians.

Oaks

Oaks comprise a large number of trees fairly easily identified as a group. Look for the **multiple buds on the twig ends**, a consistent oak characteristic. Also look for acorns in the tree or on the ground below. Mature oaks are usually large canopy trees, dominating the forest.

There are two major groups of oaks:

White Oak Group

- leaf lobes are rounded
- leaf lobes lack small spines called bristles on the lobe ends
- bark is usually dark gray or light gray
- acorns germinate immediately after dropping in autumn

The white oak group includes:

White Oak	page 108
Overcup Oak	page 109
Swamp Chestnut Oak	page 110
Chestnut Oak	page 111
Post Oak	page 112
Live Oak	page 113

Red Oak Group

- leaf lobes are pointed
- leaf lobes are bristle tipped, though it's not always obvious
- bark is usually dark gray or black
- acorns drop in autumn and germinate the following spring

The red oak group includes:

Scarlet Oak	page 114
Southern Red Oak	page 115
Bluejack Oak	page 116
Turkey Oak	page 117
Laurel Oak	page 118
Blackjack Oak	page 119
Water Oak	page 120
Willow Oak	page 121
Northern Red Oak	page 122
Black Oak	page 123

White Oak (*Quercus alba*) Beech Family, White Oak group

Identification (page 4):

Leaves: simple, alternate, toothless, lobed, leaf lobes rounded,
fall color = amber, red, orange

Flowers: small, inconspicuous

Fruit: typical acorn, acorn is bullet-shaped

Twigs: moderately stout with multiple end buds

Bark: light gray and very flaky (vertical flakes) on older trees

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree

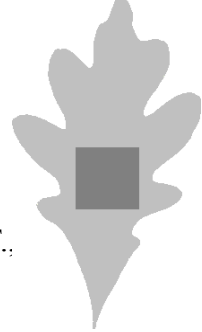
Moist, rich, well drained sites

Fire Resistant-Tolerant, Prolific Sprouter

Intermediate, Mature & Climax forests

Bottomland F, Cove F., Oak-Hickory F., Oak-Pine F.,

All Southeastern provinces



Uses (page 20):

White oak wood is hard, strong, splits easily, moderately rot-resistant (heart wood) and makes excellent coals. Modern uses include flooring, furniture, cabinets, tool handles (second only to hickory and ash), interior trim and firewood. Pores in the wood are naturally clogged with a chemical called tyloses, making the wood ideal for waterproof barrels for aging whiskey.

Indians and Pioneers made baskets from oak splits (thin strips split from straight green saplings). White oak acorns contain less bitter tannin than red oaks and were favored as food by Indians who made acorn flour and rendered acorn oil, which according to one early Virginia writer, "they keepe to supple their joynts."

America's first warship, U.S.S. Constitution had a gun deck, keel and 21-inch-thick hull of American white oak. The ship still sails today on its original white oak keel laid in 1797.

Comments:

Though intermediate in shade tolerance, white oak is the most shade tolerant oak. It can persist long in shade as a seedling until wind, fire or disease make an opening, then it grows into the opening quicker than surrounding trees to capture the canopy. A dominant tree in the mature and climax oak-hickory forest, it can survive 4 centuries.

Overcup Oak (*Quercus lyrata*), Beech Family, white oak group

Identification (page 4):

Leaves: simple, alternate, toothless, lobed, leaf lobes rounded and lacking bristles, leaf indentions often squarish in shape, fall color = maroon, amber, yellow

Flowers: small, inconspicuous

Fruit: distinctive round acorn with corky acorn cap surrounding most of the acorn body

Twigs: gray-brown with multiple end buds

Bark: light gray with thin scaly plates

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

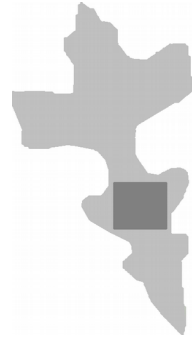
Wet to Moist soils, rivers and swamps

Fire Resistant, Moderate Sprouter

Intermediate and Mature forests

Bottomland Forest, Swamp Forest

Piedmont and Coastal Plain



Uses (page 20):

Overcup oak wood shares physical characteristics with white oak, but is of lower quality. It is used for similar purposes and also for wooden utility pallets.

Though a tree of bottomlands and swamps, overcup oak will grow on a variety of sites. It has become a fairly popular ornamental tree due to its ease of propagation, pleasant form and ability to survive even in hot dry parking lot islands.

Overcup acorns are eaten by deer, ducks, hogs, turkey & squirrels.

Comments:

Overcup acorns are adapted to float along a river after they drop, and germinate where the river and chance land them. Unlike other white oaks, whose acorns germinate immediately, overcup acorns wait until the following spring to germinate after winter flood waters recede, exposing the soil. The tree's habit of leafing out later than most trees also helps it survive late spring floods (it is dormant during flooding and so unaffected by inundation).

It is a dominant tree in the overcup oak-water hickory (*Carya aquatica*) forest found along lower Piedmont and Coastal Plain rivers where it may survive into the climax forest.

Swamp Chestnut Oak (*Quercus michauxii*), Beech Family,
white oak group

Identification (page 20):

Leaves: simple, alternate, toothed with large rounded teeth,
leaf lobes lack bristles, fall color = amber, red, yellow

Flowers: small, inconspicuous

Fruit: large acorn similar to chestnut oak

Twigs: moderately stout with multiple end buds

Bark: gray and flaky (similar to white oak, not like chestnut oak)

Ecology (page 12):

Shade Intolerant, Dominant Canopy tree

Moist sites

Fire Resistant, Moderate Sprouter

Pioneer, Intermediate and Mature forests

Bottomland F.

Piedmont and Coastal Plain, but not abundant



Uses (page 20):

Swamp chestnut oak shares modern uses with other white oaks.

The tree was once called basket oak, and Donald Peattie says, “The basket oak’s wood has the special property of splitting into fine but tough ribbons of fiber. The splitting occurs along the spring wood, Such a structure is ideal for making splints (thin wood strips split from larger pieces), and untold thousands of baskets were woven of this wood for toting cotton from the fields.” This author remembers when produce was sold in bushel and half-bushel baskets made of oak splits. One bushel is about 37 quarts.

Like all white oaks, swamp chestnut oak acorns contain less tannin than red oak acorns. This makes them relatively sweet and preferred by wildlife, and, once upon a time, Indians who boiled them in several changes of water, skimming off the oil, then ground them into flour used for bread and to thicken venison soup.

Comments:

All members of the white oak group produce acorns that mature in one year and germinate immediately after dropping in fall. Red oak acorns take two years to mature, laying on the ground all winter before germinating. Consequently they need higher levels of bitter tannin to protect them during this vulnerable time.

Chestnut Oak (*Quercus prinus*) Beech Family, white oak group

Identification (page 4):

Leaves: simple, alternate, toothed with large rounded teeth, leaf lobes lack bristles, fall color = amber, red, yellow

Flowers: small inconspicuous

Fruit: large one-inch-long acorn (largest of all Southern oaks)

Twigs: stout, red-brown with multiple end buds

Bark: gray, developing prominent flat-topped ridges separated by deep furrows

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree

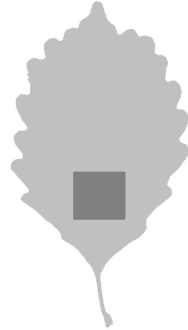
Dry slopes and ridge tops

Fire Tolerant, Vigorous Sprouter

Intermediate, Mature and Climax forests

Oak-Hickory F., Oak-Pine F.

Cumberland Plateau, Blue Ridge, upper Piedmont



Uses (page 20):

Chestnut oak wood is very hard, heavy, strong, rot-resistant, splits easily, has plugged pores (tyloses) and makes good coals. Modern uses include fence posts, railroad ties and firewood.

Indians readily ate these, the largest of the “sweet” acorns.

Pioneers considered this tree's bark the best white oak group source of tannin (for tanning leather) which was combined with hemlock bark to offset hemlock's red tannin color.

Many wild animals relish the large sweet acorns.

Comments:

Chestnut oak is also called rock chestnut oak and tanbark oak.

In Great Smoky Mountains National Park and the surrounding southern Appalachians, chestnut oak is a dominant tree in the Chestnut Oak-Hickory-Red Maple forest found growing on exposed slopes and ridge tops between 1500 and 3500 feet elevation. American chestnut was also once a major component of these forests before the chestnut blight wiped it out during the first half of the 1900's. Since then, chestnut oak and red maple have replaced the chestnut tree, though they can not replace the nutritious and abundant chestnuts that once fed the mountain farmer, his cattle and a host of wild animals.

Post Oak (*Quercus stellata*) Beech Family, white oak group

Identification (page 4):

Leaves: fairly thick, forming a rough cross shape, fall color =
amber, red, brown

Flowers: small, inconspicuous

Fruit: acorn 1/2 to 1-inch long

Twigs: stout, gray, often fuzzy with multiple end buds

Bark: gray and platy, but not flaky like white oak

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

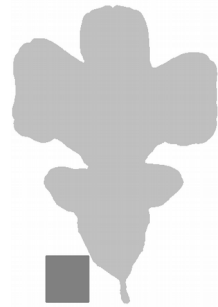
Dry nutrient-poor exposed slopes and ridge tops

Fire Resistant, Vigorous Sprouter

Intermediate, Mature and Climax forests

Oak-hickory F., Oak-Pine F.

All Southeastern provinces



Uses (page 20):

Post oak has physical characteristics and uses similar to white oak (page 108). The name implies its past use as fence posts. It also made good railroad ties and old-time split wooden roof shingles. Pioneers called this tree iron oak and ring oak. They considered the wood to be, "very durable, and esteemed the best oak for shipwork." Joshua Slocum, first man to sail single-handedly around the world from 1895 to 1898, built his ship, the *Spray* (a gaff-rigged oyster boat), with a pasture-grown post oak keel. The boat was planked with southern yellow pine planks.

Comments:

The Latin word *stellata* means star-shape and refers to the distinctive leaf shape of this tree. Some observers, however, liken the leaf shape to that of the Maltese cross.

Post oak grows very slowly. A forest-grown tree has more trunk than crown, and the crown is fairly compact. Forest-grown trees average about 2 inches diameter growth in 10 years. An open-grown post oak has a short trunk and a spreading crown with large branches, and may average 3 inches diameter growth in 10 years. Many old home sites are marked by huge post oaks measuring 30-40 inches diameter. A 40-inch diameter open-grown post oak would be around 130 years, while a 40-inch forest-grown tree could be 200 years old.

Live Oak (*Quercus virginiana*) Beech Family, white oak group

Identification (page 4):

Leaves: simple, alternate, toothless, unlobed, evergreen, leathery

Flowers: small, inconspicuous

Fruit: acorn is small and bullet-shaped, green when young & almost black when mature

Twigs: slender with multiple end buds

Bark: thick, dark gray to brown, furrowed with scaly ridges

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree

Dry sandy woods and salt marsh borders

Fire Tolerant, Vigorous sprouter

Intermediate, Mature and Climax forests

Bottomland F., Maritime F.

Coastal Plain



Uses (page 20):

Heavy, strong and tough, live oak was used for ship building in Colonial times where its curved form made good ship ribs. The U.S.S. Constitution, America's first sail-powered battleship, contained live oak cut from America's first public forest on St. Simons Island, reserved by Congress in 1789 specifically for use in ship building.

Coastal Plain Indians ate the acorns (considered the sweetest of all acorns) and made a decoction (bark boiled in water) for sore eyes. Live oak's evergreen habit, picturesque form and hardiness in the Piedmont province make it a popular ornamental there.

In 1776, while visiting Brampton, the plantation of Johnathan Bryan, on the Savannah River north of Savannah, naturalist William Bartram witnessed servants arriving after dark with, "horse loads of wild (passenger) pigeons," harvested using pine knot torches and long poles from live oak trees where they had been feeding on the tree's acorns in swamps where they wintered.

Comments:

When General James Oglethorpe arrived in Savannah in 1733, he met chief Tomacheechee in a great forest of longleaf pines, not the live oaks that now characterize old Savannah. The live oaks were planted later.

Scarlet Oak (*Quercus coccinea*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, toothless, with long narrow lobes and deep narrow indentions, lobes bristle-tipped, fall color = bright scarlet

Flowers: small, inconspicuous

Fruit: acorn, acorn tip surrounded by concentric rings, acorn cap reddish with fringes

Twigs: slender, hairless with multiple end buds

Bark: dark gray with narrow light gray stripes, lower trunk often has dead branches (poor self-pruner)

Ecology (page 12):

Shade Intolerant-Intermediate, Dominant Canopy

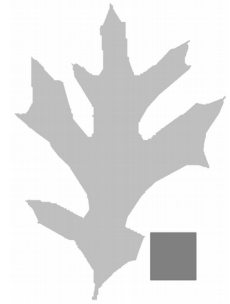
Dry sites like exposed slopes and ridge tops

Very Fire Intolerant, Very Prolific Sprouter

Intermediate and Mature forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F.

All Southeastern provinces except Coastal Plain



Uses (page 20):

The wood is reddish, strong, easily split, perishable in ground and has unplugged pores. It is of lower quality than northern red and black oak, but considered stiffer and stronger than that of white oaks. Modern uses are the same as the white oaks (page 108) with the exception of whiskey barrels, railroad ties and fence posts.

Scarlet oak leaves turn bright scarlet in autumn, making them easily visible on mountain slopes where they glorify the season.

Bright autumn colors and typical oak hardness make it a desirable ornamental tree.

Like other red oak group members, tannin-rich scarlet oak acorns are less preferred by wildlife than white oaks; however, red oaks are more consistent acorn producers than the white oaks, providing important wildlife food when the white oak crop is poor.

Comments:

In the southern Appalachian mountains scarlet oak is a dry site oak most abundant on low (1500' to 2500 ') and middle (2500' to 4500') elevation exposed slopes and ridge tops where it is usually accompanied by pines that it will outlive as succession proceeds.

Southern Red Oak (*Quercus falcata*), Beech Family, red oak group)

Identification (page 4):

Leaves: simple, alternate, toothless, bristle-tipped lobes, leaf lobes variable in shape but generally with a bell-shape (see silhouette), dead leaves often remain on tree into winter, fall color = amber, red, brown

Flowers: small, inconspicuous

Fruit: small round acorn

Twigs: gray, fuzzy with multiple end buds

Bark: black with platy ridges

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree

Dry to Moist sites

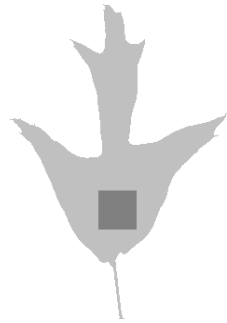
Fire Intolerant, Prolific Sprouter

Intermediate and Mature forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F.,

Maritime F., longleaf pine F.

All Southeastern provinces



Uses (page 20):

The wood is considered slightly inferior to northern red oak (page 122), but otherwise shares red oak characteristics. Modern uses include flooring, furniture, pallets, rough lumber and firewood. Cherokee Indians made a decoction (material boiled in water) of the inner bark of 5 trees including southern red oak to treat hoarseness.

John Lawson wrote in 1709, "Spanish oak rives [splits] very well into clap-boards. It is accounted durable, therefore some use to build vessels with it for the sea." Clapboards were short thin boards split by hand and used for furniture, siding and ship planks.

Comments:

In Virginia and the North this tree is called Spanish oak. The name originated in print with William Penn when he wrote of it in 1683. The origin of this name is obscure, but it may refer to the Spanish dagger-like appearance of the leaves. The Latin name *falcata* refers to an ancient sword in use during Roman times on the Iberian Peninsula.

Bluejack Oak (*Quercus incana*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, toothless, single lobed, leaf deciduous and leathery with a bristle tip, leaf bluish-gray, fall color = amber, brown

Flowers: small, inconspicuous

Fruit: acorn, 1/2-inch long

Twigs: slender, dark and fuzzy with multiple end buds

Bark: gray to black, breaking into square or rectangular plates

Ecology (page 12):

Shade Intermediate, Subcanopy or Understory tree

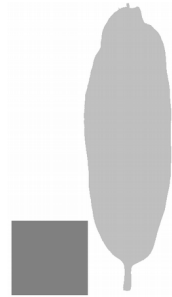
Dry sandy sites

Fire Dependent, Vigorous Sprouter

Intermediate and Mature forests

Pine-Oak F., Evergreen Oak F., Longleaf Pine F.

Coastal Plain



Uses (page 20):

Bluejack oak grows in thickets where it stabilizes sandy soil. Its acorns are eaten by deer, racoons, squirrel and rodents, and its thickets provide wildlife cover. The acorns are reported to be highly preferred by fox squirrels including the rare Sherman's fox squirrel who is dependant on the longleaf pine-scrub oak forest where bluejack oak grows. Deer, turkey and quail also eat the acorns.

Tender young bluejack oak leaves have been reportedly fatal to cattle who browsed them heavily.

Comments:

Bluejack oak is often found growing in thickets beneath longleaf pines in the longleaf pine-wiregrass forest community. Like other members of this community it is highly tolerant of fire.

The National champion bluejack oak grows (2014) in Texas. It is 51 feet tall and 2.3 feet in diameter.

Down along the Flint River in coastal plain Georgia where bluejack grows, you can also find a few cork oaks (*Q. suber*). No one knows how these oaks of Spain got there, but Hernando deSoto brought a huge herd of pigs with him in 1541 when he traveled through the region, and also cork oak acorns to feed them.

Turkey Oak (*Quercus laevis*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, toothless, 3 bristle-tipped lobes, leaves shaped like a 3-toed turkey foot, leaf stems are yellow and long and twist the leaf away from the sun when it gets too hot, fall color = amber, red, yellow

Flowers: inconspicuous

Fruit: acorn about 1 inch long

Twigs: stout with multiple end buds

Bark: black with deep furrows

Ecology (page 12):

Shade Intermediate, Subcanopy tree

Dry sandy sites

Fire Tolerant, Prolific Sprouter

Pioneer and Intermediate forests

Pine-Oak F., Evergreen Oak F.,

Longleaf Pine-Wiregrass F.

Coastal Plain



Uses (page 20):

This small tree has few commercial uses, but it is used for firewood.

The acorns are eaten by deer, turkey and other animals.

Comments:

Turkey oak grows in the Longleaf Pine-Wiregrass forest on high dry sand hills. In the past, fires, both natural and man-set, burned through this forest about every 2 years, killing hardwood seedlings that would normally "take over" the forest and exclude the pines. Longleaf pine, wiregrass (*Aristida beyrichiana*) and turkey oak are all well adapted to frequent fires, and form a pine-based climax forest (ecologists call it a fire climax forest) where fire holds normal forest succession in the pine (Pioneer to Intermediate Forest) stage.

Turkey oak and southern red oak grow together in the Pine-Oak forest, and may look similar, but turkey oak leaves almost always have just 3 lobes while southern red oak trees usually have both 3-lobed and 5-lobed leaves. Turkey oak is a small subcanopy tree, while southern red oak becomes a tall canopy tree.

Laurel Oak (*Quercus laurifolia*), Beech Family, red oak group.

Identification (page 4):

Leaves: simple, alternate, bristle-tipped, leaves deciduous but stay on tree through winter, leaves often diamond-shaped near end

Flowers: small, inconspicuous

Fruit: nearly round acorn, 3/4-inch in diameter

Twigs: slender, grayish with multiple end buds

Bark: dark gray with shallow ridges and furrows

Ecology (page 12):

Shade Intermediate to Tolerant, Dominant Canopy tree

Moist to wet sites

Very Fire Intolerant, Moderate Sprouter

Pioneer and Intermediate and Mature forests

Bottomland F., Maritime F., Cypress-Gum F.

Coastal Plain and lower Piedmont



Uses (page 20):

Laurel oak wood shares physical properties with other members of the red oak group, and is used similarly. It makes good firewood and is used for that purpose.

Laurel oak grows rapidly and has been planted along Coastal Plain streets, because of its attractive laurel-like leaves, but it is relatively short-lived and may deteriorate before it reaches 70 years.

Comments:

***Quercus hemisphaerica*, Darlington oak** (also called sand laurel oak) is very similar to laurel oak and, in fact, is often called laurel oak. Also a Coastal Plain tree with persistent leaves, it grows on dry upland sites instead of moist or wet sites. According to Dr. Claud Brown, Co-author of Trees of Georgia and Adjacent States (1990), Darlington oak leaves do not have the diamond shape often seen in Laurel oak leaves, but otherwise the trees look identical. Some authors do not recognize the two trees as separate species. Some researchers believe laurel oak is a natural hybrid between water oak and willow oak, and one is tempted to believe this oak evolved 1 "version" (sand laurel oak) adapted to sandy uplands and another (laurel oak) adapted to richer bottomlands.

Blackjack Oak (*Quercus marilandica*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, 3-lobed or unlobed triangular, lobes bristle tipped, leaves thick and leathery, leaves dark green shiny above, hairy below, fall color = amber, brown

Flowers: inconspicuous

Fruit: acorn about 3/4-inch long and fuzzy

Twigs: stout, dark gray with multiple fuzzy end buds

Bark: black and breaking into square plates

Ecology (page 12):

Shade Intolerant to Intermediate

Pioneer or understory tree (see Comments)

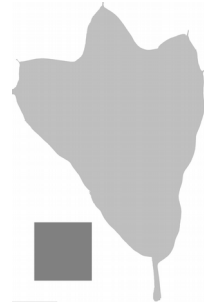
very dry rocky nutrient-poor sites

Very Fire Tolerant, Prolific Sprouter

Pioneer forest

Pine-Oak F., Oak-Pine F.

All Southeastern provinces



Uses (page 20):

Blackjack oak is used for firewood. Dr. Claud Brown says the tree's hardiness, compact growth habit and shiny green leaves might make it a good ornamental tree, but it grows slowly and is short-lived.

Comments:

Blackjack oak is not abundant in its range, but its range is large due to its ability to tolerate the worst sites in the landscape where other trees dare not tread. It often comes in as a pioneer after a fire on one of these forlorn sites, forming pure park-like stands.

Blackjack oak is one of the few trees that can tolerate the high magnesium soils of serpentine barrens. The barrens are formed where magnesium-rich ancient ocean crust was shoved up on land by plate tectonic processes where 2 continents are sutured together. Serpentine, an olive-green mineral high in magnesium is abundant here and weathers into dry nutrient-poor, magnesium-rich soils that are poisonous to most plants. Longleaf pine, shortleaf pine, pitch pine, post oak and little bluestem grass (*Schizachyrium scoparium*) are often associated with blackjack oak in serpentine barrens.

Water Oak (*Quercus nigra*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, unlobed or slightly 3-lobed, with bristle tips, leaf shape can be variable on the same tree, but mostly leaves are spatula-shaped (wider at the end than the base), fall color = yellow, amber

Flowers: inconspicuous

Fruit: acorn, rounded about 1/2-inch diameter, often with orange stripes

Twigs: slender, brushy with multiple end buds

Bark: smooth and gray, becoming slightly furrowed on older trees, trunks often have many dead limbs below the crown

Ecology (page 12):

Shade Intolerant to Intermediate, Dominant Canopy

Moist to Dry sites

Fire Intolerant, Vigorous Sprouter

Pioneer and Intermediate forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F., Bottomland

Maritime F.

All Southeastern provinces, but uncommon above 1500' elevation



Uses (page 20):

Water oak shares characteristics with other red oaks (see *Quercus rubra*, page 122), but its wood is inferior to them. Its relatively fast growth, vigorous sprouting tendency and high fuel value make it a good choice for coppice forestry where tree sprouts are cut for fuel and new sprouts quickly arise.

The tree has been planted along streets and on lawns in the South because it is easily grown, hardy and grows fast, but it drops numerous dead branches, and its small leaves and acorns are difficult to remove from lawns. It is also relatively short-lived, usually not exceeding 80 years.

Water oak is a prolific acorn producer, providing a dependable food source for many birds and animals, who survive on them when the sweeter white oak acorn crop fails.

Comments:

Water oak grows best on bottomland sites, but it is truly a cosmopolitan tree, occupying moist and dry sites too.

Willow Oak (*Quercus phellos*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate toothless, unlobed, bristle tipped, leaves very long and narrow (willow-like), leaf shape does not vary like in water oak, fall color = amber, yellow

Flowers: inconspicuous

Fruit: acorn, round 1/2-inch in diameter

Twigs: multiple end buds

Bark: gray and smooth when young, slightly furrowed when older

Ecology (page 12):

Shade Intolerant to Intermediate, Dispersed Canopy tree

Moist sites

Fire Intolerant, Vigorous Sprouter

Pioneer and Intermediate forests

Bottomland F.

Piedmont and CP (excluding Ga. and Fla. CP)

Uses (page 20):

Willow oak wood shares characteristics with other red oaks and is used for similar purposes (see *Quercus rubra*, page 122).

Willow oak is a popular landscape tree due to its ease of propagation, hardiness on a variety of sites, nice form and fast growth.

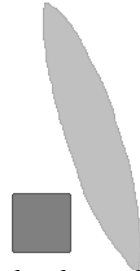
Like water oak, willow oak's acorns are eaten by a wide variety of wildlife, providing dependable food when white oaks fail.

Comments:

Willow oak is nowhere as abundant as water oak. It sticks to river bottoms and lower slopes where it grows to more than 100 feet tall and 3 feet in diameter with a clear straight trunk and lovely form.

This tree can be confused with water oak, but the long narrow unlobed leaf shape of willow oak never varies. If you see variable leaf shapes and/or spatula-shaped leaves; it's water oak.

All oaks are subject to oak galls, fruit-like tissue growths on the leaf or bark produced when a gall wasp injects its eggs there along with a chemical that causes the gall (the wasp larva's food) to develop. The oak apple gall looks like a ping pong ball-size green apple that later turns a tan color, but some galls look like pea-size red berries that accumulate on the ground below oaks in autumn.



Northern Red Oak (*Quercus rubra*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, toothless, lobed (7-11 bristle-tipped lobes), leaves wide with shallow lobe indentions, leaves hairless above and below, fall color = red, amber, yellow

Flowers: inconspicuous

Fruit: acorn, big and barrel-shaped nearly as wide as long

Twigs: stout, hairless with multiple end buds

Bark: dark gray with wide light gray vertical stripes ("ski trails")

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree

Moist well-drained sites,

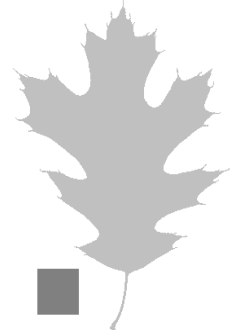
often on north-facing slopes

Fire Tolerant, Vigorous Sprouter

Intermediate, Mature and Climax forests

Oak-Pine F., Oak-Hickory F., Cove F.

All Southeastern provinces except Coastal Plain



Uses (page 20):

Northern red oak wood is reddish in color, strong, hard, heavy, easily split, lacks rot resistance and has open pores. The most valuable member of the red oak group, its uses include flooring, paneling, furniture, cabinets, trim and heavy construction lumber. Unlike white oak, northern red oak does not make good whiskey barrels or basket splits.

A more dependable acorn producer than white oaks, northern red oak with the largest of all red oak group acorns is especially valuable to wildlife when the tastier white oak crop is poor.

Comments:

Northern red oak is less shade tolerant than white oak, but more shade tolerant than all other red oak group members. It grows at higher elevations in the southern Appalachians than any other oak, where its leaves have adapted to the colder climate at higher elevations by becoming longer and narrower than those of its lowland counterpart. Early Appalachian forest researcher Robert H. Whittaker called this mountain red oak version "*Quercus borealis*".

Black Oak (*Quercus velutina*), Beech Family, red oak group

Identification (page 4):

Leaves: simple, alternate, toothless, 5-7 bristle-tipped lobes,
leaves leathery and fuzzy below along veins, lower canopy
leaves wide with shallow lobe indentions, upper canopy
leaves more narrow with deeper indentions
fall color = amber, brown

Flowers: inconspicuous

Fruit: acorn 1/2-3/4 inches long

Twigs: stout with distinctly angled fuzzy end buds, scratch the twig to see the yellow inner bark

Bark: black with platy ridges, inner bark yellow-orange

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

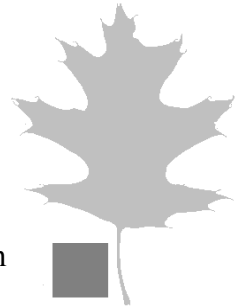
Dry sites like upper slopes and ridge tops

Fire Intolerant, Prolific Sprouter

Intermediate & Mature forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F.

All SE provinces except Ga. & Fla. Coastal Plain



Uses (page 20):

Black oak wood is slightly inferior to that of northern red oak, but it is used for the same purposes (see *Quercus rubra*, page 122).

Cherokee Indians used black oak leaves to wrap cornbread dough before boiling or baking. They made a bark tea to relieve asthma.

Pioneers made a yellow dye for clothing called quercitron from the inner bark by pounding dried bark to powder, then sifting it so the fine yellow dye particles passed through the sieve.

Pioneers also used the bark as a source of tannin, but the yellow inner bark had to be removed unless yellow leather was desired.

Comments:

In winter it is difficult to distinguish black oak from southern red oak since they have the same black ridged bark. Southern red oak often retains its leaves well into winter, and it lacks the distinctive yellow inner bark of black oak. Also, southern red oak buds lack the distinct angles of black oak buds.

It is said oaks are struck by lightning oftener than other trees, and they were associated with Thor the thunder god in ancient times.

Pines And Other Conifers

Pines with 5 needles per bundle:

only white pine page 125

Pines with 2 and 3 needles per bundle:

slash pine has mostly 2 needles per bundle, and
occasionally 3 needles per bundle page 126

shortleaf pine has 2, rarely 3 needles per bundle
and is included here with the 2-needle pines

Pines with 3 needles per bundle:

“Lob a Long Pitch into the Pond.”

Loblolly Pine	page 127
Longleaf Pine	page 128
Pitch Pine	page 129
Pond Pine	page 130

Pines with 2 needles per bundle:

“Virginia will Spruce up the Table Shortly.”

Virginia Pine	page 131
Spruce Pine	page 132
Table-mountain Pine	page 133
Shortleaf Pine	page 134

Eastern White Pine (*Pinus strobus*), Pine Family

Identification (page 4):

Leaves: leaves evergreen with **5 needles per bundle**, foliage gray-green

Flowers: pollen-filled male catkins in spring

Fruit: pine cone with thick often sugary resin-coated scales lacking prickles

Twigs: branches are arranged in distinctive whorls on the trunk

Bark: smooth and gray on young trees, becoming scaly

Ecology (page 12):

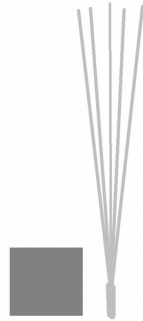
Shade Intermediate, Dominant Canopy tree

Moist to Dry sites

Fire Intolerant, No Sprouts

Pioneer, Intermediate, Mature and Climax forests

Blue Ridge, Cumberland Plateau, upper Piedmont



Uses (page 20):

White pine wood is soft, light, close grained and easily worked.

Modern uses include cabinets, paneling, furniture and toys.

Horace Kephart says the wood will scarcely burn when green due to watery sap, so it is good for campfire back logs and andirons.

Indians soaked white pine bark in water and applied it to wounds.

They also used the inner bark as a remedy for coughs and diarrhea.

During Colonial times England reserved the best American white pines for ship masts by marking them with the sign of the Kings broad arrow.

Many early American sailing ships were decorated with figureheads carved from "Pumpkin pine", very close-grained virgin white pine wood renowned for its ease of carving, and now no longer available.

Later, unscrupulous American timber barons made millions by cutting virgin New England white pines and floating them downriver to markets, repeating this process in the Lake States, then turning to the southern Appalachians where they were finally stopped by conservationists in the early 1900's.

Comments:

White pine is a long-lived pioneer tree. It colonizes open land in the S. Appalachians and then remains on into the Climax forest.

Slash Pine (*Pinus elliottii*), Pine Family

Identification (page 4):

Leaves: leaves evergreen with **2 and 3 needles per bundle**,
needles longer than loblolly but shorter than longleaf pine

Flowers: pollen-filled catkins in spring

Fruit: pine cone with shiny reddish scales and small dull prickles

Twigs: not distinctive

Bark: typical pine bark with reddish plates

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

Moist to Wet, poorly drained sites

Fire Tolerant, No Sprouts

Pioneer and Intermediate forests

Coastal Plain from South Carolina to Mississippi



Uses (page 20):

Slash pine is considered a "yellow pine" species since its wood is yellowish in contrast to the cream-colored wood of white pine.

The wood is stronger and harder than white pine, but also inferior to it as a cabinet and furniture wood. Modern uses include pulp for paper and construction lumber. Yellow pines adapt well to pressure treating with chemical preservatives, lasting for many years in ground & weather-contact projects like decks and fences.

In the days of sailing ships, slash pine trees were tapped ("slashed") for their resin which was distilled into turpentine, pitch and tar used to waterproof sailing ship hulls and decks, generating the term "naval stores" to describe this industry. Workers in the slash pine plantations were called "tar heels" because their bare feet invariably became resin-coated. Special knives were used to slash the trees, and the resulting resin was collected in clay pots. Old slash marks are still visible on some large forest trees.

Ship hulls were waterproofed by hammering tar-saturated hemp ropes between hull planks, a process called "caulking."

Comments:

Due to its fast growth slash pine was once planted in the Piedmont, but its long needles proved no match for Piedmont ice storms that coated them in heavy ice, resulting in stem breakage and disease. Now, loblolly pine is the plantation tree of Southern forestry.

Loblolly Pine (*Pinus taeda*), Pine Family

Identification (page 4):

Leaves: evergreen with **3 needles per bundle**, needles about 7 inches long

Flowers: pollen-filled catkins in spring

Fruit: cone, about hand-size with very sharp prickles

Twigs: not distinctive

Bark: typical pine bark, usually gray, sometimes reddish

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

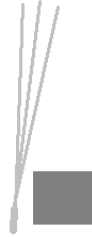
Wet to dry sites

Fire Tolerant, No Sprouts

Pioneer and Intermediate forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F., Bottomland F.

All Southeastern provinces, but absent above 1500 ' elevation



Uses (page 20):

Loblolly pine is considered a "yellow pine" species and its wood is used similarly to other yellow pines (see slash pine, page 126). It is not a prolific resin producer like slash and longleaf pines.

Modern uses include pulp for paper, plywood, telephone poles, pressure treated lumber and general construction lumber.

Loblolly's fast growth rate, large range and genetic adaptability have made it the premier pine of Southeastern forestry where genetically superior trees are planted in row on pine plantations.

When millions of worn out and gullied acres of Southern farmland were abandoned in the early 1900's, loblolly pine (also called old field pine) rapidly seeded in, stabilizing the soil and creating extensive early succession forests that gave rise to the South's forest industry, one of its most important and profitable industries today.

Comments:

The name loblolly supposedly comes from the days of sailing ships when sick sailor's rations included a soupy gruel they called "loblolly". John Lawson who traveled in the Carolinas in 1700 was served, "loblolly made of Indian corn and dried peaches," by the Congaree Indians. Later, the term was applied to low swampy spots in poorly drained Southern lowlands where loblolly pine thrives, and eventually to the tree so often found there.

Longleaf Pine (*Pinus palustris*), Pine Family

Identification (page 4):

Leaves: evergreen with **3 needles per bundle**, needles are very long (10-18 inches long)

Flowers: pollen-filled catkins in spring

Fruit: big cone (6-10 inches long), largest cone in the East

Twigs: twig end buds are long and white like "candles"

Bark: typical pine bark, usually gray

Ecology (page 12):

Very Shade Intolerant Pioneer, Dominant Canopy tree

Dry sandy hill top sites

Very Fire Tolerant, No Sprouts

Pioneer, Intermediate and fire climax forests (see below)

Coastal Plain and along large rivers in Ga. Valley & Ridge



Uses (page 20):

See slash pine (page 126) for general uses of yellow pine.

Longleaf pine wood is stronger and heavier than that of all other Southern yellow pines due to its slow growth, resulting in close growth rings and rot-resistant heartwood. It makes excellent power poles, especially if the sapwood is pressure treated.

Comments:

Longleaf pine differs from other Southern yellow pines in its unique adaptation to the very dry sandy ridges where it grows. After germination, the tree looks like a tuft of bunch grass. Called the grass stage, this 3 to 7 year period is spent sinking a long tap root down to moisture stored deep within the sandy soil.

A moderate fire during the grass stage actually helps the tree by killing dangerous fungi. Once established, the tree will break the grass stage and grow quickly as a seedling, slowing down with age. A few years after breaking the grass stage the slender tree with its grass-like tuft looks very much like a green mop!

Once covering thousands of Coastal Plain sandy ridges, the Longleaf pine-Wiregrass community consisted of large well-spaced longleaf pines with wiregrass (*Aristida beyrichiana*) beneath. Frequent fires held forest succession at bay by killing hardwood seedlings, creating a fire-induced pine climax forest. This forest had a park-like but rather monotonous appearance.

Pitch Pine(*Pinus rigida*), Pine Family

Identification (page 4):

Leaves: evergreen with **3 needles per bundle**, needles 3-5 inches long, needles stiff and twisted

Flowers: pollen-filled catkins in spring

Fruit: egg-shaped cone often attached in clusters, cones often attach directly to the tree (no stalks)

Twigs: not distinctive

Bark: typical pine bark

Ecology (page 12):

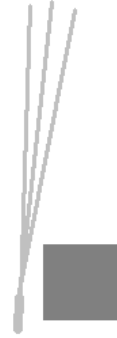
Shade Intolerant, Dispersed Canopy tree

Dry upper slope and ridge tops

Fire Tolerant, Moderate Sprouter

Pioneer and Intermediate forests

Blue Ridge, Ridge and Valley, Cumberland Plateau



Uses (page 20):

See slash pine (page 126) for general yellow pine characteristics and uses.

Indians made a vitamin rich tea from the needles of all yellow pine tree species and ate the inner bark and the pollen-filled male catkins in a food emergency to stave off hunger.

Pioneers attached resin-rich pitch pine knots to poles for torches used in hunting deer at night. They also tapped the trees for pitch glue and made charcoal from the wood for use in primitive iron furnaces.

Comments:

Pitch pine is a mountain species found growing on exposed slopes and ridge tops between 2500 feet and 4500 feet in the southern Appalachian mountains. Here, it mixes with shortleaf pine but is easily distinguished by its 3 needles per bundle as opposed to shortleaf's 2 needles per bundle. Pitch and loblolly (both 3-needle pines) are distinguished by the egg-shaped cones of pitch and the very sharp prickles of the long hand-size loblolly cones.

Pitch pine is the main tree of the New Jersey pine barrens, a dry sandy habitat once maintained as a pitch pine climax forest by fire.

Pitch is one of the few pines that produces sprouts when cut, sharing this trait with pond, Virginia and shortleaf pines.

Pond Pine (*Pinus serotina*), Pine Family

Identification (page 4):

Leaves: evergreen with **3 needles per bundle (rarely 4)**, needles often emerge directly from the tree trunk

Flowers: pollen-filled catkins in spring

Fruit: egg-shaped cone, cones are serotinous (remain unopened for years until a hot fire causes them to open and release their winged seeds)

Twigs: not distinctive

Bark: typical pine bark, usually gray

Ecology (page 12):

Shade Intolerant, Dispersed Canopy tree

wet, poorly drained bottoms

Fire Dependant (serotinous), Moderate Sprouter

Bottomland F., Maritime F., Cypress-Gum F.

Coastal Plain



Uses (page 20):

See slash pine (page 126) for general yellow pine characteristics and uses. Pond pine is mostly used for pulpwood.

The Hercules Co. in Brunswick, Georgia harvests southern yellow pine stumps (several pine species) after the tree tops have been cut.

They extract pine resins from the stumps used in products like Pine-Sol, the largest-selling household cleaner in the world.

Pioneers inhaled pine tar (several pine species) to relieve pulmonary (breathing) distress. They used "oil of turpentine" to control intestinal worms, increase urine flow, as a stimulant and as a laxative!

Comments:

Pond pine grows in pocosins (Indian for swamp-on-a-hill). While mostly wet, these sites burn readily during droughts, killing the pond pine trunks (which readily resprout) and releasing millions of seeds from the fire-dependant cones to reestablish the pine stand. Some workers consider pond a variety of pitch pine which it resembles in number of needles per bundle and cone shape. Pond and pitch pine can, in fact, hybridize where their ranges overlap. Pond differs from pitch in producing both 3 and 4 needles per bundle, and in growing on poorly drained sites like the pocosins.

Virginia Pine (*Pinus virginiana*), Pine Family

Identification (page 4):

Leaves: evergreen with **2 needles per bundle**, needles are quite short (2 inches)

Flowers: pollen-filled catkins in spring

Fruit: cone short (2 inches long), cone scales have purplish lip on the inside, prickles are sharp

Twigs: not distinctive

Bark: distinctive red scaly bark with many dead limbs on trunk

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

Dry exposed slopes and ridge tops

Fire Resistant-Intolerant, Sprouts Rarely

Pioneer and Intermediate forests

Blue Ridge, Valley and Ridge, Cumberland Plateau, upper Piedmont



Uses (page 20):

See slash pine (page 126) for general characteristics and uses of Southern yellow pines.

Virginia pine is widely grown in the Southeast for Christmas trees because of its dense branching habit, rapid growth and site adaptability. It is often planted as a buffer tree for the same reasons and because seedlings are easily and cheaply obtained from state Forestry Commission nurseries.

Cherokee Indians used a tea of Virginia pine roots for sore throat. They boiled the root, skimmed off the turpentine and used it in a plaster to "draw out sickness". They boiled buds and needles for a tea to "check bowels." They burned Virginia and other pine branches in cooking vessels, spreading the ashes over a newly kindled hearth fire to cleanse a lodge after a death occurred.

Archaeologist Scot Jones found that an excellent pitch glue used in primitive crafts is made by heating a mixture of 80 parts pine resin, 20 parts sweetgum resin and a little charcoal for temper.

Comments:

Primarily a low elevation mountain tree, Virginia pine reaches no higher on mountain slopes than around 2500 feet elevation, where it grows most often on exposed slopes and ridge tops, and is easily recognized by its flaky reddish bark and many dead branches.

Spruce Pine (*Pinus glabra*), Pine Family

Identification (page 4):

Leaves: evergreen with **2 needles per bundle**, needles are slender and twisted

Flowers: pollen-filled catkins in spring

Fruit: cone short (2 inches long) with dull prickles and no purple lip

Twigs: not distinctive

Bark: dark and scaly resembling that of a spruce tree, not a pine

Ecology (page 12):

Very Shade Tolerant, Dispersed Canopy tree

Moist to Wet sites

Fire Intolerant, No Sprouts

Pioneer, Intermediate, Mature and Climax forests

Bottomland F., Maritime F.

Coastal Plain



Uses (page 20):

See slash pine (page 126) for general yellow pine characteristics and uses. Spruce pine is mostly used for pulpwood.

Indians made a vitamin rich tea from the needles of all yellow pine tree species and ate the inner bark and the pollen-filled male catkins in a food emergency to stave off hunger.

Comments:

Spruce pine is not abundant within its range, being far outnumbered by pines like slash pine and loblolly pine that grow in similar habitats. This tree is unique among pines in being very shade tolerant like the spruce trees from which it gets its name.

Sand Pine (*Pinus clausa*), a small pine, grows in habitats similar to those of spruce pine, but it is restricted to Florida. It grows on nutrient-poor sandy soils and is the most common tree in Florida's Ocala National Forest, growing on over 200,000 acres of scrub forest called the "Big Scrub". It seeds in abundantly after fire and is managed for pulpwood production using prescribed burning as a management tool. Though sand pine has relatively thin bark that is susceptible to fire damage, its cones are actually fire dependent (serotinous), opening only after a hot fire has melted the sticky resin that glues the seed-bearing cones shut.

Table-mountain Pine (*Pinus pungens*), Pine Family

Identification (page 4):

Leaves: evergreen with **2 needles per bundle**, needles are stout stiff and twisted

Flowers: pollen-filled catkins in spring

Fruit: egg-shaped cone, thick, heavy and lopsided at the base,
prickles are very stout and recurved, cones are fire-
dependent, opening only after fire (serotinous)

Twigs: not distinctive

Bark: typical pine bark, grayish

Ecology (page 12):

Shade Intolerant, Dominant Canopy tree

Dry nutrient-poor ridge tops around 3000 '

Fire Dependent (serotinous), No Sprouts

Pioneer forest, Fire Climax forest

Pine-Oak F., Table-mountain pine fire climax F.

Blue Ridge



Uses (page 20):

See slash pine (page 126) for general yellow pine characteristics and uses. Table-mountain pine is so rare it has not been widely used for commercial purposes.

Cherokee Indians made a decoction (material boiled in water) of table-mountain pine cones, jewelweed stems, *Veronica officinalis* root and slippery elm bark they gave to pregnant women to impart health and long life to the unborn child.

Comments:

Pre-European forests were burned by natural fires or Indian-set fires on average about every 2 years. Thus, 18th-century Southern naturalist William Bartram found extensive open savannahs, park-like forests, pure pine stands and large cane brakes in Georgia and adjacent areas. When Smokey-The-Bear arrived in the 20th-century, forest fires were summarily suppressed eliminating fire-dependent trees like table-mountain pine from the forest.

Fortunately, prescribed burning is now practiced by natural resource managers and table-mountain pines are returning to our National Forests, enhancing forest diversity on the dry 3000-foot elevation mountain ridges where they formerly grew.

Shortleaf Pine (*Pinus echinada*), Pine Family

Identification (page 4):

Leaves: evergreen with **2 needles (rarely 3) per bundle**, needles are short (2 inches long)

Flowers: pollen-filled catkins in spring

Fruit: cone 2 inches long, with dull prickles and no purple lip

Twigs: not distinctive

Bark: often is pocked with small pencil lead-size holes called pitch pockets

Ecology (page 12):

Shade Intolerant Pioneer, Dominant Canopy tree

Dry exposed slopes and ridge tops

Fire Tolerant, Sprouts readily after fire

Pioneer and Intermediate forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F.

All Southern provinces except lower Ga. and Florida



Uses (page 20):

See slash pine (page 126) for general yellow pine characteristics and uses. Shortleaf pine is a commercially valuable Southern pine, and is harvested for pulp, plywood and lumber.

Horace Kephart says pine wood will barely burn unless split. Split pine burns well and actually produces more heat energy per pound than oak due to its high resin content.

The resin-rich heartwood of shortleaf and other yellow pines remains years after the outer trunk of a dead tree has rotted away.

This "fat lighter" or "lighter wood" will burn readily wet or dry, making excellent kindling, though it burns with a sooty flame.

Cherokee Indians made a tea from shortleaf needles for coughs and "hang on."

Comments:

Shortleaf pine is usually the oldest tree in a mature oak-hickory forest. It was likely one of the pioneer trees on the site, having seeded there in great numbers with other pines like loblolly. It became a dominant tree in the pioneer pine forest. As the hardwoods slowly replaced the loblolly pines, the shortleaf lived on into the century-old Intermediate forest. When this old monarch dies, as it must, count its rings to learn the forest's age.

Eastern Redcedar (*Juniperus virginiana*), Cypress Family
For Atlantic White-cedar (*Chamaecyparis thyoides*), see page 135

Identification (page 4):

Leaves: evergreen, tiny, awl-shaped

Flowers: inconspicuous

Fruit: small dark blue berry sometimes with a waxy coating

Twigs: slender and covered by the awl-shaped leaves,

Bark: reddish brown and fibrous, forming narrow strips
heartwood is dark rose-red and aromatic

Ecology Page 12):

Shade Intolerant, Dispersed Canopy or Subcanopy tree

Moist to Dry sites, prefers high calcium soils

Fire Intolerant, No Sprouts

Pioneer and Intermediate forests

Pine-Oak F., Oak-Pine F., Oak-Hickory F., Calcareous F.

All SE provinces except lower Coastal Plain



Uses (page 20):

Redcedar splits easily, makes a popping fire, is easily worked and has a dark rose-red heartwood that is extremely durable in the soil.

Indians made good rope from the inner bark. They drank a decoction (material boiled in water) to relieve coughs, and a poultice of boiled leaves for rheumatism.

Pioneers used the wood as fence posts. They split it into boards for cabin floors and roof shingles. They made cedar chests with moth-repellant aromatic properties. Ladies brushed their flatiron on a waxy redcedar bough to keep it from sticking to clothing. Redcedar has been the traditional Southern Christmas tree.

Redcedar was once the most important wood for pencils in the U.S. The Faber pencil company used it exclusively. Most of our virgin redcedar was cut for this purpose by about 1910.

The berries are on the tree year-round and are eaten by at least 50 kinds of birds including cedar waxwings. Birds are often responsible for "planting" redcedars along fences and field edges.

Comments:

Redcedar is the alternate host for cedar-apple rust, a fungus appearing as orange gelatinous masses on cedar trees in spring and later as a damaging powdery rust on apple trees.

Bald Cypress (*Taxodium distichum*), Taxodium Family

Identification (page 4):

Leaves: 1/2-inch long flat needles in opposite rows along a branchlet, but branchlets appear alternate on the twig,
needles are deciduous

Flowers: inconspicuous

Fruit: rounded leathery cone about 1 inch in diameter

Twigs: slender and reddish brown

Bark: gray or reddish, peeling into fibrous strips, the base of the trunk is usually greatly swollen and fluted with surrounding cypress "knees"

Ecology (page 12):

Shade Intermediate, Dominant Canopy tree
swamps, also moist well drained sites

Moderate Sprouter

Pioneer, Intermediate, Mature and Climax forests

Bottomland F., Cypress-Gum Swamp F.

Coastal Plain and lower Piedmont



Uses (page 20):

Bald cypress is very resistant to decay. The wood often has a greasy feel and a mildly rancid odor due to the presence of the oil that imparts decay resistance. Modern uses include docks, greenhouses, boats, railroad ties, wood shingles and exterior siding.

Indians made dugout canoes, bowls and spoons from cypress trunks. They used the heartwood for house posts and the bark for roof shingles. According to John Lawson, "...nuts which these trees bear yield a most odoriferous balsam that infallibly cures all new and green wounds."

Though a Southern swamp tree, bald cypress grows on well-drained upland sites as far north as southern Canada and is planted as an ornamental for its unique appearance and reddish brown autumn needle color.

Comments:

Cypress trees grow woody "knees" that extend vertically from the roots. Of unknown function, they are often carved into wooden novelties and sold to tourists. Monarch of the Okefenokee Swamp, bald cypress trees there can live to be 1000 years old.

Pond Cypress (*Taxodium ascendens*), Taxodium Family

Identification (page 4):

Leaves: needles very feathery in appearance, not like bald cypress, needles are deciduous

Flowers: inconspicuous

Fruit: rounded leathery cone about 1 inch in diameter

Twigs: slender, reddish brown

Bark: gray or reddish, peeling into fibrous strips, the base of the trunk is usually swollen, fluted & surrounded by "knees"

Ecology (page 12):

Shade Intermediate, Dispersed Canopy tree

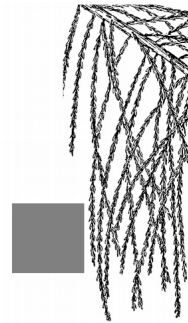
Small ponds and along streams

Moderate Sprouter

Pioneer, Intermediate, Mature and Climax forests

Bottomland F., Cypress-Gum Swamp F.

Coastal Plain



Uses (page 20):

Though its wood is generally inferior to that of bald cypress, pond cypress shares many of the same uses.

Like bald cypress, it is hardy up through New England and into southern Canada, and is sometimes planted as an ornamental.

Comments:

Pond cypress and bald cypress are nearly identical but differ in a few important respects. Pond cypress needles are linear with a very feathery appearance, while those of bald cypress look more fern-like with 2 rows of flattened needles along a short branchlet. Pond cypress is usually a smaller tree than bald cypress, and its trunk is often more twisted. Pond has flattened knees, while bald knees are conical. Finally, pond cypress grows in small slews and depressions and along stream bottoms, while bald cypress grows in large sphagnum swamps like the Okefenokee swamp.

Pond cypress and bald cypress have been known to interbreed where they occur together, producing offspring exhibiting physical characteristics intermediate between the 2 species.

Pond and bald cypress are not really members of the cypress family (Cupressaceae). Instead, they belong to the Taxodium family which they share with redwoods and giant sequoias.

Red Spruce (*Picea rubens*), Spruce Family

Identification (page 4):

Leaves: needles stiff, angle-sided and sharp pointed, and arranged all around the stem, crushed needles smell like orange rind

Flowers: inconspicuous

Fruit: papery cone that projects down on the branches and remains intact after dropping on the ground

Twigs: orange-brown with ruby red end buds

Bark: dark grayish with scaly plates

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

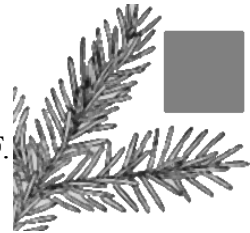
Moist mountain sites above 4500'

Fire Intolerant, No Sprouts

Intermediate, Mature and Climax forests

Red Spruce-Fraser Fir F., Northern Hardwood F.

Blue Ridge northeast of Siler's Bald, GSMNP



Uses (page 20):

Spruce wood is soft, light but strong, resinous and easily split.

Horace Kephart says spruce makes fair kindling when split, but a poor fire that pops constantly. Modern uses include pulpwood, ladder rails, wooden boats, oars, light aircraft and lumber.

Excellent resonance makes red spruce the best wood for sound boards of acoustic guitars and other acoustic string instruments. Volume and sound quality increase with age of the instrument.

Cherokee Indians used the bark for making baskets.

New England sailors made spruce beer to ward off scurvy by boiling twigs and needles, then adding molasses, honey or sugar.

Mountaineers called red spruce "He-balsam" and Fraser fir "She-balsam"(see Fraser fir, page 140).

Comments:

In the southern Appalachians red spruce once grew in mighty virgin forests beside Fraser fir (*Abies fraseri*). Most of the virgin spruce was cut during the early 1900's. In the 1960's and later, the exotic balsam wooly adelgid virtually wiped out the Fraser fir.

This made the remaining red spruce trees more vulnerable to wind throw on the high wind-swept ridge tops where it grows. Acid rain from northern factories has also damaged many red spruce trees.

Eastern Hemlock (*Tsuga canadensis*), Pine Family

For Carolina Hemlock (*Tsuga caroliniana*) see page 135

Identification (page 4):

Leaves: evergreen needles, soft flattened and blunt tipped,
oriented in 2 horizontal rows along the twig, silvery below

Flowers: inconspicuous

Fruit: small cone (3/4-inch long)

Twigs: slender, reddish-brown

Bark: reddish-brown, breaking into flat scaly plates

Ecology (page 12):

Shade Tolerant, Dominant Canopy tree

Moist acidic streams & lower slopes
between 1500 and 4500' elevation

Fire Intolerant, No Sprouts

Intermediate, Mature and Climax forests

Cove F., Acidic Cove Hemlock F.,
Northern. Hardwood F.

Blue Ridge, Cumberland Plateau



Uses (page 20):

Hemlock wood is soft, brittle, coarse-grained, tannin-rich, difficult to split, warps and checks greatly in drying. Horace Kephart says hemlock wood makes a spitting fire, but the bark is excellent tinder on a wet day, and it makes glowing coals in a hurry. Put a layer of hemlock bark over the fire to bank it for the night. Modern uses include pulpwood, crates and general construction.

Botanist Andre Michaux (1746-1802) said Indians made a poultice for burns, sores and wounds by pounding the bark into a plaster.

Pioneers used the tannin-rich bark to tan leather. They made a tea from the needles and twigs to induce sweating, and made a reddish-brown cloth dye from the "middle bark".

Comments:

Eastern hemlock has been decimated by the exotic hemlock wooly adelgid, a tiny white insect that in-mass literally sucks the sap out of the tree, killing it in a few growing seasons. Nicotine root injections have saved many individual trees, and in recent years, exotic predator beetles have been released near infested hemlock stands. Unfortunately, nearly 8000 acres of virgin hemlocks in Great Smoky Mountains have been utterly destroyed.

Fraser Fir (*Abies fraseri*), Pine Family

Identification (page 4):

Leaves: needles, soft, 1-inch long, flat, shiny above, silvery white below, arranged in rows on the twig

Flowers: inconspicuous

Fruit: resin-blotched purple cone that projects up from the branch and disintegrates into scales while on the tree

Twigs: stout, end buds are resinous

Bark: thin, smooth gray with resinous blisters

Ecology (page 12):

Very Shade Tolerant, Dominant Canopy

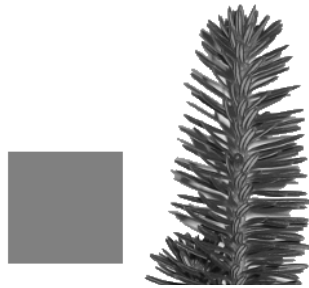
Mountains above 6000 feet elevation

Fire Intolerant, No Sprouts

Intermediate, Mature and Climax forests

Red-Spruce Fraser F.

Blue Ridge



Uses (page 20):

Fraser fir wood is lighter, softer and weaker than that of spruce. It splits easily, is resinous with resinous knots that can chip an axe.

Kephart says Fraser fir makes a quick popping fire soon spent.

Fraser fir is one of America's favorite Christmas trees for its excellent form and aroma, and it is widely grown in commercial plantations where it is protected from the balsam wooly adelgid.

Mountaineers called Fraser fir "She-balsam" because its trunk has swollen resin-filled blisters. They called red spruce "He-balsam".

Legend says Scottish botanist John Fraser and French botanist Andre Michaux were traveling together in the southern Appalachians in the late 1700's. Fraser talked so much that Michaux let him ride ahead on the trail just to be rid of him. In this way Fraser, instead of Michaux, was the first person to discover and describe Fraser fir, which he named for himself!

Comments:

Since the 1960's, the Asian exotic balsam wooly adelgid insect has destroyed 90% of mature Fraser firs. Many young trees still grow, but by about age 20 they too will be attacked and killed. The National Park Service maintains a plantation of native Fraser fir, protected from the adelgid, to help save the tree's genetic diversity.

Carolina Hemlock (*Tsuga caroliniana*), Pine Family

This rare tree is similar to Eastern hemlock (page 139), but it is smaller, has slightly larger cones and its needles emerge all around the twig instead of in 2 horizontal rows.

Carolina hemlock is found in the North Carolina Blue Ridge and scattered places in Virginia and Georgia.

Carolina hemlock also falls victim to the hemlock wooly adelgid.

Atlantic White-cedar (*Chamaecyparis thyoides*), Cypress Family

This rare tree has foliage similar to Eastern redcedar (page 135) except it is blue-green in color, more flattened and has small glandular dots.

Atlantic white-cedar grows in the Florida panhandle and along the Coastal Plain north of North Carolina.

The wood is resistant to decay and is sought after as a premier wood for construction of laminated wooden sailboat hulls. It is also planted as an ornamental for its evergreen habit, hardiness and good form.

Cabbage Palm (*Sabal palmetto*) Palm Family

Cabbage palm is a fan palm with long-tipped palm-shaped leaves extending from a long central stalk. It is evergreen but not a conifer.

Cabbage palm is native to the Florida peninsula and the Georgia and South Carolina lower Coastal Plain.

Indians and Pioneers ate the large end bud of the tree like cabbage. The tree is killed by this practice. Cabbage palm is widely used as an ornamental within its range.

Appendix I Shade Tolerance, **Moisture Preference And Typical** **Canopy Position For Major Southeastern Trees**

Typical canopy position is indicated by one of the following letters in parenthesis after each tree name:

D	Dominant Canopy Tree
d	Dispersed Canopy Tree
S	Subcanopy Tree
U	Understory Tree

There are three tables; one for each of the following:

Coastal Plain	page 143
Piedmont	pages 144 & 145
Blue Ridge, Cumberland Plateau, V. & R.	pages 146-148

Shade tolerance is shown in each table with shade intolerant pioneer trees in row 1, shade intermediate trees in row 2 and shade tolerant trees in row 3.

Moisture preference for each tree is shown by its presence in one of three columns of the table, with the first column representing the most moist, rich forest type/s and the last column representing the driest, poorest forest type/s.

Trees are ranked in the tables based on canopy position and relative abundance of the tree in the forest type.

By descending each column on a given page of the table, the reader can get a good idea of the trees in each moisture category dominating the forest as forest succession proceeds over time, with shade intolerant pioneers dominating early in succession, shade intermediates dominating in middle succession, and shade tolerants dominating in the late succession mature and climax forests.

COASTAL PLAIN PROVINCE (0-500')	Cypress-Gum F. & Bottomland F.	Maritime F. & Pine-Oak F.	Longleaf Pine- Wiregrass F. & Evergreen Oak F.
Shade Intolerant Pioneer Trees	Sweetgum(D) Loblolly P.(D) Slash P.(D) Yellow Pop.(D) B. Willow(d) B. Cherry(d) E. Redcedar(d) Pond P.(d) S. Catalpa(d) Hazel Alder(-)	Sweetgum(D) Loblolly P.(D) Slash P.(D) Longleaf P.(D) B. Willow(d) Cabbage Palm.(d) E. Redcedar(d) Sassafras(d) Laurel Cherry(d)	Longleaf P. (D) Sassafras(d)
Shade Intermediate Trees	Bald Cypress (D) Red Maple(D) Pond Cypress(D) Water O. (D) Live O. (D) Laurel O.(D) Sycamore(D) R. Birch(D) Willow O.(d) Overcup O.(d) Green Ash(d)	Live O. (D) S. Red O.(D) Red Maple(D) Water O. (D) Darlington O.(d) White O.(D) Post O. (D) Black Oak(d) Winged Elm(d-S) Blackgum(S) Bluejack O.(U)	Water O. (D) Live O. (D) Turkey O.(S) Darlington O.(d) Blackjack O.(d) Bluejack O.(U)
Shade Tolerant Trees	Bitternut H.(D) Beech(D) Mockernut H.(D) Sugarberry(d) Persimmon(d) Spruce Pine(d) Sweetbay M.(S) Hornbeam(U) R. Mulberry(U) H. Hornbeam(U) PawPaw(U) S. Sugar M.(U)	Mockernut H.(D) Bitternut H.(D) Pignut H.(D) Red Hic.(D) S. Magnolia(d) A. Holly(S) Red Bay(U) Yaupon H.(U) Redbud(U) F. Dogwood(U)	PawPaw(U)

PIEDMONT PROVINCE (500-1500')	Bottomland F. & Cove F.	Oak-Hickory F.	Pine-Oak F.
Shade Intolerant Pioneer Trees	Sweetgum(D) Y. Poplar(D) Loblolly P.(D) B. Willow(d) Hazel Alder(d) B. Cherry (d) B. Walnut(d) E. Redcedar(d) E. Cottonwd.(d) S. Catalpa(d)	Loblolly P.(D) Sweetgum(D) Y. Poplar(D) Shortleaf. Pine(D) B. Cherry(d) E. Redcedar(d) Sassafras(d) Ga. Hackberry B. Locust(d)	Shortleaf Pine(D) Loblolly P.(D) Va. Pine(D) Sassafras(d)
Shade Intermediate Trees	Red Maple(D) Sycamore(D) Water O. (D) White Oak(D) N. Red Oak(D) Green Ash(d) River Birch (d) Willow O.(d) Slippery. Elm(d) Silver M.(d) Overcup O.(d) A. Elm(d) Boxelder(S) Bigleaf Mag.(S)	White Oak(D) Red Maple(D) S. Red Oak.(D) Scarlet Oak(D) N. Red Oak(D) Black Oak(d) Winged Elm(d-S) Blackgum(d-S) Sourwood(S)	S. Red Oak.(D) Scarlet Oak(D) Red Maple(D) Post O. (D) Black Oak(d) Blackjack O.(d) Chestnut O.(d) Sourwood(S) Blackgum(S)
Shade Tolerant Trees	M. Hickory(D) Beech(D) Red H.(D) Bitt. Hickory(d) Persimmon(d) Silverbell(d) Sugarberry(d) Basswood(d)	M. Hickory(D) Pig. Hick.(D) Red H.(D) Beech(D) Persimmon(d) A. Holly(S) F. Dogwood(U) Redbud(U)	Pig. Hick.(D) Sand H.(d) A. Holly(S) F. Dogwood(U)

PIEDMONT PROVINCE <i>CONTINUED</i>	Bottomland F. & Cove F.	Oak-Hickory F.	Pine-Oak F.
Shade Tolerant Trees continued from previous page	Shagbark H.(d) Hornbeam(U) F. Dogwood(U) R. Mulberry(U) H. Hornbeam(U) Redbud(U) Pawpaw(U) Chalk Maple(U) S. Sugar Map.(U) Fringetree(U)	Pawpaw(U) Serviceberry(U) Chalk Maple(U) Paint. Buckeye(U)	

**See next page for Blue Ridge, Cumberland Plateau and Valley
and Ridge provinces.**

**BLUE RIDGE , CUMBERLAND PLATEAU AND VALLEY
AND RIDGE PROVINCES**

LOW ELEV. ZONE (1500- 2500')	Bottomland F. & Cove F.	Oak-Hickory F & Acidic Hemlock F.	Pine-Oak F.
Shade Intolerant Pioneer Trees	Y. Poplar(D) B. Cherry (d) B. Walnut(d) Sweetgum(d) Cuc-tree(d) B. Willow(d)	Y. Poplar(D) B. Locust(D) Va. Pine(D) B. Cherry(d)	B. Locust(D) Shrt. Pine(D) Va. Pine(D) Pitch Pine(d) Sassafras(d)
Shade Intermediate Trees	Black Birch(D) Red Maple(D) W. Pine(D) Sycamore(d) Green Ash(d) River Birch (d) White Oak(d) N. Red Oak(d) Shingle. Oak(d) Hackberry(S) Boxelder(S)	Black Birch(D) Red Maple(D) White Pine(D) White Oak(D) N. Red Oak(D) Scar. Oak(D) Chest. Oak(D) Blackgum(d) Fraser Mag.(S) Sourwood(S)	Chest. Oak(D) Scar. Oak(D) Black Oak(d) Red Maple(d) White Pine(d) Blackgum(d) Sourwood(S) Chestnut(U)
Shade Tolerant Trees	Hemlock(D) Sug. Maple(D) Beech(D) Buckeye(D) Basswood(d) Bitt. Hickory(d) Silverbell(d) Hornbeam(U) Hazel Alder(U) R. Rhodo.(U) F. Dogwood(U) Umbr. Mag.(U)	Hemlock(D) M. Hickory(D) Beech(D) A. Holly(S) Serviceberry(U) R. Rhodo.(U) Mt. Laurel(U) F. Dogwood(U) Witch Hazel(U) Chestnut(U)	Pig. Hick.(D) Mt. Laurel(U) F. Dogwood(U) Blueberry(U)

MID. ELEV. ZONE (2500- 4500')	Bottomland F. & Cove F.	Oak-Hickory F & Acidic Hemlock F.	Pine-Oak F.
Shade Intolerant Pioneer Trees	Y. Poplar(D) Cuc-tree(d) B. Cherry (d)	Y. Poplar(D) B. Locust(d) Pitch Pine(D) B. Cherry(d)	Pitch Pine(D) Tm. Pine(D) Black Locust(d) Sassafras(d)
Shade Intermediate Trees	Black Birch(D) Red Maple(D) White Ash(D) Y. Birch (d) N. Red Oak(d)	Black Birch(D) Red Maple(d) White Oak(d) N. Red Oak(D) Scar. Oak(D) Chest. Oak(D) Blackgum(d) Fraser Mag.(S) Sourwood(S)	Scar. Oak(D) Chest. Oak(D) Red Maple(D) Blackgum(d) Sourwood(U) Chestnut(U)
Shade Tolerant Trees	Hemlock(D) Sug. Maple(D) Buckeye(D) Basswood(D) Silverbell(D) Beech(d) Bitt. Hickory(d) Serviceberry(U) R. Rhodo.(U) F. Dogwood(U) Str. Maple(U) A. Dogwood(U)	Hemlock(D) M. Hickory(D) Beech(D) Sliverbell(d) A. Holly(S) Serviceberry(U) R. Rhodo.(U) Mt. Laurel(U) F. Dogwood(U) Witch Hazel(U) Chestnut(U) Str. Maple(U)	Pig. Hick.(D) Silverbell(S) Mt. Laurel(U) Blueberry(U)

HIGH. ELEV. ZONE (4500- 6500')	Creeks, Draws & Ravines	Protected Slopes & Ridge Tops	Exposed Slopes & Ridge Tops
Shade Intolerant Pioneer Trees	B. Cherry (d) Pin Cherry(S)	B. Cherry(d) Pin Cherry(S)	B. Cherry(d) Pin Cherry(S)
Shade Intermediate Trees	Y. Birch (D) Red Maple(d) N. Red Oak(d)	Y. Birch(D) Red Maple(d) N. Red Oak(D)	N.Red Oak(D) Y. Birch (D) Red Maple(d)
Shade Tolerant Trees	Beech(D) Buckeye(D) Silverbell(d) R. Spruce(D) Sug. Maple(d) Hemlock(d) Serviceberry(U) R. Rhodo.(U) Str. Maple(U) Mt. Maple(U) A. Dogwood(U) Mt. Holly(U)	Beech(D) Buckeye(D) Silverbell(d) R. Spruce(D) Sug. Maple(d) Hemlock(d) Serviceberry(U) R. Rhodo.(U) Str. Maple(U) Mt. Maple(U) A. Dogwood(U) Mt. Holly(U)	R. Spruce(D) Fraser Fir(D) Mt. Ash(S) Sliverbell(d) Serviceberry(U) Mt. Holly(U) Mt. Maple(U) Hobblebush(U) Cat. Rhodo.(U) Witch Hazel(U) Minnibush(U)

Appendix II Exotic Tree Key

Trees With Deciduous Compound Alternate Leaves

Leaves once compound, small tree with large feathery pink flowers or bean pod fruit-----**Mimosa (*Albizia julibrissin*)**

Leaves once compound, medium tree with brown twisted wing-shaped fruit with the seed in the middle of the wing, leaflet underside has 2 tiny glands.

-----**Tree-of-Heaven (*Ailanthus altissima*)**

Leaves twice compound (leaflets have leaflets), medium tree with many small pinkish flowers or green to yellow round fruit 1/2-inch in diameter.

-----**Chinaberry (*Melia azedarach*)**

Trees With Deciduous Simple Opposite Leaves

Leaves medium-size, eye-shaped, flowers honeysuckle-like or numerous bright red glossy berries, a shrub.

-----**Bush Honeysuckle (*Lonicera spp.*)**

Trees With Deciduous Simple Alternate Leaves

Leaves very large 3-lobed like fig leaves, tree trunk green-----**China Parasol-Tree (*Firmiana spp.*)**

Leaves medium-size, wide eye-shaped with long green leaf stems, tiny yellow flowers on a long slender stalk or splitting brown pods containing a 3-lobed fruit.

-----**Tallowtree (*Triadica sebifera*)**

Leaves green above but silvery scaly below, silvery flowers or red berry-like fruit covered with tiny brown scales, stem may have thorns.

-----**Elaeagnus (*Elaeagnus spp.*)**

Leaves medium-sized, wide eye-shaped, glossy above, numerous white flowers on tree before leaves in early spring or 1/2-inch diameter brownish berry-like fruit, small to medium-sized tree with numerous thorns.

-----**Bradford Pear (*Pyrus calleryana*)**

Leaves long and narrow, toothed and bristle-tipped, fruit an acorn with a very frilly cap, tree medium to large-size.

-----**Sawtooth Oak (*Quercus acutissima*)**

Trees With Evergreen Simple Opposite Leaves

Leaves small, eye-shaped, numerous small white flowers or small dark blue berries, hedge-forming shrub.

-----**Chinese/European Privet (*Ligustrum spp.*)**

Coniferous Trees

Dawn Redwood (*Metasequoia glyptostroboides*)

a deciduous (drops leaves in autumn) conifer
ornamental from China that looks like baldcypress
does not produce cypress knees
leaf twigs have an opposite arrangement

Bibliography

- Banks, William H., Jr. 2004. *Plants of the Cherokee*. Gatlinburg, TN.: Great Smoky Mountains Association.
- Brown, Claud L. and L. Katherine Kirkman. 1990. *Trees of Georgia and Adjacent States*. Portland, OR.: Timber Press.
- Coon, Nelson. 1974. *The Dictionary of Useful Plants*. Emmaus, PA.: Rodale Press
- Dirr, Michael A. *The River Trail A Self-guided Tour To The Indigenous And Adapted Trees Of The University Of Georgia Botanical Garden*
- Edwards, Leslie, Jonathan Ambrose and L. Katherine Kirkman. 2013. *The Natural Communities of Georgia*. Athens, GA. University of Georgia Press.
- Fowells, H.A. 1965. *Silvics of Forest Trees of the United States*. USDA, Forest Service. Agr. Handb. 271. Washington, D.C.: U.S. Gov't. Printing Office.
- Gledhill, D. 1990. *The Names of Plants*. Cambridge and New York.: Cambridge University Press.
- Hageneder, Fred. 2005. *The Meaning of Trees*. San Francisco, CA.: Chronicle Books.
- Harlow, William M. 1957. *Trees of the eastern and central United States and Canada*. New York.: Dover Publications.
- Harper, Francis. 1998. *The Travels of William Bartram*. Athens, GA.: University of Georgia Press.
- Harrar, Ellwood S. and J. George Harrar. 1962. *Guide to Southern Trees*. New York.: Dover Publications.
- Holmes, Sandra. 1974. *Trees Of The World*. New York, NY.: The Ridge Press
- Hudson, Charles. 1976. *The Southeastern Indians*. Knoxville, TN.: University of Tennessee Press.
- Hudson, Charles. 1997. *Knights of Spain, Warriors of the Sun*. Athens and London.: University of Georgia Press.
- Jones, Scott. 2008. *A View to the Past, Experience and Experiment in Primitive Technology*. Charleston, SC.: Scott Jones.
- Kephart, Horace. 1917. *Camping and Woodcraft*. New York: Macmillan Publishing Co.
- Krochmal, Arnold & Connie. 1973. *A Guide to Medicinal Plants of the United States*. New York.: Quadrangle/The New York Times Book Co.

- Lawson, John. 1709. *A New Voyage To Carolina*.
ReadaClassic.com
- Martin, Alexander. Zim, H. S., Nelson, A. L. 1951. *American Wildlife & Plants, A Guide To Wildlife Food Habits*. New York, NY.:Dover Publications Inc.
- Martin, Laura C. 1992. *The Folklore of Trees and Shrubs*. Chester, CT.: Globe Pequot Press.
- Miller, James H. 2003. *Nonnative Invasive Plants of Southern Forests*. Asheville, NC: USDA Southern Research Station
- Panshin, A.J. and Carl de Zeeuw. 1980. *Textbook of Wood Technology*. New York.: McGraw-Hill Book Co.
- Peattie, Donald C. 1977. *A Natural History of Trees of Eastern and Central North America*. Boston, MA.: Houghton Mifflin Co.
- Peterson, Lee A. 1977. *A Field Guide to Edible Wild Plants*. Boston, MA.: Houghton Mifflin Company
- Spornick, Charles D., Alan R. Cattier, and Robert J. Greene. 2003. *An Outdoor Guide to Bartram's Travels*. Athens and London.: University of Georgia Press.
- Underhill, Roy. 1981. *The Woodwright's Shop, Exploring Traditional Woodcraft*. New York.: MJF Books.
- Walker, Laurence C. 1996. *Forests, A Naturalist's Guide To Woodland Trees*. University of Texas Press.
- Wells, Warren R. 1958+-. *A New Look At Some Old Things*. Cincinnati, OH. *Hamilton County Park District*
- Wharton, Charles H. 1978. *The Natural Environments Of Georgia*. Atlanta, Georgia.: Georgia Department of Natural Resources,
- Wigginton, Eliot. 1972. *The Foxfire Book*. Garden City, NY.: Anchor Books.
- Williams, Dan D. 2011. *The Forests of Great Smoky Mountains National Park*. Athens, GA.: Possum Publ.
- Williams, Dan D. 2013. *The Rocks of Georgia, A full-color photo guide to Georgia's rocks*. Athens, GA. *Possum Publications*

Alphabetical Index

Abies fraseri.....	140
Acer floridanum.....	39
Acer leucoderme.....	40
Acer negundo.....	33
Acer pensylvanicum.....	41
Acer rubrum.....	42
Acer saccharinum.....	44
Acer saccharum.....	43
Acer spicatum.....	45
Acidic Cove Hemlock Forest.....	19
Acidic Soil.....	13
Aesculus octandra.....	51
Aesculus parviflora.....	53
Aesculus pavia.....	52
Aesculus sylvatica.....	54
Ailanthus altissima.....	149
Albizia julibrissin.....	149
Alder.....	88
Allegheny Serviceberry.....	86
Alnus serrulata.....	88
Alternate-leaf dogwood.....	46
Amelanchier arborea.....	85
Amelancier laevis.....	86
American Beech.....	98
American Chestnut.....	99
American Elm.....	73
American Holly.....	90
American Hornbeam.....	81
American Sycamore.....	106
Aralia spinosa.....	37
Ash.....	34
Asimina parviflora.....	58
Asimina triloba.....	58
Atlantic White-cedar.....	141
Bald Cypress	136
Basswood	97
Bay.....	67
Beech.....	98
Betula alleghaniensis.....	75
Betula lenta.....	76

Betula nigra.....	77
Bigleaf Magnolia	63
Birch.....	76
Bitternut Hickory.....	23
Black Birch.....	76
Black Cherry.....	93
Black Locust.....	30
Black Oak.....	123
Black Walnut.....	32
Black Willow.....	87
Blackgum.....	70
Blackjack Oak.....	119
Blocky bark.....	10
Blue Ridge.....	18
Bluejack Oak.....	116
Bottlebrush Buckeye.....	53
Bottomland Forest.....	19
Boxelder Maple.....	33
Bradford Pear.....	150
Buckeye.....	53
BUCKEYES.....	50
Buckthorn.....	83
Bush Honeysuckle.....	149
Cabbage Palm.....	141
Calcareous (calcium-rich) Forest.....	19
Carolina Buckthorn.....	83
Carolina Hemlock.....	141
Carolina Laurelcherry.....	94
Carolina Silverbell	84
Carpinus caroliniana.....	81
Carya cordiformis.....	23
Carya glabra.....	24
Carya illinoensis.....	29
Carya ovalis.....	25
Carya ovata.....	26
Carya pallida.....	27
Carya tomentosa.....	28
Castanea dentata.....	99
Catalpa.....	55
Catalpa bignonioides.....	55
cedar.....	135

Celtis laevigata.....	80
Celtis occidentalis.....	78
Celtis tenuifolia.....	79
Cercis canadensis.....	66
Chalk maple.....	40
Chamaecyparis thyoides.....	141
Cherry.....	93
Chestnut	99
Chestnut Oak.....	111
China Parasol-Tree.....	149
Chinaberry.....	149
Chinese/European Privet.....	150
Chionanthus virginicus.....	49
Climax Forest.....	17
Coastal Plain.....	18
Coniferous Trees	150
Cornus alternifolia.....	46
Cornus florida.....	47
Cornus stricta.....	48
Cottonwood.....	96
Cove Forest.....	19
Cucumbertree.....	64, 68
Cumberland Plateau.....	18
Cypress	136
Cypress-Gum Swamp.....	19
Darlington oak.....	118
Dawn Redwood.....	150
Deciduous	149
Developing Forest.....	15
Devil's Walkingstick.....	37
Diospyros virginiana.....	69
Dispersed canopy tree.....	14
dogwood.....	46
Dominant canopy tree.....	14
Downy Serviceberry.....	85
Dry Site Tree.....	12
Eastern Cottonwood.....	96
Eastern Hemlock.....	139
Eastern Hophornbeam.....	82
Eastern redbud.....	66
Eastern Redcedar	135

Eastern White Pine.....	125
Elaeagnus.....	149
Elaeagnus spp.....	149
Elderberry.....	37
Elm.....	72
Evergreen	150
Evergreen Oak Forest.....	19
Fagus grandifolia.....	98
Fibrous bark.....	10
Fir.....	140
Fire Cherry).....	95
Fire Dependent.....	14
Fire Intolerant.....	14
Fire Resistant.....	14
Fire Tolerance Rating.....	14
Fire Tolerant.....	14
Firmiana spp.....	149
Florida maple.....	39
Flowering Dogwood.....	47
Forest Succession.....	15
Fraser Fir.....	140
Fraser Magnolia.....	62
Fraxinus americana.....	35
Fraxinus pennsylvanica.....	34
Fringetree.....	49
Georgia Hackberry.....	79
Gleditsia triacanthos.....	31
Green Ash.....	34
Hackberry.....	78
Halesia carolina.....	84
Hazel Alder.....	88
Hemlock	139
HICKORIES.....	22
Hickory.....	23
Holly.....	89
Honeylocust.....	31
Honeysuckle.....	149
Hophornbeam.....	82
Hornbeam.....	81
How To Identify A Tree Using This Book.....	11
Ilex montana.....	89

<i>Ilex opaca</i>	90
<i>Ilex vomitoria</i>	91
Intermediate Forest.....	16
Introduction.....	3
<i>Juglans nigra</i>	32
<i>Juniperus virginiana</i>	135
Laurel Oak	118
Laurelcherry.....	94
<i>Ligustrum</i> spp.....	150
<i>Liquidambar styraciflua</i>	101
<i>Liriodendron tulipifera</i>	103
Live Oak.....	113
lobes and teeth.....	6
Loblolly Pine.....	127
locust.....	30
Longleaf Pine.....	128
Longleaf Pine-Wiregrass Forest.....	19
<i>Lonicera</i> spp.....	149
<i>Magnolia</i>	60
<i>Magnolia acuminata</i>	64, 68
<i>Magnolia fraseri</i>	62
<i>Magnolia grandiflora</i>	60
<i>Magnolia macrophylla</i>	63
<i>Magnolia tripetala</i>	65
<i>Magnolia virginiana</i>	61
MAGNOLIAS.....	59
Major Southeastern Provinces and Forest Types.....	18
Maple.....	39
Maritime Forest.....	19
Mature Forest.....	16
<i>Melia azedarach</i>	149
<i>Metasequoia glyptostroboides</i>	150
<i>Mimosa</i>	149
Mockernut Hickory.....	28
Moist Site Tree.....	12
<i>Morus alba</i>	105
<i>Morus rubra</i>	104
Mountain Holly.....	89
Mountain Maple.....	45
Mountain-Ash.....	36
Mulberry.....	104

Native Shrubs With Compound Leaves.....	37
Neutral to slightly Basic Soil.....	13
Northern Hardwood Forest.....	19
Northern Red Oak.....	122
<i>Nyssa sylvatica</i>	70
Oak.....	108
Oak-Hickory Forest.....	19
Oak-Pine Forest.....	19
Old-mans-beard.....	49
opposite versus alternate leaves.....	5
<i>Ostrya virginiana</i>	82
Overcup Oak.....	109
<i>Oxydendrum arboreum</i>	92
Painted Buckeye.....	54
Palm.....	141
Paulownia	56
<i>Paulownia tomentosa</i>	56
Pawpaw.....	58
Pear.....	150
Pecan.....	29
Peeling Bark.....	10
<i>Persea borbonia</i>	67
Persimmon.....	69
<i>Picea rubens</i>	138
Piedmont.....	18
Pignut Hickory	24
Pin Cherry	95
Pine.....	125
Pine-Oak Forest.....	19
Pines And Other Conifers.....	124
Pines with 2 and 3 needles per bundle:.....	124
Pines with 2 needles per bundle.....	124
Pines with 3 needles per bundle.....	124
Pines with 5 needles per bundle.....	124
<i>Pinus echinata</i>	134
<i>Pinus elliotii</i>	126
<i>Pinus glabra</i>	132
<i>Pinus palustris</i>	128
<i>Pinus pungens</i>	133
<i>Pinus rigida</i>	129
<i>Pinus serotina</i>	130

Pinus strobus.....	125
Pinus taeda.....	127
Pinus virginiana.....	131
Pioneer Forest	16
pioneer trees.....	15
Pitch Pine.....	129
Platanus occidentalis.....	106
Platy bark.....	10
Pond Cypress.....	137
Pond Pine.....	130
Poplar.....	103
Populus deltoides.....	96
Post Oak.....	112
Privet.....	150
Prunus caroliniana.....	94
Prunus pensylvanica.....	95
Prunus serotina.....	93
Pyrus calleryana.....	150
Quercus acutissima.....	150
Quercus alba.....	108
Quercus coccinea.....	114
Quercus falcata.....	115
Quercus hemisphaerica.....	118
Quercus incana.....	116
Quercus laevis.....	117
Quercus laurifolia.....	118
Quercus lyrata.....	109
Quercus marilandica.....	119
Quercus nigra.....	120
Quercus phellos.....	121
Quercus prinus.....	111
Quercus rubra.....	122
Quercus stellata.....	112
Quercus velutina.....	123
Quercus virginiana.....	113
Red Bay.....	67
Red Buckeye.....	52
Red Elm.....	74
Red Hickory.....	25
Red Maple.....	42
Red Mulberry.....	104

Red Spruce.....	138
Red Spruce-Fraser Fir Forest.....	19
redbud.....	66
Redcedar.....	135
Rhamnus caroliniana.....	83
Rhus spp.....	37
Ridged bark.....	10
River Birch.....	77
Robinia pseudoacacia.....	30
Royal Paulownia	56
Sabal plametto.....	141
Salix nigra.....	87
Sambucus canadensis.....	37
Sand Hickory.....	27
sand laurel oak.....	118
Sassafras.....	102
Sassafras albidum.....	102
Sawtooth Oak.....	150
Scarlet Oak.....	114
Serviceberry.....	85
shade intermediate tree.....	16
Shade Intermediate tree.....	12
shade intolerant tree.....	15
Shade Intolerant tree.....	12
Shade Tolerance.....	12
Shade tolerant tree.....	16
Shade Tolerant tree.....	12
Shagbark Hickory.....	26
Shaggy/Flaky/Scaly bark.....	10
Shortleaf Pine.....	134
Silver Maple.....	44
Silverbell	84
Simple versus compound leaves.....	4
Slash Pine.....	126
Slippery Elm.....	74
Smooth bark.....	10
Soil Moisture Preference.....	12
Soil PH and Soil Nutrients.....	13
Sorbus americana.....	36
Sourwood.....	92
Southeastern forest types by province.....	19

Southern Catalpa.....	55
Southern Magnolia.....	60
Southern Red Oak.....	115
Southern Sugar Maple.....	39
Spruce.....	138
Spruce Pine.....	132
Striped Maple.....	41
Subcanopy tree.....	14
Sugar Maple.....	43
Sugarberry.....	80
Sumac.....	37
Swamp Dogwood.....	48
Sweetbay Magnolia.....	61
Sweetgum.....	101
Sycamore.....	106
Table-mountain Pine.....	133
Tallowtree.....	149
Taxodium ascendens.....	137
Taxodium distichum.....	136
Tilia heterophylla.....	97
Tree Bark Types.....	10
Tree Ecology And Forest Succession	12
Tree Identification	4
Tree-of-Heaven.....	149
Trees With Compound Leaves.....	22
Trees With Deciduous Compound Alternate Leaves.....	149
Trees With Deciduous Simple Alternate Leaves.....	149
Trees With Deciduous Simple Opposite Leaves.....	149
Trees With Evergreen Simple Opposite Leaves.....	150
Trees With Lobed Leaves.....	100
Trees With Opposite Leaves.....	38
Trees With Simple Alternate Toothed Leaves.....	71
Trees With Simple Alternate Toothless Leaves.....	57
Triadica sebifera.....	149
Tsuga canadensis.....	139
Tsuga caroliniana.....	141
Tulip Poplar.....	103
Turkey Oak.....	117
Ulmus alata.....	72
Ulmus americana.....	73
Ulmus rubra.....	74

Ultramafic Soil.....	13
Umbrella Magnolia.....	65
Understory tree.....	14
Valley and Ridge.....	18
Vertical Position in the Forest Canopy.....	14
Virginia Pine.....	131
Water Oak.....	120
Wet Site Tree.....	12
White Ash.....	35
White Mulberry.....	105
White Oak.....	108
White Pine.....	125
White-cedar.....	141
Willow.....	87
Willow Oak.....	121
Winged Elm.....	72
Yaupon Holly.....	91
Yellow Birch.....	75
Yellow Buckeye.....	51
Yellow Poplar.....	103